

QUASI-RANDOM MATCHES: EVIDENCE FROM DUAL LABOR MARKETS*

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Abstract

A fast-growing literature studies how sorting into specific jobs, firms, or locations affects workers. However, a key challenge is the non-random sorting of workers. We propose an identification strategy that exploits the *timing* of worker-firm matching, by interacting high-frequency information on contract durations on the labor supply and transitory fluctuations in job creation on the labor demand side. We apply this method to address a central question in *dual labor markets*: how do different contract types—fixed-term vs. permanent—affect workers’ careers? We find that transitory variation in the opening of permanent contracts is highly predictive of individual contract upgrade probabilities. Securing a permanent position translates into higher employment and earnings growth in the short run. However, despite lasting gains in work experience, the long-run wage advantage is statistically indistinguishable from zero, and the estimate is too imprecise to establish whether the initial earnings gain persists or fades. We argue that the negative effect of open-ended contracts on worker mobility may help to explain the absence of a detectable long-run wage return to the additional experience.

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1 Introduction

A fast-growing literature studies how sorting into particular jobs, firms, or locations affects workers. For example, there has been much interest in the observation that pay premia vary across firms (Abowd et al., 1999), the mechanisms that generate such variation (Manning, 2021; Card et al., 2018), and its implications (Card et al., 2013). A natural question then is whether jobs also differ in their *dynamic* implications—if workers learn more and enjoy faster earnings growth in some jobs while being “stuck” in others. Recent studies suggest that earnings growth varies systematically across firms (Arellano-Bover and Saltiel, 2021; Pesola, 2011), regions (Roca and Puga, 2017), and jobs (Kambourov and Manovskii, 2009; Gathmann and Schönberg, 2010; Garcia-Louzao et al., 2023).

The key challenge when studying such questions is the non-random sorting of workers to jobs. For example, firms paying higher wages might attract better applicants, and workers in urban labor markets might differ from those in rural areas. To address this selection problem, the literature often adopts a fixed effects strategy: by tracking workers across firms, researchers can decompose wages into time-constant differences between individuals (individual fixed effects) and match-specific components (such as firm fixed effects, as in Abowd et al. 1999). While this strategy is widespread across settings, there is an obvious tension: if workers or firms differ in their *level* of pay, they might also differ in wage *growth*, which the fixed effects would not capture.

We propose an alternative strategy that exploits the *timing* of worker-firm matching. Specifically, we isolate quasi-random variation in matches by interacting high-frequency information on (i) the duration of contracts on the supply side of the labor market and (ii) transitory fluctuations in job creation on the demand side. We apply this method to address a central question in “dual” labor markets: how do different contract types—fixed-term or open-ended contracts—affect workers’ careers? A common concern is that fixed-term contracts may discourage firms from providing training or other investments to their workers (Cabrales et al. 2017; Albert et al. 2005). While we focus on the consequences for workers, this problem has important aggregate implications, and the prevalence of fixed-term contracts is one suspected reason for low labor productivity in countries characterized by dual labor markets (Cahuc et al. 2016).¹

Our application focuses on Spain. With the highest rate of temporary employment in the OECD until 2021, at nearly 25% of dependent employment (see Figure 1), and as much as 90% of new contracts being fixed-term (until a recent labor reform in 2022), the country provides an interesting context. We also exploit comprehensive matched employer–employee data from Social Security records, which follow workers longitudinally and provide detailed information on the type and duration of each employment contract.

¹Labor market duality has also been linked to outcomes such as fertility (Auer and Danzer, 2016; Lopes, 2020; Nieto, 2022) and internal migration (Llull and Miller, 2018).

We first provide evidence using a standard fixed effects approach, estimating an earnings equation that allows for time-constant differences between individuals and different rates of worker experience gained in fixed-term or open-ended contracts. Consistent with recent evidence by [Garcia-Louzao et al. \(2023\)](#), we find that earnings growth is higher for workers with more experience in open-ended contracts: evaluated at ten years of experience, an additional year of fixed-term experience raises earnings by 3.0%, against 4.5% for a year of open-ended experience. Adding worker fixed effects attenuates the gap but does not remove it (4.6% versus 5.6%). These patterns are highly non-linear, and the gap is much greater for experienced than for young, inexperienced workers. An intuitive interpretation of these findings is that fixed-term contracts slow skill acquisition and wage growth (i.e., differences in returns to experience). However, they could also be due to different workers experiencing different wage growth *irrespective* of contract type (i.e., selection).

A key piece of evidence to distinguish between these competing interpretations is an event study graph studying wage growth around contract switches. For example, [Card et al. \(2013\)](#) show that workers who switch from low-to higher-paying firms tend to experience similar wage growth as those who make the reverse switch (“parallel pre-trends”), suggesting that worker-firm matching is sufficiently random in a dynamic sense. However, we show that the parallel trends assumption does not hold in dual labor markets: workers who switch into an open-ended contract as opposed to another fixed-term contract experienced higher wage growth even *before* they entered their new contract. The difference is sizable: conditional on age and time fixed effects, earnings of workers who switch to an open-ended contract increase by 7 log points, on average, over the 15 months before the switch, whereas earnings growth is negligible among workers who instead move to another fixed-term contract. This observation suggests that matching workers to contract types is not random in a dynamic sense: the differences in wage growth between fixed-term and open-ended contracts primarily reflect heterogeneity between workers rather than differences in returns between contract types.

The selection problem is, therefore, more severe for contract types than for firms ([Card et al., 2013](#)) or regions ([Roca and Puga, 2017](#); [Card et al., 2025](#)). Conversion to an open-ended contract typically occurs within the firm, allowing employers to condition decisions on information that is largely unobserved to the researcher and potentially use temporary contracts as a screening device ([Faccini, 2014](#); [Bentolila et al., 2023](#)). Conversion within a firm can also amount to a promotion, with the decision closely tied to recent performance. Moreover, higher-ability workers may sort into higher-quality fixed-term contracts to begin with, which can serve as genuine stepping stones toward permanent employment. These mechanisms all imply that workers who eventually convert may already be on different earnings trajectories before conversion, making the pre-conversion divergence in outcomes precisely what selection would predict.

Our paper contributes to two distinct strands of the literature. Methodologically, it relates to recent work that extends the standard two-way fixed effects framework to accommodate richer forms of heterogeneity. For example, [Roca and Puga \(2017\)](#) allow city-specific experience to interact with worker fixed effects, showing that workers differ in the returns they obtain from employment in larger cities. Similarly, [Arellano-Bover and Saltiel \(2021\)](#) document substantial heterogeneity in learning across firms and classify employers into latent skill-learning types that are not captured by observable characteristics. More broadly, this literature emphasizes that labor market outcomes depend on interactions between workers and their economic environments rather than on worker and employer characteristics alone.

Our approach applies this insight to contract types but adopts a different identification strategy. We do not model heterogeneity and dynamic selection directly; instead, we exploit quasi-random variation in access to permanent contracts. We interact predetermined variation in the expiration dates of fixed-term contracts with transitory fluctuations in the availability of open-ended jobs, generating exogenous variation in contract type. This strategy allows us to estimate heterogeneous returns to permanent employment while circumventing the dynamic selection problem that arises when contract conversions reflect unobserved worker performance.

Conceptually, our strategy is similar to studies that analyze the effects of labor market conditions at the entry on worker careers—“graduating in a recession”—([Oreopoulos et al., 2012](#); [Kahn, 2010](#)), in particular, recent work by [Arellano-Bover \(2024\)](#) on the selection of workers into different firm types. However, instead of relying on yearly variation in the labor market entry of recent graduates, we exploit high-frequency information on contract durations. The precision of administrative employment records, allow us to match the precise month when the individual’s contract is about to end with transitory variation in job openings at the province level. Our approach faces the usual challenges in establishing instrument relevance and validity. The upside, however, is that we do not have to specify the functional form of individual heterogeneity and dynamic selection.

Under our identifying assumptions, the IV estimates recover a Local Average Treatment Effect ([Angrist et al., 1996](#)). Although this parameter need not generalize to the full population, it is directly policy relevant as it captures the effect of permanent contracts on workers whose contract status responds to fluctuations in the availability of open-ended jobs. Compliers are positively selected on baseline earnings and are substantially less likely to be employed at young firms. They are also statistically indistinguishable from the full sample in terms of education, age, and gender.

We begin by establishing instrument relevance. The leave-one-out measure of newly created open-ended jobs in a province strongly predicts whether workers whose fixed-term contracts expire in a given month transition into permanent employment. We then assess the plausibility of the independence and exclusion restrictions. Independence requires

that, from the worker’s perspective, exposure to unusually favorable conditions in the market for open-ended jobs at the time of contract expiration is effectively random. Consistent with this assumption, the instrument displays little systematic relationship with observable worker and firm characteristics.

By contrast, the exclusion restriction is unlikely to hold mechanically without further controls. The flow of new open-ended contracts is naturally correlated with aggregate and local business cycle conditions, raising the concern that observed wage growth may reflect favorable macroeconomic conditions rather than the causal effect of starting in an open-ended contract. Our strategy isolates the high-frequency timing of job transitions from lower-frequency economic fluctuations. We exploit month-to-month variation in the composition of available contracts while flexibly controlling for broader local and aggregate labor market conditions. Identification comes from differences in labor market opportunities faced by otherwise similar workers whose fixed-term contracts expire at different points in time.

To our knowledge, this paper is the first to use quasi-random variation in timing to address endogenous sorting into contract types. More broadly, the approach illustrates how the high-frequency information contained in administrative records can be used to generate credible variation in labor market opportunities. While administrative data are often valued for their scale and longitudinal coverage, they also provide exceptionally precise information on contract start and end dates. We exploit this high-frequency variation for identification.

Beyond its methodological contribution, the paper speaks directly to the large literature on dual labor markets ([Bentolila et al., 2020](#); [Boeri and Garibaldi, 2024](#)). Beginning in the 1980s, many European countries liberalized the use of fixed-term contracts in an effort to increase hiring flexibility and reduce structural unemployment. While these reforms may have lowered firms’ adjustment costs, they also created a sharp divide between workers with stable employment relationships and those cycling through temporary jobs.

A central concern in this literature is that fixed-term contracts may slow human capital accumulation and delay career progression. Temporary workers receive less firm-provided training ([Cabrales et al., 2017](#); [Bratti et al., 2021](#)), face greater uncertainty regarding conversion to permanent employment, and may experience slower wage growth as a result ([Gagliarducci, 2005](#)). Whether temporary contracts ultimately serve as stepping stones or dead ends appears to depend on the institutional environment. In labor markets with relatively low employment protection, temporary jobs can facilitate transitions into stable employment ([Bentolila et al., 2020](#)). By contrast, in countries such as Spain and Italy, where temporary employment is widespread and protection gaps across contract types are substantial, temporary jobs often lead to repeated spells of precarious employment rather than durable labor market attachment ([Güell and Petrongolo, 2007](#); [García-Pérez and Muñoz-Bullón, 2011](#); [Garcia-Louzao et al., 2023](#)).

Despite extensive evidence on transitions between contract types, credible estimates of the causal effect of obtaining a permanent contract remain scarce because workers who convert differ systematically from those who do not. By addressing this selection problem, our paper provides direct evidence on the extent to which permanent contracts shape workers' earnings trajectories and long-run labor market outcomes.

Using our instrumental variables strategy, we find that obtaining an open-ended contract generates substantial short-run earnings gains. These gains operate primarily through enhanced employment stability: workers who transition to a permanent contract are 17.6 percentage points more likely to be employed 12 months later, whereas workers in the control group frequently cycle through periods of non-employment between fixed-term jobs. As a result, treated workers accumulate roughly five additional months of work experience during the first year after conversion, and that experience is disproportionately experience under open-ended contracts, relative to workers who do not immediately transition into permanent employment.

Despite these initial gains, the additional experience does not translate into higher pay. The total earnings effect itself persists: five years after conversion it stands at 22.4% of baseline earnings and remains statistically significant. Decomposing it, however, shows that it operates almost entirely through employment rather than through wages. The employment margin contributes 0.170 of the 0.224 total, while the within-firm and between-firm components of wage growth are 0.058 and -0.004 , and neither is distinguishable from zero. Some convergence in the control group is expected as these workers gradually obtain permanent contracts themselves (Booth et al., 2002), but the data show that treated workers accumulate 13 additional months of total work experience over the same period, with a disproportionate increase in experience in permanent employment, without a corresponding wage advantage. Early access to permanent contracts therefore alters employment trajectories substantially while generating no measurable wage return to the additional experience, even though the wage estimates are not precise enough to rule out a modest return.

One potential explanation for the fading long-run earnings gains is that permanent contracts alter the rate at which workers convert accumulated experience into external labor market opportunities. Workers who transition into open-ended contracts experience a substantial and persistent reduction in labor market mobility. They are significantly less likely to switch province or industry and more likely to remain within the firm where conversion occurs. In a labor market with heterogeneous match quality, this decline in reallocation may slow the transition toward higher-productivity employers, attenuating the returns to experience over time.

This interpretation is consistent with canonical models of wage dynamics in which earnings growth reflects both within-job human capital accumulation and between-job reallocation toward better matches (Postel-Vinay and Robin, 2002; Cahuc et al., 2006). Re-

cent structural work on dual labor markets emphasizes that employment protection generates two opposing forces: increased stability and tenure accumulation within matches, and reduced job-to-job mobility that limits access to higher-wage firms (Cabrales et al., 2017; Busch et al., 2025). Our empirical design allows us to assess the net implication of these forces. We find that conversion into a permanent contract increases employment stability and experience accumulation, while simultaneously reducing mobility along both geographic and sectoral margins. However, this reduction in mobility is not sufficient to explain the long-run earnings patterns: the shortfall is concentrated among workers whose baseline matches exhibit weak earnings-growth trajectories, while workers in high-growth firms retain substantial earnings gains even though their mobility falls by less. Five years after conversion, workers converted in firms with above-median earnings growth gain 46.0% of baseline earnings, against a statistically insignificant 17.8% for workers converted in below-median-growth firms. We also find differences in employment of 28.1 pp and 15.7 pp, respectively, implying that, on average, workers in low-earnings-growth firms experience an earnings loss relative to workers who are not initially upgraded to a permanent contract. The two groups accumulate almost identical amounts of additional experience over this period (1.27 versus 1.18 years), yet only one group receives a corresponding earnings gain. This contrast is difficult to reconcile with explanations based on contract type alone, such as a low return to permanent-contract experience, weaker incentives, or compressed pay, since each would predict a lower return to experience regardless of the retaining firm.

These patterns point to an interpretation in which mobility constraints interact with match-specific wage growth schedules. Permanent contracts generate persistent attachment to the conversion firm, and the consequences of this attachment depend on the underlying growth potential of the match. When firms exhibit steep internal wage trajectories, reduced mobility is largely inconsequential or even reinforcing, as workers remain on favorable growth paths. In contrast, when firms offer limited wage progression, reduced reallocation prevents workers from accessing better external matches, leading to declining relative earnings over time.

Finally, we provide evidence on behavioral responses consistent with changes in perceived income stability. Workers who obtain a permanent contract are more likely to have an additional child, consistent with delayed family formation under uncertainty. The response is concentrated among women, where the probability rises steadily after conversion and peaks about two years later before narrowing; for the full sample it is positive but not statistically significant. The timing suggests that workers interpret contract conversion as a discrete improvement in employment security rather than a temporary shock. This pattern aligns with evidence that employment protection affects household timing decisions through perceived income stability rather than contemporaneous income changes (Adsera, 2004; Rica and Iza, 2005; Guner et al., 2024).

We assess the results to a range of robustness checks that address both measurement concerns and the validity of the identifying assumptions. We first validate the instrument by comparing the MCVL-based measure of newly created open-ended contracts with aggregate Social Security population data. Despite relying on a 4% sample, the MCVL series closely tracks aggregate dynamics, both in levels and after removing time effects, indicating that it captures meaningful variation in contract creation across provinces and months. We also show that the results are robust to alternative definitions of the instrument constructed at different levels of aggregation, including national, regional, and sectoral measures.

To assess the plausibility of the exclusion restriction, we conduct a placebo test restricting the sample to workers who remain in fixed-term employment. In this group, exposure to fluctuations in open-ended job creation has no systematic effect on either employment or earnings, suggesting that the instrument is neither proxying for general business cycle conditions nor operating on wages through improved outside options. Additional robustness checks show that the results are stable across alternative sample restrictions and specification choices, including focusing on terminations due to plausibly exogenous reasons and varying tenure windows. We also instrument employment and contract status jointly; there the earnings effect loads onto the employment margin and the direct contract effect is no longer distinguishable from zero, reinforcing the decomposition in Section 6. Finally, augmenting the baseline specification with more demanding controls for aggregate and local economic conditions—such as high-dimensional fixed effects and leave-one-out measures of local outcomes—yields only modest attenuation of the estimates.

The paper is organized as follows: Section 2 provides a background of the institutional framework, Section 3 introduces the main data source, Sections 4 and 5 discuss the selection problem and our identification strategy, respectively, and Section 6 analyzes the effect of contract upgrade in workers’ career trajectory. Section 7 provides additional robustness checks, and Section 8 concludes.

2 Institutional Background

Following the democratic transition, Spain’s labor-market institutions underwent significant changes. The Workers’ Statute (*Estatuto de los Trabajadores*), enacted in 1980, established open-ended contracts as the standard employment arrangement and restricted fixed-term contracts to specific circumstances, such as seasonal employment and short-term replacements. The Statute also introduced severance regulations, setting compensation for temporary contracts at 8 days of wages per year of service, compared to up to 45

days for permanent ones. In practice, the use of fixed-term contracts remained limited.²

The dual structure of the Spanish labor market emerged with the enactment of *Law 32/1984*, which liberalized the use of temporary contracts. Fixed-term contracts, previously restricted to seasonal or temporary activities, were opened to general use and grew rapidly. Firms could thereafter choose between permanent and temporary contracts, but the two differed markedly in their associated labor costs and legal exposure. Open-ended contracts entailed substantial dismissal costs, up to 45 days of wages per year of tenure in cases of wrongful dismissal, and exposed firms to litigation risk and procedural uncertainty, making separations both costly and relatively infrequent. Temporary contracts, by contrast, expired at a predetermined date, required limited severance upon expiration, and were not subject to judicial review, making them a substantially more flexible margin of adjustment.³

Despite successive reforms, the gap between the two contracts persisted. The 1994 reform restricted the use of temporary contracts to seasonal activities and eased dismissal conditions for permanent employees, but employers continued hiring temporary workers for non-seasonal roles (Bentolila et al., 2012; García-Pérez et al., 2019). The perceived ineffectiveness of these measures prompted further reforms in 1997 and 2001, neither of which substantially changed the underlying asymmetry. Not until 2012 did severance payments for permanent employees reduced across all open-ended contracts while temporary contract termination compensation rose, narrowing the gap from both sides.⁴ Yet the reform did not eliminate market segmentation. The share of workers on fixed-term contracts remained above 20% thereafter, and part of the subsequent decline reflects employment losses during the Great Recession rather than a fundamental change in firms' hiring practices. The distinction between the two types of contracts remained important throughout the period. The most recent reform, enacted in December 2021 and fully effective since March 2022, does appear to have reduced the use of fixed-term contracts substantially (Conde-Ruiz et al., 2023), but it falls outside our sample period.⁵

Over the past decade, approximately one-third of Spanish workers held a fixed-term

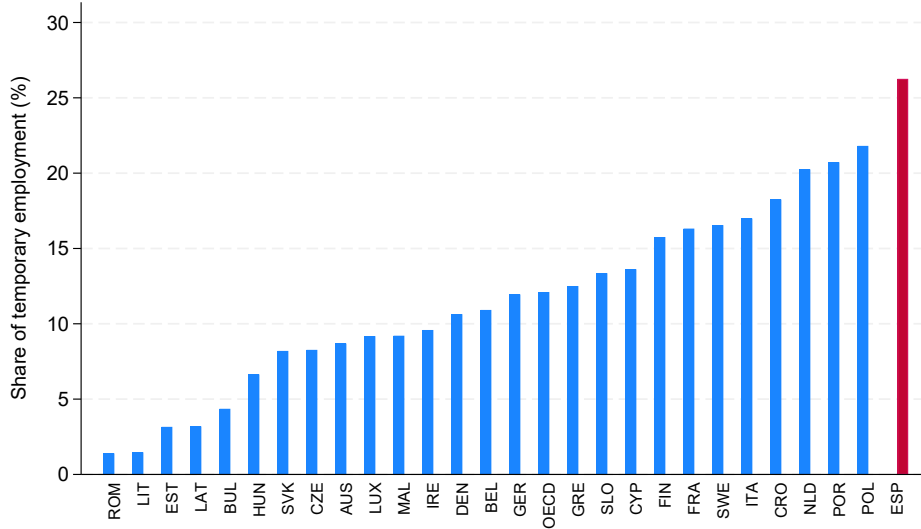
²During the early years of the transition (1975–1982), policymakers sought to prioritize stable employment. Over time, however, growing labor market pressures made it increasingly difficult to sustain a system based exclusively on open-ended hiring, contributing to the 1984 liberalization that led to the expansion of temporary contracts (Galacho, 2006).

³If terminated before their scheduled expiration date, temporary contracts are subject to the same employment protection as permanent contracts and may therefore be challenged through judicial review.

⁴The 2012 reform raised compensation for terminating a temporary contract from 8 to 12 days of wages per year worked, while reducing compensation for *unfair dismissals* of permanent contracts from 45 to 33 days. Compensation for *fair* separations of permanent contracts remained unchanged at 20 days per year worked (see Dolado 2017). The reform also eliminated interim wages during judicial processes and introduced a new open-ended contract for firms with fewer than 50 employees.

⁵The reform abolished project-based contracts and replaced them with a single temporary contract for structural needs, the *circumstances of production* contract, with a maximum duration of six months, extendable to one year by collective bargaining agreements, and tied to demand peaks such as Christmas or agricultural harvests. It also promoted *intermittent-permanent* contracts, a highly flexible modality that has expanded considerably since.

Figure 1: Proportion of workers in temporary contracts by country, 2019



Notes: Share of dependent employees in temporary employment, OECD countries. Source: OECD, Employment and Labour Market Statistics.

contract in any given year, while temporary contracts accounted for as much as 90% of new hires. Despite a decline following the Great Recession (Appendix Figure B.1.1), Spain recorded the highest share of temporary employment among OECD countries until the 2021 reform, when the share stayed close to 25% of dependent employment (Figure 1).⁶

Duality is not restricted to low-skilled or seasonal work. Table C.1.1 shows that the share of high-skilled workers employed on fixed-term contracts rose over the early part of the period before stabilizing, indicating that exposure to temporary employment extends to also high skilled workers. Temporary employment is also rarely a stepping stone to stability: for many workers it leads to another temporary job or a spell of non-employment (Bentolila et al., 2020; Güell and Petrongolo, 2007; García-Pérez and Muñoz-Bullón, 2011). This persistence is most pronounced where the gap in employment protection between contract types is large, as in Spain and Italy.

Because the difference in employment protection runs in one direction, worker preferences over contract types are monotone. Holding wages constant, an open-ended contract strictly dominates a fixed-term contract: it provides protection against dismissal, eliminates the risk of contract expiration, and facilitates access to credit, housing, and household formation. This distinguishes our setting from those typically considered in the two-way fixed effects literature, where workers differ in their preferences over firms or locations and such heterogeneity can attenuate positive sorting on ability (Card et al., 2013, 2025). Here, workers have a common ranking over contract types.

⁶The 2022 labor market reform reduced temporary employment in Spain (Appendix Figure B.1.3), although existing evidence suggests more limited effects on job duration and short-term labor market transitions (Conde-Ruiz et al., 2023).

A fixed-term contract specifies its expiration date when it is signed. Contract duration is predetermined and known to both parties. Duration is also legally constrained. During our sample period, workers who accumulated more than twenty-four months of employment within a thirty-month period under two or more temporary contracts with the same employer were entitled to permanent status. Consequently, the month in which a contract expires is determined largely by its signing date and prevailing duration conventions, not by labor market conditions. At expiration, the firm chooses whether to terminate the relationship, renew the temporary contract, or convert it into a permanent one. Section 5 builds on these characteristics.

3 Data

Our main data source is the 2006–2021 waves of the Continuous Sample of Working Lives (Muestra Continua de Vidas Laborales, MCVL), a 4% non-stratified random sample of individuals with any relationship with Spain’s Social Security system in a given year. The data reconstruct complete employment histories from 1967 onward. The unit of observation is a change in an individual’s labor market status—transitions across jobs, changes in occupation, or updates to contract conditions within a firm. Sampled individuals are followed in all subsequent waves until they exit the system (e.g., due to death, migration, or prolonged inactivity). The sample is refreshed each wave with new entrants, whose complete employment histories are also observed.

The MCVL records the exact start and end dates of each contract spell, as well as updates to contract and occupational status within spells.⁷ This allows us to measure workers’ labor market trajectories at a granular level, and is central to our identification strategy. We aggregate the data to a monthly panel. Following [Roca and Puga \(2017\)](#), in the presence of overlapping spells we first drop those that end before an ongoing spell and then assign each worker-month to a single “main job,” defined as the job with the highest earnings within that month. This procedure yields a consistent panel structure with precise measures of tenure and accumulated experience. It also allows us to track changes in contract type, earnings, and firm mobility across main jobs.

The Social Security records are matched with annual information from the municipal population registry (*Padrón Continuo Municipal*) and income tax records from 2006 onward. Together, these sources provide rich demographic and firm-level information. We observe each worker’s date of birth, gender, educational attainment, and country of birth. While occupation is not directly recorded, we construct five occupational-skill groups using the ten contribution categories reported by employers to the Social Security

⁷Because we observe exact entry and exit dates for each employment spell, we can compute the precise number of days worked and identify the exact timing of contract changes. Employment durations are based on administrative payroll records and include paid leave and holidays.

Administration, which are closely linked to the skill requirements of each job.

Contract-level information includes contract type (fixed-term or open-ended) and part-time status. At the establishment level, we observe the province in which the firm is located and its employment size. For each job, we also observe the sector of economic activity at the two-digit level, as well as whether the worker is self-employed or employed in the private or public sector.

The MCVL provides earnings from two sources: Social Security contribution bases, which are top- and bottom-coded with thresholds that vary by contribution group and are updated annually, and income tax records, which are available from 2006 onward and are uncensored. We use tax-record earnings whenever available, deflating both sources to 2009 prices, and rely on contribution-base earnings before 2006 and for workers in Navarre and the Basque Country, which administer separate tax systems. Results are not sensitive to this choice of earnings measure, as the top-coding primarily affects observations at the upper end of the earnings distribution. Finally, where the analysis requires daily wages, we construct them by dividing monthly earnings by the exact spell duration recorded in the data.

3.1 Sample Restrictions

The MCVL provides earnings histories from the 1980s, but reliable contract-type information begins in January 1998. We use 24 monthly lags of open-ended and fixed-term contract creation, which requires data from January 1998 and sets the start of the estimation period at January 2000. We end the data in February 2020, before the March 2020 state of alarm and the associated labor market disruption. To allow for 24 months of post-entry outcomes and use complete calendar years, we restrict entry events to January 2000 through December 2017.

We restrict to native workers aged 18–49. The age ceiling limits confounding from early retirement, and the nativity restriction addresses concerns about incomplete labor market histories among foreign-born workers. We further restrict to workers affiliated with the general Social Security regime and the special regimes for agriculture, sea workers, and mining. This excludes self-employed workers, who fall outside our analysis by construction since open-ended contracts do not apply to self-employment.

The baseline sample covers private-sector employment only. Contract duration and access to permanent positions in the public sector are governed by statutory requirements, including competitive examinations, that differ fundamentally from the upgrade process in the private sector. Nevertheless, prior public-sector experience, whether under fixed-term or permanent contracts, enters the experience measure used in the analysis. Finally, we exclude Ceuta and Melilla due to insufficient sample sizes, leaving 50 provinces.

4 Selection into permanent positions

We begin with the descriptive relationship between contract type and earnings growth. To establish comparability with prior studies on returns to experience (Roca and Puga, 2017; Garcia-Louzao et al., 2023; Arellano-Bover and Saltiel, 2021), we estimate the contribution of contract-specific experience to earnings growth using Mincer-type regressions:

$$\ln w_{it} = f(\exp_{it}^{FT}, \exp_{it}^{OE}, \exp_{it}) + X'_{it}\boldsymbol{\Omega} + \mu_{r(i,t)} + \psi_t + \varepsilon_{it}, \quad (1)$$

where $\exp_{it}^{FT}$, $\exp_{it}^{OE}$, and $\exp_{it}$ denote the experience that worker i has accumulated until period t in fixed-term contracts, open-ended contracts, and overall, respectively. In our main specification, we allow the returns to contract-specific experience to vary flexibly by including linear contract-specific experience and their interactions with overall experience. X_{it} is a vector of time-varying individual and job characteristics; $\mu_{r(i,t)}$ and ψ_t denote province and year-month fixed effects, respectively; and ε_{it} is the error term. See Appendix A.1 for further details.

Results are presented in Appendix Table A.1.1. Wage returns to experience are similar across contract types early in a worker’s career but diverge thereafter. After ten years of experience, an additional year under a fixed-term contract raises earnings by approximately 3.0 percent, compared with 4.5 percent under an open-ended contract. Including worker fixed effects attenuates the estimated gap, but the qualitative pattern persists. This is consistent with Garcia-Louzao et al. (2023), who show that the gap cannot be explained by unobserved firm heterogeneity or match quality.⁸

These estimates are consistent with a central concern in dual labor markets: temporary contracts may provide fewer on-the-job training opportunities, leading to slower skill accumulation and wage growth (Cabrales et al., 2017). Worker fixed effects address only part of the endogeneity from non-random sorting into contract types. They absorb level differences in ability across workers but not heterogeneity in earnings growth rates, so any correlation between growth potential and contract type survives within-worker estimation.

To assess this possibility directly, we examine whether workers who transition to open-ended and fixed-term contracts exhibit parallel earnings dynamics prior to conversion, when all workers are still employed under fixed-term contracts. Following an event-study design similar to Card et al. (2013) and Card et al. (2025), we take a sample of workers in their final month of a fixed-term contract, classify them by the contract type they transition into, and trace earnings over a symmetric 15-month window before and after

⁸Garcia-Louzao et al. (2023) also instrument for contract type using regional variation in open-ended hiring subsidies. Our identification strategy exploits a different source of variation, drawing on high-frequency administrative records.

the transition.⁹

Panel (a) of Figure 2 plots average earnings by transition type.¹⁰ Workers who eventually obtain an open-ended contract already follow a steeper earnings trajectory before the transition: over the 15-month pre-transition window they experience earnings growth of 9.4 percent, compared with 6.2 percent for workers who remain in fixed-term employment. There is no discrete change in earnings growth at the moment of transition, either relative to the worker’s own pre-transition trend or relative to workers remaining in fixed-term employment.

These patterns suggest an alternative interpretation of the Mincer regression results: rather than permanent contracts offering higher returns to experience, workers with potentially higher returns are more likely to secure such contracts. They also suggest that selection into *contract type* is a more severe problem than selection into *firms* or *regions*, where this type of event-study test is typically passed.¹¹

To control for age effects and aggregate time variation, we further estimate an event-study specification. For each worker, we define the final month of a fixed-term contract as event time $h = 0$ and index all preceding and subsequent months relative to this date.¹² We classify workers as transitioning from a fixed-term to an open-ended contract (FT→OE, $C_i = 1$) or to another fixed-term contract (FT→FT, $C_i = 0$). Our baseline specification considers a balanced panel of workers for whom we observe fifteen periods (months) before and after the event.¹³ We denote by y_{ith} the log earnings of individual i at year-month t and event-time h . We then estimate the following regression:

$$y_{ith} = \sum_{k \neq -1} \alpha_k^{FT-OE} \cdot \mathbf{I}[k = h] \cdot \mathbf{I}[C_i = 1] + \sum_{k \neq -1} \alpha_k^{FT-FT} \cdot \mathbf{I}[k = h] \cdot \mathbf{I}[C_i = 0] \\ + \sum_j \beta_j \cdot \mathbf{I}[j = age_{it}] + \sum_q \gamma_q \cdot \mathbf{I}[q = t] + \lambda_C + \nu_{ith}, \quad (2)$$

where we include a full set of event-time indicators (first and second terms on the right-hand side), age dummies (third term), year-by-month dummies (fourth term), and a fixed effect for the type of contract transition. The event-time indicator for $h = -1$ is omitted,

⁹The estimation sample includes 949,481 transition events observed between 1998 and 2020, for which we observe 15 months with positive earnings both before and after the transition. Of these, 227,990 correspond to transitions into open-ended contracts and 721,491 to transitions into another fixed-term contract.

¹⁰Appendix Figure B.2.1 further distinguishes transitions to open-ended contracts within the same firm from those involving a move to a different employer.

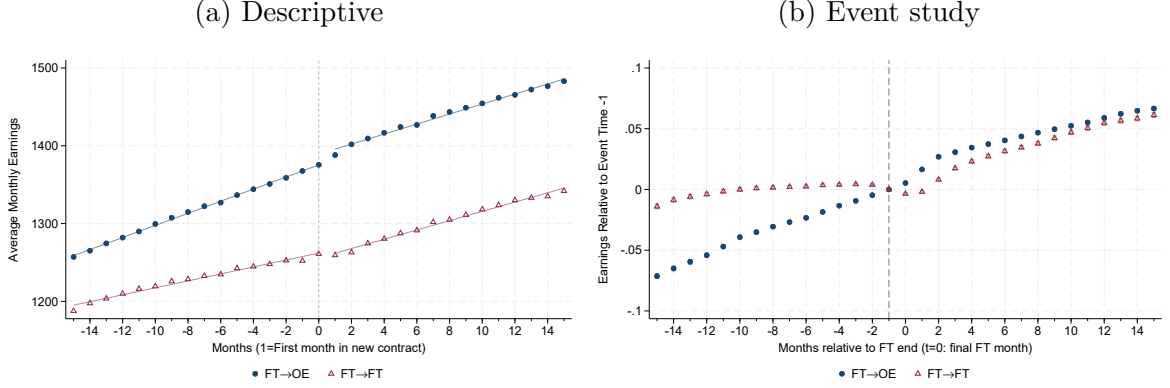
¹¹For example, Card et al. (2025) presents a similar exercise to study the causal effect of places on earnings, showing that workers’ trajectories only differ *after* changing locations.

¹²Workers may experience multiple contract transitions over their careers, so individuals whose fixed-term contracts expire can appear multiple times in the sample; on average, each worker contributes 2.4 transition events. Appendix Figure B.2.3 shows that restricting the analysis to each worker’s first observed transition yields results consistent with the main text’s qualitative conclusion.

¹³We allow for short unemployment spells before or after switching contracts, but only include periods with non-zero earnings in the regression. Consequently, the event time (h), which covers 15 months before and after the end of the fixed-term contract, may differ from standard calendar months (t).

so coefficients are interpreted relative to earnings in the month immediately preceding the expiration of the fixed-term contract. Age dummies flexibly control for life-cycle earnings profiles, which is particularly relevant given that workers transitioning to open-ended contracts tend to be older than those remaining in fixed-term employment. Year-by-month dummies absorb aggregate time variation, including business cycle fluctuations, so that identification relies on within-period comparisons across workers.

Figure 2: Evolution of earnings before and after switching to a new contract



Notes: Panel (a) plots average earnings by event-time month for workers transitioning into open-ended (OE) or fixed-term (FT) contracts. Panel (b) plots event-time coefficients from equation 2. The sample is balanced and covers January 1998 to February 2020, with earnings tracked over a 15-month window before and after the contract change. Monthly earnings are extrapolated from daily earnings for months of partial employment. Short unemployment spells between contracts are permitted, so observation windows need not coincide with calendar months. The omitted category is $h = -1$. All specifications in the event study include age and year-by-month fixed effects; standard errors are clustered at the individual level.

Panel (b) of Figure 2 reports event-study estimates. Pre-transition earnings paths differ sharply by transition type: conditional on controls, workers who transition into an open-ended contract experience earnings growth of approximately 7 log points over the 15-month pre-transition period, while those who remain on fixed-term contracts show no meaningful growth. Transitions to open-ended contracts are also associated with an increase of approximately 2 log points in the level of earnings in the first month, and one that is more pronounced among workers who change firms (Appendix Figure B.2.2). Post-transition trajectories, by contrast, are similar across groups. A natural interpretation is ability sorting: high-productivity workers are more likely to transition into open-ended contracts and also exhibit faster earnings growth independent of contract type. This is the pre-transition divergence anticipated above: individual fixed effects absorb level differences in ability but not heterogeneity in growth rates, so the bias survives within-worker estimation.

Two features of contract-type transitions make this selection problem more severe than analogous selection into firms or locations. Transitions to open-ended contracts frequently occur within firms and reflect richer employer information about worker productivity than does the matching of workers to new employers. Consistent with this, Appendix

Figure B.2.2 shows that workers who obtain an open-ended contract within the same firm exhibit substantially stronger pre-transition earnings growth than both workers remaining in fixed-term contracts and those transitioning to open-ended positions at other firms. Worker preferences over contract types are also relatively uniform: given the greater job stability and employment protection of open-ended contracts, most workers prefer them to temporary contracts (Section 2). Unlike firm or geographic mobility, there is little scope for preference heterogeneity to offset the sorting of higher-ability workers into permanent positions (Card et al., 2013, 2025).

5 Identification

We propose an instrumental variables strategy to address the endogeneity of transitions from fixed-term to permanent contracts. Our source of exogenous variation combines worker-level heterogeneity in the expiration timing of fixed-term contracts with transitory fluctuations in the availability of new open-ended positions across time and local labor markets. The identifying assumption is that, conditional on fixed effects and business-cycle controls, temporary regional fluctuations in open-ended contract creation are orthogonal to unobserved determinants of individual workers' transitions to permanent employment. Identification comes from the interaction between predetermined contract expiration dates and unpredictable month-to-month variation in local hiring conditions. Transitory employment fluctuations predominantly reflect changes in labor demand rather than labor supply (Blanchard and Katz, 1992). A temporary increase in the availability of permanent positions can raise the probability of contract upgrading. It directly expands workers' opportunities to obtain a permanent position, either with their current employer or elsewhere, while job-to-job transitions by other workers create additional vacancies that can be filled by fixed-term workers whose contracts are about to expire.

Illustration.— Consider two workers employed in identical 12-month fixed-term contracts in the retail sector of Almería, whose contracts expire one month apart. Worker A's contract ends in July 2007, whereas Worker B's ends in August 2007. Suppose that Almería experiences a temporary increase in the creation of open-ended contracts in August 2007. Worker B is consequently exposed to a higher arrival rate of permanent positions at the time of contract expiration and is more likely to transition to an open-ended contract, either with the current employer or elsewhere. A cross-sectional analogue compares workers whose fixed-term contracts expire in the same month but in different provinces. For example, if Almería experiences a temporary increase in open-ended contract creation in August 2007 while neighboring Murcia does not, workers whose contracts expire in Almería face a higher probability of transitioning to permanent employment. Appendix Figure B.6.1 illustrates this source of variation by plotting monthly open-ended contract creation in the two provinces over the full sample period.

Exploiting the high frequency of our data, we can precisely match the month when an individual’s fixed-term contract is set to expire with the number of new contracts at the regional level starting the next month. We use these new contracts to capture the labor demand available to the worker, even when the worker began searching for a job before the contract expired.¹⁴ We argue that, conditional on year and calendar-month fixed effects, the amount of labor demand a worker encounters at contract expiration is plausibly orthogonal to the worker’s unobserved characteristics. We provide evidence supporting this assumption below.

Our approach is conceptually related to prior work on how labor market conditions shape worker careers, including research on entering the labor market during a recession (“graduating into a recession”), e.g. (Kahn, 2010; Hershbein, 2012; Wachter and Bender, 2006; Altonji et al., 2016; Schwandt and Von Wachter, 2019) and recent work on compositional changes in labor demand (Arellano-Bover, 2024). These studies, however, examine yearly fluctuations in labor market conditions affecting entire cohorts. We instead use high-frequency administrative data to isolate individual-level labor demand shocks while *controlling* for overall business cycle conditions.¹⁵

We formalize this source of variation in the following first-stage equation, estimated for the period 2000–2017:

$$p_{i,t+1} = \sum_{k=-24}^{24} \alpha_k OEC_{-i,r,t+k} + \sum_{k=-24}^{24} \gamma_k FTC_{-i,r,t+k} + \mathbf{X}'_{it} \theta + \mu_r + \psi_t + \phi_s + \epsilon_{it}, \quad (3)$$

where t refers to the exact month the worker’s fixed-term position ends. Here, $p_{i,t+1}$ indicates whether the worker starts an open-ended contract in $t+1$, the month after their fixed-term contract ends.¹⁶ The variable $OEC_{-i,r,t+k}$ is constructed as the log of the total count of new open-ended positions in period $t+k$ in worker i ’s baseline province r —defined as the province of employment under the expiring fixed-term contract—excluding worker i herself (leave-one-out), and we allow contract upgrades to depend on the leads and lags of this measure. Appendix Figure B.5.1 supports the linear specification in equation (3) by plotting the underlying conditional expectation using the binscatter approach of Cattaneo et al. (2024).

As motivated above, we use the first lead, $OEC_{-i,r,t+1}$, as our instrumental variable.

¹⁴Workers may search for new positions *before* their current contract expires. Similarly, firms may post job openings some time before filling them. Regardless of the timing of this job search process, most workers will only start a new contract after their current one ends. Our instrument will, therefore, be relevant despite the (unobserved) dynamics of the underlying job search process, as we confirm below.

¹⁵Our identifying variation relies on the fine-grained information available in the MCVL. Appendix Figure B.4.1 shows that this variation is not driven by a limited set of firms; instead, similar dynamics are present also among small firms.

¹⁶ $p_{i,t+1} = 1$ if the worker is employed on an open-ended contract in the calendar month following the expiration of the fixed-term contract, and $p_{i,t+1} = 0$ if the worker is unemployed or employed on another fixed-term contract.

At the individual level, X_{it} includes controls for gender, baseline labor market experience and its square, and interactions between age categories and educational attainment. We additionally include baseline-province fixed effects μ_r , year and month fixed effects ψ_t , and baseline-sector fixed effects ϕ_s . Conditional on these controls and fixed effects, the instrument captures transitory regional fluctuations in the arrival of new open-ended contracts that, as argued above, are plausibly exogenous from the worker’s perspective (“instrument independence”). We provide supporting evidence for this assumption below. To further absorb business-cycle and seasonal variation in hiring, we control for the full set of leads and lags of the log number of new fixed-term contracts, $FTC_{-i,r,t+k}$. Appendix Table C.1.2 reports descriptive statistics for the estimation sample and the instrument.

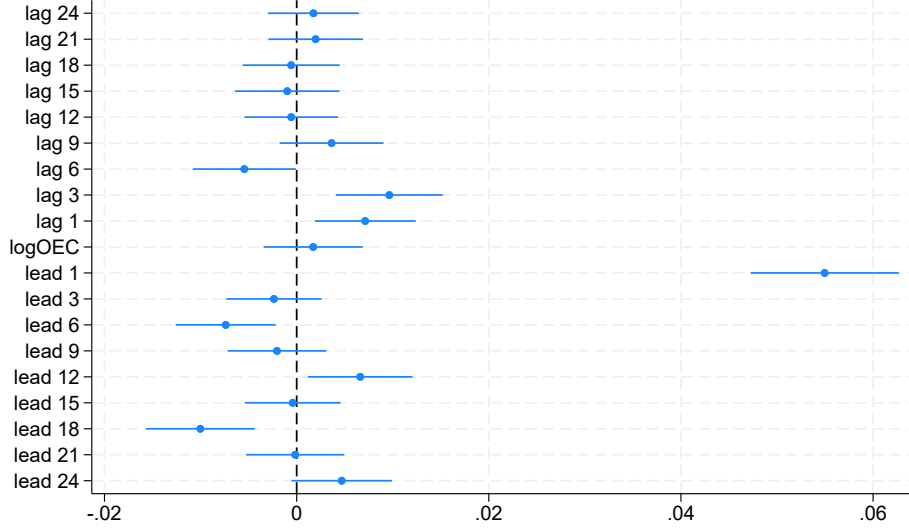
Under our identification assumptions, the first lead, captured by coefficient α_1 , should be the strongest predictor of a worker’s transition to a permanent position. Including the full set of leads and lags serves two purposes. It isolates transitory fluctuations in labor demand around contract expiration by allowing for broader labor market conditions that may affect the probability of a contract upgrade. This lets us assess whether α_1 retains predictive power after accounting for these dynamics; under our identification argument, we expect it to be larger in magnitude than the other coefficients. The specification also absorbs general business-cycle conditions that may affect both hiring and contract transitions, reducing concerns that such conditions could threaten the exclusion restriction.

Identification relies on three conditions. First, conditional on controls and fixed effects, the leave-one-out first lead of log new open-ended contracts must predict the probability of conversion at the end of a fixed-term spell. Second, conditional on the same controls and fixed effects, the instrument must be independent of unobserved determinants of worker outcomes and affect outcomes only through contract status. These conditions identify the IV estimand. Third, interpreting this estimand as a local average treatment effect (Angrist et al., 1996) for workers whose conversion decisions respond to the instrument requires monotonicity: the instrument must affect the probability of conversion in the same direction for all workers. We discuss these conditions in turn and characterize the population of compliers to whom our estimates apply.

5.1 Instrument Relevance

Relevance Condition. Figure 3 reports the first-stage estimates from equation (3), estimated on a sample of workers in the last month of a fixed-term contract, testing whether they are more likely to transition to an open-ended contract in period $t + 1$. Among the leads and lags of new open-ended contracts in the province, the first lead is the strongest predictor of a worker’s transition to permanent employment. The first-stage F -statistic for the excluded instrument is 195.2, well above conventional thresholds, so

Figure 3: Effect of new open-ended contracts in the region on permanent contract status



Notes: This figure presents estimates of equation (3) for workers observed in the last month of a fixed-term contract at event period $h = 0$, restricted to workers with tenure between 0.3 and 2 years. Coefficients measure the effect of leads and lags of log new open-ended contracts in the baseline province on the probability of converting to an open-ended contract at $t + 1$; the sample mean of the conversion dummy is 0.254. All specifications include year and month fixed effects, province fixed effects, sector fixed effects, gender, age-by-educational-attainment interactions, experience and experience squared, a part-time contract indicator, and leads and lags of log new fixed-term contracts in the baseline province. Standard errors are clustered at the province $\times$ year level.

weak-instrument bias is not a concern in this setup. The lack of strong correlation with the other leads and lags is consistent with the instrument capturing transitory shocks on job market matches unrelated to general business cycle trends. The observation that the coefficient is positive further supports that transitory fluctuations in employment reflect shifts in labor demand rather than labor supply—consistent with [Blanchard and Katz 1992](#).

This specification defines the leads and lags at the *regional* level; the instrument can be constructed analogously using national-level or baseline-industry counts of open-ended contract creation. Appendix Figure [B.6.2](#) shows similar patterns under these alternative specifications, so relevance holds regardless of whether the instrument is measured at the national, regional, or industry level. Relevance also persists when we exclude months with high job seasonality (Appendix Figure [B.7.1](#)).

5.2 Instrument Exogeneity

The exclusion restriction requires that regional variation in open-ended contract creation affects worker outcomes only through the probability of transitioning to permanent employment. Conditional on province, sector, year, and month fixed effects, identification exploits residual month-to-month fluctuations in permanent hiring at the time of contract

expiration. The fixed effects absorb persistent differences across provinces and sectors as well as common time trends, while the leads and lags of fixed-term contract creation flexibly account for the dynamic evolution of local labor-market conditions and job creation. The identifying variation, therefore, comes from transitory fluctuations in the availability of permanent jobs rather than from broader changes in regional economic conditions.

Two threats to the exclusion restriction remain. Exposure to the instrument may be correlated with worker or firm characteristics if transitory increases in permanent hiring occur disproportionately among certain workers or firms. Firms may also respond to local labor-market conditions by changing the timing of contract expiration or the types of permanent positions they create, linking the instrument to the composition of workers or jobs. The following subsections examine these two concerns in turn.

Instrument Independence. Conditional on fixed effects and controls, the number of new permanent positions available at the time a worker’s fixed-term contract expires is, from the worker’s perspective, plausibly as good as random. This assumption cannot be tested directly, but Appendix A.2 reports balance regressions showing that, conditional on year, month, sector, and province fixed effects, the instrument is uncorrelated with worker characteristics (age, experience, and educational attainment), firm characteristics (age and size), and contract characteristics (part-time status and tenure). The instrument is standardized so that coefficients can be interpreted as deviations from the mean; a few baseline sectors show statistically significant coefficients, which is why we always include sector fixed effects in the specification, but these coefficients are small in magnitude. Workers exposed to different realizations of the instrument are otherwise observationally similar.

Beyond the sources of heterogeneity already absorbed by the fixed effects, time-varying unobserved factors could still confound identification. To address this concern, Section 6 tests for differential pre-trends across workers exposed to different values of the instrument. We find no statistically significant differential pre-trends for any outcome; in particular, as shown below, workers exposed to different values of the instrument exhibit similar wage trajectories prior to contract expiration. The balance tests and the absence of differential pre-trends provide little evidence that instrument exposure is systematically related to worker characteristics or pre-existing outcome dynamics.

New Open-Ended Contract Composition. An additional concern is that the instrument may be correlated with the characteristics of newly created open-ended contracts. In particular, firms posting a large number of permanent contracts in a given month may offer positions that differ systematically from those posted in other months. To address this concern, we examine whether the instrument is related to key observable characteristics of newly created open-ended contracts, including average tenure, required skill level, and starting earnings. Appendix Table A.2.1 reports these estimates. Across all specifications, we find no statistically significant relationship between the instrument and

any of these measures. This suggests that variation in the instrument reflects changes in the number of available open-ended contracts rather than systematic differences in their observable composition.

Exogeneity of Contract Termination Date. Our identification strategy relies on two elements: fluctuations in new open-ended contract creation and the exact timing of fixed-term contract expiration. A potential concern is a direct link from the former to the latter—that is, whether the number of newly created permanent contracts could influence termination dates for some workers. For example, firms might repeatedly renew fixed-term contracts while waiting for economic conditions to improve, creating a feedback effect from labor demand (the number of new permanent contract openings) to labor supply (the number of workers with recently expired temporary contracts). Such feedback would not necessarily be problematic: it could weaken the first stage by diluting demand shocks across more workers, but the first stage is strong (Figure 3). Feedback effects pose a problem only if they are *selective*, affecting some worker types more than others. If more talented workers were more likely to exit fixed-term employment early to move into a permanent position, worker ability would be correlated with the instrument, contaminating our estimates of the effect of contract status on worker careers.

Several features of the empirical strategy mitigate these concerns. At the time of our analysis, legal limitations on the consecutive renewal of temporary contracts restricted such arrangements to a maximum of two years.¹⁷ We also limit the sample to workers whose contracts terminated after at least 0.3 years, reducing the influence of very short contracts, which are generally considered lower quality and are common in Spain (Bentolila et al., 2020).¹⁸ Balance regressions show no meaningful correlation between the instrument and observable worker characteristics, including education (Appendix A.2). Finally, if selection were driving the results, we would expect it to appear in pre-treatment trends, but the IV estimates show no evidence of such pre-trends (Figure 4).

5.3 Characterizing Complifiers

Monotonicity Condition. The instrument plausibly satisfies the monotonicity condition. An increase in newly created open-ended contracts expands the set of permanent employment opportunities available to workers at the time of contract expiration, weakly increasing their probability of obtaining a permanent position. It is difficult to conceive of

¹⁷Under Article 15.5 of the Workers’ Statute, an employee who works more than 24 months within a 30-month window, continuously or not, in the same or different positions, for the same firm or business group, across two or more fixed-term contracts, automatically acquires permanent employee status. The 2021 reform reduced these thresholds to 18 and 24 months, respectively. Results restricting the sample to events near the legal two-year limit are broadly consistent with the main findings, as discussed in Section 7, though less precisely estimated given the smaller number of such observations.

¹⁸Appendix Figure B.1.2 shows that a substantial share of fixed-term contracts terminate near the one-year mark. Appendix Section B.12 reports results under alternative tenure restrictions; estimates for contracts near the legal maximum are similar to the baseline.

a mechanism through which additional permanent vacancies would systematically reduce a worker’s likelihood of contract conversion, making the presence of “defiers” unlikely in this setting.¹⁹

Local Average Treatment Effect. Under the independence, exclusion, and monotonicity assumptions discussed above, the IV estimates identify a Local Average Treatment Effect (LATE) (Angrist et al., 1996): the effect of permanent contract transition for workers whose conversion probability responds to transitory changes in the local availability of open-ended positions, that is, workers at the margin of conversion. Although this parameter need not coincide with the average effect of conversion for all fixed-term workers, it is economically and policy relevant, since it identifies the effect for workers whose contract status is most responsive to changes in permanent hiring opportunities. This margin is directly relevant for policies that expand the availability of open-ended positions, such as hiring subsidies and other incentives for permanent employment of the type studied in Ciani and De Blasio (2015) and Elias and Redondo (2025).

The IV identifies a local effect for workers whose treatment responds to the instrument (Heckman et al., 2006; Angrist and Pischke, 2009). Because the instrument is continuous, compliers are marginal workers whose conversion probability responds to small fluctuations in local labor demand. Following Marbach and Hangartner (2020), we residualize the instrument with respect to the baseline controls and construct a binary approximation based on the sign of the residual. This decomposition invokes the same assumptions as LATE identification, and standard errors are bootstrapped with 1,000 replications. Table C.5.1 reports the results.

On worker characteristics, compliers sit between the two comparison groups and resemble always-takers more closely than never-takers. They are more educated than never-takers (0.448 versus 0.383) and somewhat less educated than always-takers (0.509), and the same intermediate pattern holds for gender. They are not distinguishable from either group by age: the share below thirty is 0.603 among compliers, 0.568 among always-takers, and 0.581 among never-takers. The one worker dimension on which compliers stand out is baseline pay. The share with above-median initial earnings is 0.576 among compliers, well above the never-taker share of 0.477 and above the sample mean of 0.500.

The sharpest distinction is firm age. Compliers are substantially less likely to work in young firms than either comparison group, 0.273 against 0.413 for always-takers and 0.435 for never-takers, and both gaps are significant at the one percent level. Compliers are also more likely than never-takers to work in large firms (0.515 versus 0.455), though they are indistinguishable from always-takers on this margin (0.521). On sectoral exposure to fixed-term work they again fall between the two groups, at 0.500 against 0.435 for

¹⁹Additional suggestive evidence comes from first-stage coefficients that are consistently positive and quantitatively similar across subsamples (see Appendix Section C.3). A related argument is used by Bhuller et al. (2020).

always-takers and 0.606 for never-takers.

The LATE is identified from workers in established firms, positively selected on baseline earnings, whose transition to permanent employment is sensitive to short-run fluctuations in open-ended contract availability. Conversion at the margin appears to occur in firms with the organizational capacity to translate transitory increases in permanent hiring into actual contract upgrades, rather than in young or high-turnover employers.

6 Results

We exploit regional fluctuations in the availability of new open-ended contracts to isolate exogenous variation in workers' probability of transitioning from fixed-term to permanent positions. The estimation sample consists of workers observed in the final month of a fixed-term contract between 2000 and 2017, with tenure in that contract between 0.3 and 2 years ($N = 701,761$).²⁰ We follow workers from 60 months before to 60 months after contract expiration. We first present reduced-form estimates of the effects of contract-upgrade opportunities on labor market outcomes over both horizons. Under the exclusion restriction, we then implement a two-stage least squares (2SLS) strategy to estimate the causal effect of transitioning to permanent employment on workers' career trajectories.

The IV specification is:

$$y_{i,t+h} = \delta_1^h p_{i,t+1} + \sum_{k \neq 1, k=-24}^{24} \beta_k^h \log OEC_{-i,r,t+k} + \sum_{k=-24}^{24} \gamma_k^h \log FTC_{-i,r,t+k} + X_{it}'\theta + \mu_r + \psi_t + \phi_s + \epsilon_{it}, \quad (4)$$

where $y_{i,t+h}$ is worker i 's outcome in month $t+h$, with $h < 0$ denoting months before contract expiration and $h > 0$ months after, and $p_{i,t+1}$ is an indicator equal to one if the worker holds an open-ended contract in $t+1$, the month immediately following expiration. Workers who do not transition either enter a new fixed-term contract or move into non-employment. The excluded instrument is $\log OEC_{-i,r,t+1}$, the leave-one-out log of newly created open-ended contracts in province r in month $t+1$. The specification also includes $\log OEC_{-i,r,t+k}$ for all $k \neq 1$ and the 24 leads and lags of newly created fixed-term contracts, so identification rests on contract availability in the transition month conditional on availability in the neighboring months.

The vector X_{it} contains worker characteristics measured at baseline: gender, labor-market experience and its square, a part-time indicator, and interactions between age

²⁰The sample ends in 2017 to limit the loss of observations near the end of the data period (February 2020), given that outcomes are tracked over a 60-month window following contract expiration. Table C.1.2 in the Appendix reports descriptive statistics. We do not restrict the number of contract expirations per worker, so workers may appear multiple times in the sample. The sample comprises 307,166 unique workers, with an average of 2.28 observations per worker.

categories and educational attainment. We include year, month, province, and sector fixed effects, estimate equation (4) separately for each horizon $h = -60, \dots, 60$, and cluster standard errors at the province $\times$ year level.

Earnings are measured both contemporaneously and cumulatively, relative to the final month of the expiring contract. Employment outcomes include employment status, the probability of holding an open-ended contract, and cumulative experience, both overall and in open-ended positions. Mobility outcomes capture firm switches and transitions across sectors and provinces following expiration. For each outcome, we report two sets of estimates. We first present reduced-form estimates of the effect of a transitory increase in the number of new open-ended contracts in the worker’s province in the month following expiration. Under the assumption that the instrument is independent of potential outcomes conditional on the controls in equation (4), these estimates identify the sign and timing of the response to the induced variation in contract status. We then report 2SLS estimates in Table 1, which identify, for workers whose contract status is shifted by the instrument, the local average treatment effect under the conditions presented in Section 5.

6.1 Reduced form evidence

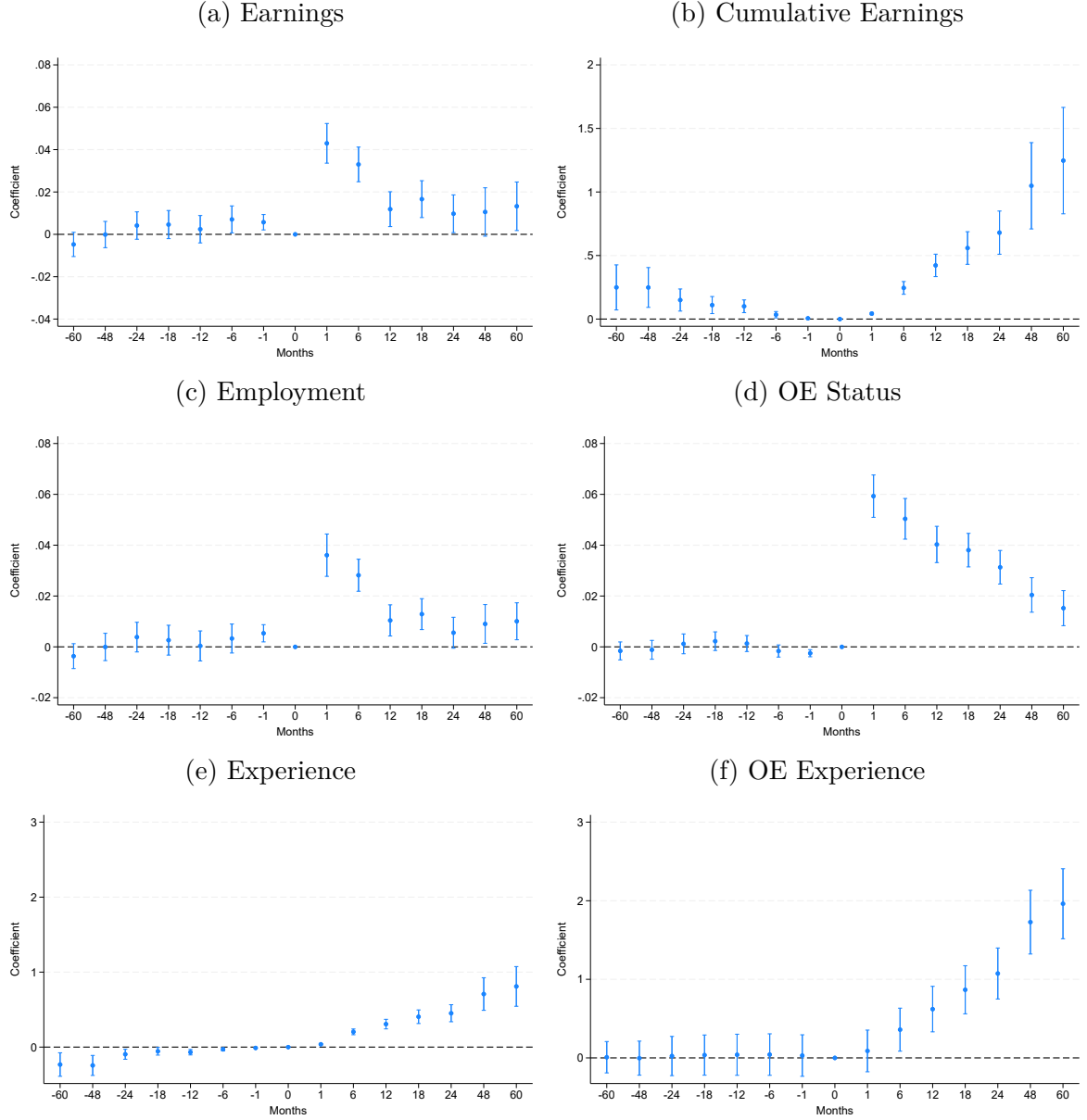
Figure 4 reports the reduced-form coefficients from 60 months before to 60 months after the expiration of the fixed-term contract. The pre-expiration coefficients are small for every outcome, as are those for the mobility outcomes in Figure 5. Workers who face more favorable open-ended hiring conditions at the time of contract expiration are observationally similar, before expiration, to workers who face less favorable conditions. This is the test for differential pre-trends anticipated in Section 5.2, and its outcome complements the balance regressions reported there.

6.1.1 Earnings

Panel (a) of Figure 4 reports the effect on relative monthly earnings h periods after termination, measured at $t + h$ relative to baseline earnings in month t , the final month of the expiring contract. The outcome is defined as $(E_{t+h} - E_t)/E_t$, where E_t represents monthly earnings in period t , so the coefficients measure changes in monthly earnings as a fraction of baseline earnings. Panel (b) reports cumulative earnings from t to $t + h$, also normalized by baseline monthly earnings.

The effect is positive at $h = 1$: the coefficient is 0.043, so a 10 log point increase in the number of new open-ended positions raises earnings by 0.43 percent of baseline earnings in the month after expiration. It attenuates over time, falling to 0.013 at five years, or 0.13 percent of baseline earnings, where it is significant at the 5 percent level. Part of this decline reflects catch-up in the comparison group: workers who remain in fixed-term

Figure 4: Effect of New Open-Ended Contracts on Worker Outcomes (Reduced Form)



Notes: The sample consists of workers in the last month of a fixed-term contract at event period $h = 0$ with tenure between 0.3 and 2 years. Sample period: 2000–2017. Coefficients measure the effect of $\log OEC_{-i,r,t+1}$ on each outcome. All specifications control for leads and lags of log new open-ended and fixed-term contracts, province, year and month fixed effects, gender, interactions of age and educational attainment, experience, and experience squared, and contract part-time indicator. Standard errors are clustered at the province $\times$ year level.

contracts at $h = 0$ transition into open-ended positions later on, and those transitions are themselves associated with higher earnings, narrowing the gap with workers who transitioned at $h = 1$. As we show below, catch-up accounts for only part of the decline.

Although the effect on *current* earnings fades, the early gains cumulate. As shown in Panel (b), a 10 log point increase in open-ended contract creation raises cumulative earnings over the following 60 months by 12 percent of baseline monthly earnings.

6.1.2 Employment

Panels (c) to (f) of Figure 4 report effects on employment trajectories. Panel (c) examines the likelihood of being employed at $t + h$ and Panel (d) the probability of being employed under an open-ended contract. Panels (e) and (f) report cumulative experience: total experience, measured as days worked since the termination of the fixed-term contract, and experience in open-ended contracts, both converted into years. Appendix Figure B.10.2 reports the effect on part-time employment, which is not statistically significant at any horizon.

As illustrated in Panel (c), a temporary increase in the number of open-ended contracts raises the probability of employment in the short run. As with earnings, this effect diminishes over time, though after 60 months exposed workers remain somewhat more likely to be employed. A favorable draw of upgrade opportunities provides a temporary boost, with limited long-run consequences for employment and contract status. This is consistent with the selection patterns documented in Section 4: contract status is largely determined by the worker’s own “type,” while fluctuations in labor market conditions affect the timing of transitions rather than whether they ultimately occur.

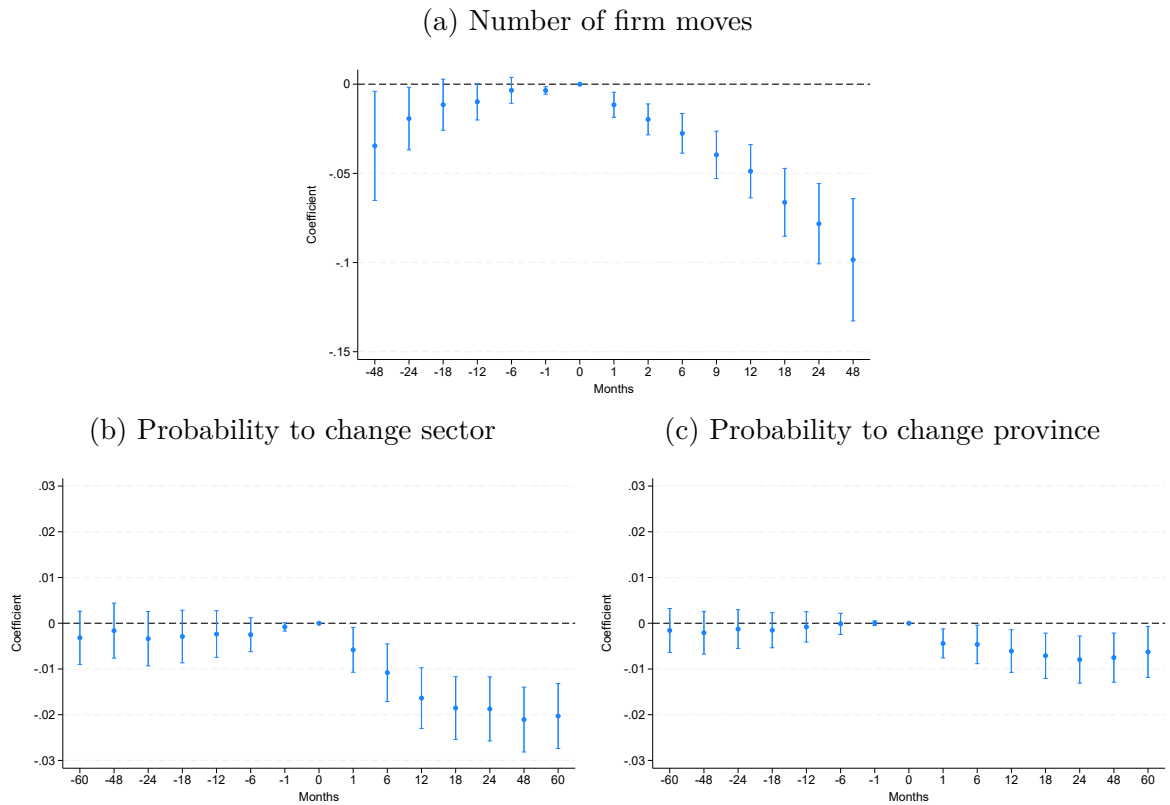
The overall pattern mirrors that for earnings, with sharp initial gains that fade over time, but the employment effect decays more gradually. Experience, by contrast, does not attenuate at all. As shown in Panels (e) and (f), exposed workers accumulate lastingly more work experience, and disproportionately experience under open-ended contracts. Section 6.2 translates these reduced-form differences into a substantial long-term effect of contract type on accumulated experience.

This tension motivates the remainder of the section. If returns to experience are positive, a lasting gain in experience should show up as a lasting gain in earnings. Yet the earnings effect fades within a few years, so the additional experience acquired under a permanent contract appears not to be rewarded in the way the standard argument implies.

6.1.3 Sectoral and Regional Mobility

A key advantage of permanent over temporary contracts is job security, and that stability also shapes workers’ mobility decisions. Figure 5 assesses how the opportunity to upgrade

Figure 5: Effect of new open-ended contracts in the region on workers' mobility



Notes: Baseline sample restrictions and empirical specification are described in Figure 4 notes. In Panel (a), the outcome is the number of firm switches between period t and $t + h$. In Panels (b) and (c), the outcome is an indicator that equals one if a worker has ever changed sector or province between period t and $t + h$. Missings are coded as zero. Confidence intervals are constructed using standard errors clustered at the province $\times$ year level.

into a permanent contract affects the number of firm switches, the likelihood of moving to a different sector, and the likelihood of moving to a different region. The number of firm moves is measured as the number of firm switches before ($h < 0$) or after ($h > 0$) the expiration of the fixed-term contract. For sectoral and regional mobility, we consider an indicator equal to one if the worker has ever changed sector or province between t and $t + h$.

As shown in Panel (a), a temporary increase in the number of open-ended contracts results in a marked reduction in firm mobility. Workers on permanent contracts are more likely to stay with their current firm than those in temporary positions, whose contracts cannot be extended indefinitely (see Section 2). An increase in the number of permanent contracts also reduces the probability that workers switch to a different sector (Panel b) or relocate to another province (Panel c). The estimates for sectoral mobility remain highly significant over the longer term, whereas those for regional mobility are less pronounced after 5 years.²¹

Reduced mobility has opposing effects on earnings growth. Greater stability in employment relationships may encourage firms to provide training or investments in their workers (Cabrales et al., 2017; Albert et al., 2005). Mobility, however, is also how workers climb the “firm ladder” (Burdett and Mortensen, 1998) and search for positions that better match their skills or interests. Read alongside the absence of a long-run wage return documented below, the estimates in Panels (a) to (c) are consistent with the second channel dominating: the security derived from permanent employment deters workers from pursuing alternative career paths, so that job stability, desirable as it is in itself, may come at the cost of career progression.

As a further robustness check, we augment the baseline specification with the leave-one-out provincial mean of the outcome variable, $\bar{Y}_{-i,r,t+h}$, constructed from all workers aged 18 to 49 irrespective of contract timing. This control conditions on residual local economic variation, so that any effect attributed to $p_{i,t+1}$ is net of contemporaneous provincial conditions.²² Adding this control leaves the estimates essentially unchanged. Appendix Figures B.9.1 and B.9.2 report the results.

6.2 2SLS Estimates

Table 1 reports the IV estimates.²³ On the earnings, employment, and experience margins, the short-run effects are large and precisely estimated. Twelve months after ex-

²¹Appendix Figure B.10.1 explore alternative definitions of sectoral and regional mobility. In Figure B.10.1, the sample is restricted, for each period, to workers employed in period $t + h$.

²²There is no mechanical link between $y_{i,t+h}$, defined for workers around contract expiration, and $\bar{Y}_{-i,r,t+h}$, which reflects aggregate outcomes: the full sample used to construct the latter comprises an average of 311,150 workers per month, against 3,249 in the estimation sample. Appendix Table C.1.3 reports descriptive statistics for the full sample.

²³Appendix Table C.2.1 reports corresponding OLS estimates for Equation (4).

Table 1: Effect of Permanent Contracts on Worker Careers

	(1)	(2)	(3)	(4)	(5)	(6)
	Earnings	Cum. Earnings	Employment	Experience	Change Region	Change Sector
<i>Panel A: Short-term effects (12 months)</i>						
$p_{i,t+1}$	0.201*** (0.067)	7.123*** (0.637)	0.176*** (0.049)	0.431*** (0.036)	-0.103*** (0.039)	-0.276*** (0.051)
Obs.	701,761	701,761	701,761	701,761	701,761	701,761
Mean Dep.	-0.223	7.739	0.696	6.574	0.118	0.354
<i>Panel B: Long-term effects (60 months)</i>						
$p_{i,t+1}$	0.224** (0.097)	21.030*** (3.222)	0.170*** (0.062)	1.137*** (0.169)	-0.106** (0.047)	-0.342*** (0.056)
Obs.	701,761	701,761	701,761	701,761	701,761	701,761
Mean Dep.	-0.264	39.204	0.563	9.110	0.201	0.553

Notes: IV estimates of Equation (4). $p_{i,t+1}$ indicates whether worker i holds an open-ended contract in $t + 1$, the month immediately following fixed-term contract expiration. Sample and controls as in Figure 4. First-stage F -statistic: 195.2. Standard errors clustered at the province $\times$ year level in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

piration, conversion raises total monthly earnings by 0.201, or 20.1 percent of baseline earnings (Column 1), cumulative earnings by an amount equivalent to 7.1 months of baseline pay (Column 2), the probability of employment by 17.6 percentage points (Column 3), and accumulated labor market experience by 0.43 years, or roughly five months (Column 4). Both magnitudes are large relative to the sample means reported below each panel. The employment gain equals a quarter of the mean employment rate at that horizon.

Column 1 measures *total* earnings, which combines whether the worker is employed at all with how much she earns conditional on employment. This distinction allows us to decompose earnings growth into within-firm growth, changes associated with moving across firms, and earnings losses associated with non-employment. Table 2 reports the decomposition; implementation details and the corresponding reduced-form paths appear in Appendix Section A.3.

The decomposition attributes almost all of the earnings effect to the employment margin. One month after expiration, the total effect of 0.725 divides into 0.608 from the employment margin, 0.089 from between-firm wage growth, and 0.028 from within-firm wage growth. The employment margin accounts for 84 percent of the total at $h = 1$, 88 percent at twelve months, and 76 percent at five years, and it is the only component that remains statistically significant beyond the first months after expiration. The between-firm component is significant only at $h = 1$, consistent with a brief window in which some converted workers move to better-paying employers before their mobility declines. The within-firm component—wage growth for workers who remain at the firm in which they converted—is not statistically distinguishable from zero after the first six months, and

Table 2: Effects of Permanent Contracts on Worker Careers (Between-Within Decomposition)

	Horizon h (months)						
	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$	$h = 48$	$h = 60$
Total earnings	0.725*** (0.069)	0.557*** (0.065)	0.201*** (0.067)	0.280*** (0.072)	0.164** (0.074)	0.178* (0.095)	0.224** (0.097)
Between-firm component	0.089*** (0.028)	0.037 (0.036)	0.010 (0.039)	0.021 (0.047)	0.017 (0.051)	0.006 (0.074)	−0.004 (0.083)
Within-firm component	0.028*** (0.006)	0.044** (0.021)	0.015 (0.024)	0.042 (0.037)	0.053 (0.038)	0.020 (0.064)	0.058 (0.076)
Employment margin	0.608*** (0.058)	0.475*** (0.050)	0.176*** (0.049)	0.217*** (0.051)	0.094* (0.051)	0.152** (0.063)	0.170*** (0.062)

Notes: IV estimates from equation (4). The dependent variable in each row measures earnings growth since period 0, decomposed into within-firm growth, between-firm growth, and non-employment. Sample restrictions and controls are identical to those in Figure 4. Robust standard errors clustered at the province $\times$ year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the estimates are too imprecise to rule out a modest gain: at sixty months it stands at 0.058 with a standard error of 0.076.

Over the longer run the experience advantage persists. By 60 months, converted workers have accumulated 1.14 additional years of total experience (Column 4), and, as Appendix Table C.2.3 shows, some 2.8 additional years of experience under open-ended contracts. The second figure exceeds the first because conversion does not simply add employment: it substitutes roughly 1.62 years of fixed-term experience for open-ended experience. Converted workers, therefore, end the observation window with both more experience and a different composition of experience—precisely the combination that the literature on contract-specific returns predicts should command a wage premium (Garcia-Louzao et al., 2023).

The earnings estimates do not detect such a premium. The total earnings effect at 60 months, 0.224 (Column 1), is if anything slightly larger than at twelve months, but the decomposition in Table 2 shows that it continues to operate through employment: the employment margin contributes 0.170 of the 0.224, while the within- and between-firm components are 0.058 and −0.004 and neither is distinguishable from zero. Under constant returns to experience the prediction runs the other way: as the experience gap widens, the wage gap conditional on employment should widen with it. Instead, the conditional components remain flat at every horizon.

Columns (5) and (6) of Table 1 quantify the mobility reductions shown in the reduced form. Conversion lowers the probability of having changed sector by 27.6 percentage points within the first year and by 34.2 percentage points at five years, against sample means of 0.35 and 0.55 respectively. The probability of having changed province falls by 10.3 percentage points at twelve months and by 10.6 percentage points at five years,

against means of 0.12 and 0.20. For compliers, in other words, conversion removes the greater part of the sectoral and geographic reallocation that would otherwise have taken place within five years.²⁴

We benchmark the realized conditional earnings path against a constant-returns counterfactual. Applying the estimates of [Garcia-Louzao et al. \(2023\)](#)—annual returns of 0.05 log points per year of open-ended experience and 0.0421 per year of fixed-term experience, together with a fixed-term wage penalty of 0.0359—to our IV estimates of experience accumulation and contract status implies a wage premium, conditional on employment, of 5.1 percent at twelve months and 8.2 percent at five years.²⁵ Figure 6 plots the realized conditional-on-employment earnings path against this benchmark: the two coincide over the first year and then separate, with the realized path flat while the benchmark rises with accumulated experience.

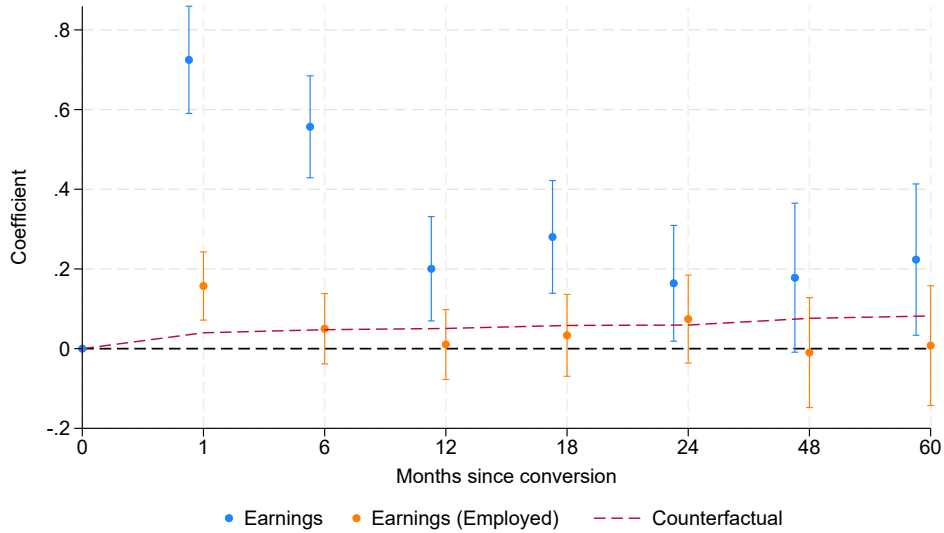
Convergence in the comparison group accounts for part of the pattern, but cannot fully explain it. Workers who do not upgrade at $h = 1$ transition into permanent positions over subsequent months (Panel (f) of Figure 4), accumulate open-ended experience of their own, and narrow the gap with early upgraders. The convergence is nonetheless incomplete: converted workers retain a statistically significant advantage in accumulated open-ended experience at 60 months, and the counterfactual in Figure 6 already incorporates the observed convergence in contract status while still predicting a substantial premium. What requires explanation is not that the comparison group catches up, but that converted workers, despite persistently richer employment histories, do not convert that advantage into higher wages.

The mobility results point to a mechanism. In search models with on-the-job search, wage growth operates through two margins: human capital accumulated within the job, and reallocation toward higher-paying firms or better matches across jobs ([Postel-Vinay and Robin, 2002](#); [Cahuc et al., 2006](#)). A lasting reduction in mobility mutes the second margin. This cannot by itself account for the pattern in Table 1, however, if forgone mobility imposed a uniform cost, it would depress the wages of all converted workers equally. Section 6.3 examines whether the cost instead depends on the match in which workers are retained.

²⁴Figure B.10.3 examines occupational mobility. Consistent with the main results, workers more likely to convert to permanent contracts are less likely to switch occupations following fixed-term contract expiration.

²⁵The counterfactual change in log earnings is $\Delta \ln w = \beta_{OE} \Delta Exp^{OE} + \beta_{FT} \Delta Exp^{FT} + \delta \Delta FT$, where $\beta_{OE} = 0.05$ and $\beta_{FT} = 0.0421$ are the annual returns to open-ended and fixed-term experience and $\delta = -0.0359$ is the fixed-term wage penalty, all taken from [Garcia-Louzao et al. \(2023\)](#). The three experience and status terms are the IV estimates from Table C.2.3. At 12 months, $\Delta Exp^{OE} = 0.873$, $\Delta Exp^{FT} = 0.431 - 0.873 = -0.442$, and $\Delta FT = -0.680$, giving $\Delta \ln w = 0.04945$, or $e^{0.04945} - 1 \approx 5.1\%$. At 60 months, the estimated counterfactual is 8.2 percent.

Figure 6: Effect of Permanent Contracts on Earnings



Notes: IV estimates of the effect on earnings from equation (4). The figure reports three measures: (i) earnings growth relative to period 0; (ii) earnings relative to the employed baseline; and (iii) counterfactual earnings paths using estimates from [Garcia-Louzao et al. \(2023\)](#). Sample restrictions and controls are identical to those in Figure 4. Confidence intervals use standard errors clustered at the province $\times$ year level.

6.3 Mechanism: Lock-In and Firm Growth

The long-run estimates in Table 1 admit two main interpretations. One attributes the missing returns to the contract itself. Converted workers accumulate more experience, but that experience is weakly rewarded, so the flat long-run profile reflects a genuinely low return. This interpretation predicts *uniformity*: since the low return attaches to the contract rather than the employer, the missing return should not vary with the characteristics of the initial employer.

The second interpretation attributes the missing returns to the composition of the employers. A permanent contract raises the value of remaining in the current match and the cost to outside employers of poaching the worker, tying her subsequent career to the firm in which she is converted. In search models with on-the-job search, wage growth runs through two margins: human capital accumulated within the job, and reallocation toward better-paying firms or better matches across jobs ([Postel-Vinay and Robin, 2002](#); [Cahuc et al., 2006](#)). Conversion attenuates the second margin (see Figure 5), leaving within-firm progression as the main source of wage growth.

If firms differ in their potential earnings growth ([Arellano-Bover and Saltiel, 2021](#)), this interpretation predicts the missing return should appear where internal progression is weakest and be absent where the retaining firm offers an internal ladder to climb. Averaging across workers would then mask large gains in some matches against losses in others, and the consequences of employment protection would turn on where workers become

protected rather than on the contract itself. A substantial long-run return in a subset of matches would be difficult to reconcile with a uniformly low return to experience under permanent contracts, or with uniform effort responses or uniform wage compression.

We proxy the earnings growth potential of the initial firm by the average monthly earnings growth of its MCVL workforce in the year the worker’s fixed-term contract expires, so that the measure is determined prior to subsequent outcomes. We residualize workers’ monthly earnings growth on age, education, and gender,²⁶ thereby accounting for differences in workforce composition, and average the residuals within firm using a leave-one-out measure that excludes each worker’s own trajectory. This classification is undefined for workers who are the only MCVL employee observed in their firm, and dropping them leans the estimation sample toward larger firms.²⁷ We then estimate Equation (4) separately below and above the median of the firm-growth distribution (Table 3), which leaves 207,042 and 206,128 observations. The instrument predicts conversion strongly in both groups, with first-stage coefficients of 0.074 and 0.063.

Figure B.13.1 shows the reduced-form dynamics by firm-growth group. In both groups, the pre-expiration coefficients are small, and earnings jump after expiration. The trajectories then separate: in high-growth firms the gain remains positive five years out and grows slightly after the second year, the pattern implied by the counterfactual path in Figure 6, while in low-growth firms it declines to indistinguishable from zero. The gap opens with the horizon, not at expiration, as the worker’s attachment to the initial employer builds over time.

Table 3 quantifies this difference. Twelve months after expiration, the estimated effects are similar across the two groups: conversion increases monthly earnings by 20.4 percent among workers below the median and by 30.3 percent among those above the median (Columns 1 and 2). The other margins are equally close: employment rises by 20.9 and 18.6 percentage points, cumulative earnings by amounts equivalent to 7.9 and 6.5 months of baseline pay, and experience by 0.47 and 0.38 years, roughly six and five months.

At the five-year horizon, the earnings estimates diverge. Workers converted above the median retain a gain of 46.0 percent of baseline earnings, significant at the one percent level (Column 4), while the estimate below the median is 0.178, smaller than its own twelve-month value and indistinguishable from zero (Column 3). The divergence is not driven by how much experience the two groups accumulate, which is nearly identical at 1.27 and 1.18 additional years, a difference of about one month. The employment margin moves with earnings, though less sharply. At twelve months the employment effects are close, 0.209 below the median and 0.186 above it, but by sixty months they

²⁶Specifications that residualize using year fixed effects, and that restrict the sample to firms with at least five observed workers per year, yield similar patterns (Tables C.6.2 and C.6.3).

²⁷Re-estimating the pooled specification on the subsample for which the measure is defined yields slightly larger point estimates while leaving the qualitative conclusions unchanged (Table C.6.4).

have separated: the below-median estimate falls to 0.157 and is significant only at the ten percent level, while the above median estimate rises to 0.281 and is significant at one percent. Part of the long-run earnings gap is about employment itself, not pay conditional on employment. The employment gap of 0.124, although imprecisely estimate, points the same way as the earnings result: conversion at a high-paying match both raises pay and secures a firmer attachment to employment, while conversion at a low-paying match does neither five years out. The two groups accumulate the same additional experience, and only one group is paid for it.

Mobility falls in both groups, and by more in the low-growth group. By 60 months the reduction in the probability of having changed sector is 51.4 percentage points below the median against 30.6 above it, and regional mobility falls by 19.3 percentage points below the median against an insignificant 3.7 above it.

The same ordering emerges when firms are partitioned by size. Among workers in large firms, the long-run earnings effect is 34.1 percent of baseline earnings, while the employment effect is 23 percentage points. Among workers in small firms, the corresponding estimates are 8.3 percent and 9.2 percentage points, neither statistically distinguishable from zero (Appendix Table C.6.1). The gradient tracks the internal opportunities for earnings growth that larger firms are known to offer (Arellano-Bover, 2024).

The long-run return varies with the retaining firm. Converted workers accumulate nearly identical additional experience regardless of where they start, yet a measurable earnings gain emerges only among those converted in firms with strong earnings growth. That is what the second interpretation predicts and what the first cannot accommodate. The firm, then, is an important determinant of the payoff to conversion: once workers convert, their subsequent earnings depend on the opportunities available within that firm, while the loss of outside options limits reallocation to better matches. The earnings cost of attachment therefore depends on the initial match; its insurance value need not.

6.4 Family Formation as a Behavioral Response

If workers treat permanent employment as a durable improvement in income stability rather than a transitory shock, behavioral responses should appear along margins sensitive to long-run income expectations. Family formation is one such margin, and the fertility response provides evidence on whether workers perceive the OE as conferring genuine permanent stability, a maintained assumption of the lock-in mechanism.

Figure B.14.1 reports reduced form estimates from our instrument on the effect of conversion on the probability of having an additional child, controlling for baseline fertility.²⁸ Panel (a) shows that conversion raises the probability of an additional child

²⁸The MCVL records cohabitants rather than children directly. We classify a cohabitant as a child when the age difference with the worker is at least 18 years.

Table 3: Employment Lock-In and Earnings Divergence: Heterogeneity by Firm Growth

	Panel A: $h = 12$ months		Panel B: $h = 60$ months	
	Below median	Above median	Below median	Above median
	(1)	(2)	(3)	(4)
<i>First stage</i>				
log OEC _{-i,r,t+1}	0.074*** (0.007)	0.063*** (0.006)	0.074*** (0.007)	0.063*** (0.006)
<i>Earnings and employment</i>				
Earnings	0.204** (0.101)	0.303*** (0.104)	0.178 (0.141)	0.460*** (0.142)
Cumulative earnings	7.876*** (1.023)	6.530*** (1.020)	23.196*** (5.192)	27.757*** (4.940)
Employment	0.209*** (0.072)	0.186** (0.084)	0.157* (0.086)	0.281*** (0.089)
Experience (years)	0.473*** (0.059)	0.382*** (0.060)	1.267*** (0.249)	1.176*** (0.250)
<i>Mobility</i>				
Change sector	-0.327*** (0.081)	-0.305*** (0.088)	-0.514*** (0.087)	-0.306*** (0.087)
Change region	-0.122** (0.060)	-0.099 (0.068)	-0.193*** (0.072)	-0.037 (0.079)
Mean dep. (earnings)	-0.195	-0.234	-0.211	-0.294
Observations	207,042	206,128	207,042	206,128

Notes: IV estimates of equation 4. Firms are classified according to their leave-one-out average monthly earnings growth, with the median threshold computed on the analysis sample. Growth is residualized by removing interactions on age, education and gender. Below-median firms are at or below the 50th percentile; above-median firms are above the 50th percentile. Sample restrictions and controls are as in Figure 4. Robust standard errors clustered at the province $\times$ year level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of Permanent Contracts on Earnings-Growth Quartiles

Panel A: Short term effects (12 months)					
	Earnings Q_4	Earnings Q_3	Earnings Q_2	Earnings Q_1	Employment
	(1)	(2)	(3)	(4)	(5)
$p_{i,t+1}$	-0.122** (0.048)	0.112** (0.051)	0.293*** (0.051)	-0.283*** (0.053)	-0.201*** (0.056)
Obs.	488,108	488,108	488,108	488,108	701,761

Panel B: Long term effects (60 months)					
	Earnings Q_4	Earnings Q_3	Earnings Q_2	Earnings Q_1	Employment
	(1)	(2)	(3)	(4)	(5)
$p_{i,t+1}$	-0.043 (0.056)	0.042 (0.063)	0.167*** (0.064)	-0.165** (0.065)	-0.193*** (0.070)
Obs.	395,310	395,310	395,310	395,310	701,761

Notes: IV estimates based on equation (4). Panels correspond to horizons $h \in \{12, 60\}$ months. Within each panel, columns (1)–(4) report the effect on the probability of falling into the top through bottom quartile of the earnings-growth distribution, conditional on employment in month h ; workers not employed in month h are excluded from these four regressions at that horizon. Column (5) reports the effect on the probability of being non-employed in month h and is estimated on the full sample. Sample restrictions and controls are identical to those in Figure 4. Standard errors, clustered at the province $\times$ year level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

within 24 months, though the estimate is not statistically significant. Panel (b) reports a similar pattern for workers younger than 30. The clearest response appears in Panel (c): among female workers, the probability of an additional child rises steadily after conversion and peaks 24 months after the expiration of the fixed-term contract. The gap narrows thereafter and loses significance, consistent with control-group workers obtaining stable employment with a delay and adjusting fertility accordingly.

The pattern is consistent with the stability interpretation. A literature on labor market duality documents that fixed-term contracts delay entry into parenthood (Adsera, 2004; Rica and Iza, 2005; Guner et al., 2024), and the estimated response is large relative to baseline rates. Its concentration in the period immediately after conversion indicates that workers advance plans deferred under employment uncertainty. The timing of the female response, peaking two years after conversion and then converging, is what one would expect if the control group eventually attains comparable stability rather than if the treated group were responding to a transitory windfall.

6.5 Distributional Effects

The estimates so far capture the mean of the earnings-growth distribution, but are silent about its spread. Two contracts may generate the same expected earnings growth while exposing workers to very different risks. Reallocation across employers is a source of risk as much as of opportunity: a worker who changes firms may land a substantially

better match or a substantially worse one. If conversion forecloses reallocation, it should compress the distribution from *both* ends.

Table 4 reports IV estimates of the effect of conversion on the probability of being in each quartile of the earnings-growth distribution, conditional on employment at the relevant horizon. Twelve months after contract expiration, conversion shifts workers away from both tails of the distribution (Panel A). Converted workers are 28.3 percentage points less likely to be in the bottom quartile (Column 4) and 12.2 percentage points less likely to be in the top quartile (Column 1). At the same time, they are 29.3 percentage points more likely to be in the second quartile (Column 3) and 11.2 percentage points more likely to be in the third (Column 2). All four estimates are statistically significant.

Five years after expiration, the shift away from the bottom of the distribution remains. Converted workers are 16.5 percentage points less likely to be in the bottom quartile and 16.7 percentage points more likely to be in the second quartile, both statistically significant (Panel B). At the upper end of the distribution, the estimates are smaller and not statistically different from zero: -0.043 for the top quartile and 0.042 for the third quartile.

Permanent employment provides insurance against poor earnings outcomes, protecting workers from the low earnings that can result from a bad match or a spell out of work. The level estimates in Section 6.3 do not capture this benefit, and, as discussed there, it need not depend on the earnings trajectory of the retaining firm. The protection comes at a cost: the negative and statistically significant effect on the top quartile at twelve months indicates that conversion also reduces the chance of a large earnings gain following a favorable move. Both effects follow from the same reduction in reallocation. The modest long-run wage return is a change in the distribution of earnings, not simply the absence of a gain.

7 Robustness Checks and Additional Results

Social security records

One potential concern with our measure of new open-ended contracts, $\log OE_{-i,r,t+1}$, is that it is based on a 4% random sample of the workers registered with Social Security, which could reduce precision in capturing the actual number of open-ended contracts during each period. However, we argue that the MCVL dataset is sufficiently rich to capture meaningful variation in openings of open-ended contracts by month and province, key factors for constructing our instrument. This section compares our MCVL-based measure with measures from Social Security registry data to address this concern.

We use two time series from Social Security records: the monthly count of affiliates by province and contract type, beginning in January 2009, and the data on newly created open-ended contracts per month over the same period. The affiliate data enables us

to track variations in the number of affiliates as a proxy for the opening of open-ended contracts over time and across provinces. Additionally, we examine the data on new open-ended contracts, which directly captures the dynamics of each contract type, but lacks regional variation, relying solely on the time dimension. Both analyses show that our MCVL-derived measure of new open-ended contracts closely aligns with Social Security population data, supporting its validity. The analysis uses data from Social Security records, accessible online via PX-Web.²⁹ This dataset provides detailed monthly information on the average number of affiliates by province and contract type from 2009 to the present. However, despite its richness, the relatively short time coverage means that it cannot be used in the main analysis. Using these data, we compute monthly counts of full-time affiliates under fixed-term and open-ended contracts and compare them with the series of newly created open-ended contracts constructed from the MCVL.

Figure B.3.1 in the Appendix presents the number of new open-ended contracts derived from the MCVL alongside the count of new OEs from the Social Security registry from January 2009 to March 2020. Panel (a) shows a strong correlation between the two series. While we observe minor discrepancies, these are expected as the MCVL restricts to 4% of the population, which introduces some noise. Nevertheless, the overall trends are consistent, indicating that the MCVL provides a reliable representation of open-ended contract dynamics compared to the full Social Security data. Panel (b) shows the residuals from a regression of each time series in Panel (a), controlling for year and month-fixed effects. This approach helps to reduce volatility in both series and supports a strong correlation between them.

As an additional robustness check, Table C.4.1 uses our instrument as described in the main text: the logarithm of new open-ended contracts by province and month. This is compared to the actual number of affiliates in open-ended contracts by month and province. The analysis covers the period from January 2009 to March 2020. The results indicate a strong alignment between the two series, as evidenced by the high R^2 values across all specifications. This indicates that our measure effectively captures a significant portion of the variation associated with monthly and provincial fluctuations in creating open-ended contracts.

Alternative instrument measures

Our baseline specification examines the leads and lags in the number of new open positions at the *regional* level. This same approach can be extended to analyze new openings for permanent positions at national or more granular levels, such as industry-specific data or combining industry and province variation. This introduces an important consideration regarding the appropriate level of disaggregation that accurately reflects the relevant labor market for workers and influences the worker's chances of securing a

²⁹<https://w6.seg-social.es/PXWeb/pxweb/es/> "Afiliados R. GENERAL por sexo, tipo de contrato y jornada, provincial."

permanent position. Previous work by [Marinescu and Rathelot \(2018\)](#) and [Manning and Petrongolo \(2017\)](#) show that workers tend to focus on job opportunities nearby and are discouraged by the distance in job vacancies. This motivates our preference for exploiting the variation at the specific month of a fixed-term contract’s expiration within the same province of the previous job. However, we also present evidence in this section that our findings are robust across alternative instrument definitions. As shown in Figure [B.6.2](#) in the Appendix, we find similar patterns in these alternative specifications.³⁰ The instrument is, therefore, relevant, irrespective of whether we measure it at the national, regional (baseline), or industry level. Moreover, these findings are robust to excluding from the dataset months of potentially high job-seasonality (see Appendix [B.7](#)).

Exclusion restriction

Next, we present further evidence that supports the validity of our instrument, showing that it is not systematically related to economic conditions. As mentioned earlier, a key aspect of our identification strategy relies on whether our instrument reflects random fluctuations orthogonal to the business cycle. We exhaustively control for economic trends and cycle conditions in our model through time-fixed effects, leads and lags of our instrument, and new fixed-term contracts. Despite these controls, one might still be concerned about unobserved factors that could challenge our identification assumptions. A related concern is more specific to our setting: a transitory local burst in open-ended hiring could raise a worker’s earnings even absent her own conversion—for instance, through improved outside options or bargaining position—in which case the instrument would violate the exclusion restriction on the wage margin rather than merely proxying the business cycle. If either concern were relevant, we would expect a positive effect on employment and on earnings irrespective of which contract type a worker found in event period $h = 1$. To alleviate this concern, we estimate the following placebo test. Following the specification in equation [\(4\)](#), we estimate the reduced-form effect of the instrument on employment and on earnings after removing, at each horizon, workers who hold an open-ended contract in $t + h$. The retained sample therefore contrasts workers who remain in fixed-term employment with workers who are not employed, and excludes by construction the margin through which the instrument is meant to operate. If our instrument captured general business cycle conditions, or if it affected wages through channels other than conversion itself, we would also expect a positive impact on employment and earnings for these workers. Appendix Figure [B.11.1](#) shows no significant or systematic response for either outcome: the employment coefficient (Panel a) is statistically indistinguishable from zero, as is the earnings coefficient (Panel b). The absence of a detectable earnings response among stayers is evidence against wage-margin leakage through improved outside options

³⁰Additionally, the different instrument alternatives show similar predictive power. The R^2 values for the national, province, sectoral, sector-by-province, and autonomous communities instruments are 0.0947, 0.0948, 0.0949, 0.0832, and 0.0950 respectively.

or bargaining, and, together with the employment result, rules out a pure business-cycle interpretation of the instrument.

Exogenous reason of dismissal and tenure restrictions

Section 5.2 detailed the sample restrictions used in our estimation, focusing on workers employed in the expiring fixed-term contract for at least 0.3 years to eliminate extremely short contracts. In this subsection, we extend our analysis with two additional exercises. First, we limit the sample to spells where the reason for dismissal is unrelated to firm-specific factors. Second, we find similar first-stage results under alternative tenure restrictions.

First, we exploit the fact that the MCVL records the reason for termination of each employer–employee spell. To refine our analysis, we further restrict the sample to spells where the dismissal reason is linked explicitly to the expiration of a fixed-term contract.³¹ Then, we repeat the first stage, where we study the effect of our instrument on the probability of holding an open-ended contract. The corresponding figure is provided in Section B.8 of the appendix, while the results are presented in Figure B.8.1. This exercise provides evidence that our instrument affects the probability of transitioning to an open-ended contract following fixed-term contract termination.

As described in Section 5.2, we restrict the sample to those workers in the last month of a fixed-term contract with a tenure of 0.3–2 years. In Appendix B.12, we extend our first-stage and the reduced-form analysis by considering the tenure window within to 0.4–2 years and alternative tenure restriction in that range. The results remain qualitatively similar, showing a positive effect on short-term earnings growth and a higher likelihood of continued employment. The former, however, dissipates over time.

Heterogeneity

Figure B.15.1 characterizes heterogeneity in the effects of OE conversion across worker types. Short-run earnings gains are largest for men, workers without secondary education, and those with low baseline earnings, the same groups that exhibit the largest employment and cumulative earnings gains. The gradient by baseline earnings is particularly informative. Low-earning workers benefit primarily through higher employment, consistent with their greater baseline risk of non-employment, whereas workers with above-median earnings experience larger wage gains conditional on employment.

Long-run heterogeneity instead emerges along the mobility margin. OE conversion significantly reduces sector and province switching among workers with high baseline earnings, while the corresponding estimates for low earners are small and statistically insignificant. This pattern is consistent with the lock-in mechanism: workers with stronger initial labor market attachment have more to lose from relinquishing the employment

³¹We use Code 54 (involuntary dismissal) and Code 93 (end of fixed-term contract). While the literature on the earnings consequences of job loss typically relies on mass-layoff events, such events are rare in our data. Of the total observations in our first-stage sample, only 2,286 are associated with a mass-layoff event.

protection associated with permanent contracts and therefore face weaker incentives to reallocate across firms, whereas low earners retain stronger incentives to search despite conversion.

These results suggest that conversion operates through different channels across the earnings distribution. For workers with weak initial labor market attachment, the primary benefit is improved employment stability. For workers with stronger attachment, conversion primarily affects subsequent mobility decisions by increasing the value of remaining with the initial employer. Appendix Section C.3 reports subgroup-specific first-stage estimates, confirming strong instrument relevance across all groups.

Two Endogenous variables

Finally, we explore whether part of the earnings effect documented in Section 6 is driven by the employment consequences of holding an OE contract. Specifically, we extend the specification to control not only for OE status in period $t + 1$ but also for employment status in $t + 1$, as shown in the following equation:

$$\begin{aligned}
 y_{it+h} = & \beta_1 p_{i,t+1} + \gamma_1 emp_{t+1} + \sum_{k \neq 1, k=-24}^{24} \beta_k \log OEC_{-i,r,t+k} \\
 & + \sum_{k \neq 1, k=-24}^{24} \gamma_k \log TNC_{-i,r,t+k} + X'_{it} \theta + \mu_r + \psi_t + \phi_s + \epsilon_{it},
 \end{aligned} \tag{5}$$

where the specification follows Equation 4, but additionally includes employment status in $t + 1$. Since employment status may itself be endogenous, we instrument for both endogenous variables using two instruments: the first lead of new OE contracts and the first lead of total contracts. The inclusion of total contracts as an instrument is justified by its correlation with employment status in $t + 1$ (relevance condition). Both the leave-out measures of new OE contracts and total contracts are constructed at the month-autonomous community level to ensure sufficient variation when instrumenting for two endogenous variables.

Table C.7.1 reports the results. Here the instrument is defined at the month $\times$ autonomous community level rather than the month $\times$ province level, since the province-level variation lacks power in this specification. The first stages in Columns (1) and (2) show that each instrument loads primarily on its intended margin. New OE contracts move OE status roughly 5.6 times more than employment (0.090 versus 0.016), while total new contracts move employment roughly 7 times more than OE status (0.120 versus 0.017). The ratio 0.016/0.090 implies that about 18 percent of the OE contracts induced by the instrument represent transitions out of non-employment, with the remaining 82 percent switches from a temporary to a permanent contract among workers who would have been employed in either case.

That same ratio determines how the estimates move once employment is instrumented

separately. At twelve months, Column (4) puts the return to employment at 0.222 and the additional return to that employment being permanent at 0.092, against the pooled estimate of 0.133 in Column (3). Because the system is exactly identified, the gap between the two OE coefficients is mechanically the cross-loading ratio times the employment coefficient, $0.178 \times 0.222 = 0.040$, so roughly 30 percent of the one-year effect operates through employment rather than contract type. Treating the first-stage ratio as known, this component has a standard error of about 0.016 and is therefore estimated far more precisely than a direct comparison of Columns (3) and (4) would suggest, since the two coefficients are strongly positively correlated.

At sixty months the employment margin has dissipated. The employment coefficient falls to 0.004 with a standard error of 0.110, and the OE coefficient is unchanged at 0.109 relative to 0.110 in Column (5). The implied mediated component is bounded within roughly ± 0.04 against a baseline effect of 0.11, so employment accounts for at most about one third of the five-year effect and the point estimate is essentially zero. The confidence interval on the employment coefficient at +60 nonetheless contains the twelve-month estimate, so the evidence speaks to the share of the long-run effect that is mediated rather than to the long-run return to employment as such. The pattern is that the short-run earnings gain is largely extensive margin, whereas what persists at five years is attributable to the contract itself.

Additional controls

We further extend our baseline specification to more tightly control for business-cycle fluctuations. While the baseline already includes year and month fixed effects to capture aggregate and seasonal variation in labor market outcomes, a more demanding specification incorporates year $\times$ month fixed effects and, alternatively, controls for the aggregate leave-one-out average of the outcome variable, $\bar{Y}_{-i,r,t+h}$. This measure is constructed using the full sample of workers aged 18 to 49, irrespective of the timing of their contract expiration, ensuring that there is no mechanical relationship between $y_{it} + h$ for treated workers and $\bar{Y}_{-i,r,t+h}$ for the broader labor market. Together, these controls absorb shocks common to workers within a given month and hold constant local economic conditions, thereby isolating the transitory variation in the availability of open-ended positions that is orthogonal to general business-cycle dynamics.

The results of these robustness exercises for the reduced-form estimates are reported in Appendix Figures B.9.1 and B.9.2. They show only a modest attenuation in the estimated coefficients relative to the baseline, with no change in the qualitative conclusions.

Finally, Table C.2.2 presents additional specifications that incorporate a richer set of fixed effects, including occupation $\times$ month, occupation $\times$ year, sector $\times$ month, and sector $\times$ year. The estimates remain stable, reinforcing our results are not driven by differential trends across occupations or sectors.

8 Discussion & Conclusion

The matching of workers to firms, jobs and contract types has important implications both for individual careers and aggregate outcomes. However, it is difficult to provide causal evidence on this question, as workers sort non-randomly into jobs. The key challenge is to disentangle whether differences in career trajectories are due to unobserved heterogeneity on the supply side or whether they reflect true causal effects from job or other attributes on the demand side.

By examining the Spanish context as a case study, we investigate how different types of contracts affect workers’ careers. Consistent with recent evidence by [Garcia-Louzao et al. \(2023\)](#), workers who spent more time in open-ended contracts experienced higher earnings growth than workers who instead spent time in fixed-term positions. However, such differences in earnings growth may reflect not only differences in returns between permanent and temporary contracts but also heterogeneity between workers.

A crucial test to discriminate between these explanations is the pattern of earnings growth *before* workers enter a permanent contract. Using an event study approach, we reject the assumption of “parallel pre-trends”, as workers who switch from a fixed-term into an open-ended contract experience high earnings growth even before that switch: their earnings grow by 7 log points over the 15 months preceding the switch, while earnings growth is negligible for workers who switch to another fixed-term contract instead. A fixed effects approach accounting for time-constant wage differences between workers, as typically used to account for the selection of workers into firms or regions, is therefore not sufficient to address selection into contract types.

We, therefore, propose a novel identification strategy to address the non-random sorting of workers into jobs. Using matched employer-employee data, we isolate quasi-random variation in worker-firm matches by interacting high-frequency information on the duration of contracts on the labor supply side and transitory fluctuations in job creation on the demand side. Our proposed instrumental variable is uncorrelated to workers’ characteristics and past employment history but highly predictive of their probability of securing a permanent position. This allows us to study the causal effect of entering a permanent contract for “compliers”, i.e., workers on the margin of finding a permanent contract and whose contract status is sensitive to labor market conditions.

We find that workers securing a permanent contract experienced a large gain in earnings in the short run. These earning gains are primarily due to more stable employment relationships; while workers in permanent contracts are employed uninterruptedly, workers in the control group tend to experience breaks in their employment status when switching from one fixed-term contract to the next. As a result, workers in permanent contracts gain more work experience, especially more experience in open-ended positions, than workers who do not find a permanent position as soon as their fixed-term contract

ends.

However, these initial earnings gains shrink over time. As a qualitative pattern, this is not surprising, as it reflects a catching-up process in the control group: some workers who initially did not find a permanent position become increasingly likely to find such position as time goes by; and once they do, their employment relationships and therefore earnings stabilize. What is surprising is where the surviving gain comes from. The total earnings effect remains positive and statistically significant after five years, at 22.4% of baseline earnings, but it operates through employment rather than through pay: the employment margin accounts for 0.170 of the 0.224 total, while the within-firm and between-firm components of wage growth are both indistinguishable from zero. Benchmarking our own estimates against published contract-specific returns to experience, the additional experience acquired by treated workers would imply a wage premium of 8.2% five years after conversion; the realized path, conditional on employment, is flat. The wage return to the additional experience is therefore not detectable, and the estimates are not precise enough to rule out a modest one. The striking finding is that treated workers accumulate substantially more work experience without a corresponding increase in wages: five years after entering a permanent contract, they have accumulated 13 more months of work experience than the control group, with a larger share in permanent contracts.

The explanation is not that experience acquired under a permanent contract is intrinsically less valuable, but that conversion ties the worker to the firm in which it occurs. Workers who secure a permanent contract become markedly less mobile: they are less likely to change industry or region, whereas workers who remain on fixed-term contracts continue to move between employers, sectors, and regions. Reduced mobility limits one of the two main sources of wage growth, leaving wage progression within the incumbent firm as the primary remaining margin. This account makes a sharp prediction, which the data confirm: the lower return should arise only when the firm retaining the worker provides limited scope for internal wage growth. Five years after conversion, workers converted in firms with above-median earnings growth gain 46.0% of baseline earnings, compared with a statistically insignificant 17.8% for workers converted in below-median-growth firms, even though the two groups accumulate almost identical additional experience (1.27 versus 1.18 years). The same experience is acquired in both cases, but its earnings return differs substantially. Explanations based on contract type alone, such as a lower return to permanent-contract experience, weaker incentives, or greater wage compression, cannot account for this pattern, since each predicts a lower return regardless of the firm in which the worker remains.

Consistent with this, conversion compresses the distribution of earnings growth from both tails: it shields workers from poor earnings realizations while removing the upside that a fortunate move can deliver. The protection against bad outcomes is the one effect that remains significant at five years, when the effects on the level of earnings no longer

are. What a permanent contract buys these workers is a narrower earnings path, not a higher one.

The positive correlation between experience in permanent contracts and wages reflects selection to an important extent—securing a permanent contract is, in significant part, a consequence of favorable career progression rather than its cause. But the causal effect that survives this correction is neither zero nor uniform. Converting marginal workers raises their employment and experience wherever it occurs; whether it raises their wages depends on where it occurs.

These findings have implications for policy. Neglecting the dynamic selection issue may lead to suboptimal policy recommendations, especially in segmented labor markets where such selection can be easily confounded with the labor market structure. In Spain and other European labor markets, the high prevalence of fixed-term contracts has been assessed as a potential cause for low productivity growth ([Bentolila et al., 2020](#); [Dolado et al., 2016](#)). However, our findings suggest that shifting workers from fixed-term to permanent contracts does not by itself yield long-term benefits on wages, a proxy for productivity: it does so where the retaining firm offers scope for internal progression, and not otherwise. The margin on which such policies operate is informative in this respect, as the workers whose contract status responds to the availability of permanent jobs are disproportionately young and employed in larger, established firms. What is more, shifting workers into permanent positions reduces workers’ geographic and inter-industry mobility, reinforcing another structural problem of European labor markets ([Blanchard and Katz, 1992](#)).

While our focus here is on dual labor markets and the selection into contract types, the methodology we propose can be applied more generally. The key idea is to exploit two advantages of administrative registers, namely their high frequency, such that we know when exactly a worker’s contract ends, and their large size, such that we can measure fluctuations in local labor market conditions. Because many administrative registers offer these same advantages, our method can be applied broadly to study (dynamic) selection in the matching of workers to firms, jobs, and contracts.

References

- ABOWD, J. M., F. KRAMARZ, AND D. N. MARGOLIS (1999): “High Wage Workers and High Wage Firms,” *Econometrica*, 67, 251–333.
- ADSERA, A. (2004): “Changing fertility rates in developed countries. The impact of labor market institutions,” *Journal of population economics*, 17, 17–43.
- ALBERT, C., C. GARCÍA-SERRANO, AND V. HERNANZ (2005): “Firm-provided Training and Temporary Contracts,” *Spanish Economic Review*, 7, 67–88.
- ALTONJI, J., L. KAHN, AND J. SPEER (2016): “Cashier or Consultant? Entry Labor Market Conditions, Field of Study, and Career Success,” *Journal of Labor Economics*, 34, S361–S401.
- ANGRIST, J. D., G. W. IMBENS, AND D. B. RUBIN (1996): “Identification of causal effects using instrumental variables,” *Journal of the American statistical Association*, 91, 444–455.
- ANGRIST, J. D. AND J.-S. PISCHKE (2009): *Mostly harmless econometrics: An empiricist’s companion*, Princeton university press.
- ARELLANO-BOVER, J. (2024): “Career Consequences of Firm Heterogeneity for Young Workers: First Job and Firm Size,” *Journal of Labor Economics*, 42, 549–589.
- ARELLANO-BOVER, J. AND F. SALTIEL (2021): “Differences in on-the-job learning across firms,” *Working Paper*.
- AUER, W. AND N. DANZER (2016): “Fixed-term employment and fertility: Evidence from German micro data,” *CESifo Economic Studies*, 62, 595–623.
- BENTOLILA, S., A. CABRALES, AND M. JANSEN (2023): “Does Dual Vocational Education and Training Pay Off?” .
- BENTOLILA, S., P. CAHUC, J. J. DOLADO, AND T. LE BARBANCHON (2012): “Two-Tier Labour Markets in the Great Recession: France Versus Spain,” *The economic journal*, 122, F155–F187.
- BENTOLILA, S., J. J. DOLADO, AND J. F. JIMENO (2020): “Dual Labour Markets Revisited,” *Oxford Research Encyclopedia of Economics and Finance*.
- BHULLER, M., G. B. DAHL, K. V. LØKEN, AND M. MOGSTAD (2020): “Incarceration, recidivism, and employment,” *Journal of political economy*, 128, 1269–1324.
- BLANCHARD, O. AND L. KATZ (1992): “Regional Evolutions,” *Brookings Papers on Economic Activity*, 23, 1–76.

- BOERI, T. AND P. GARIBALDI (2024): “Temporary Employment in Markets with Frictions,” *Journal of Economic Literature*, 62, 1143–85.
- BOOTH, A. L., M. FRANCESCONI, AND J. FRANK (2002): “Temporary Jobs: Stepping Stones or Dead Ends?” *The economic journal*, 112, F189–F213.
- BRATTI, M., M. CONTI, AND G. SULIS (2021): “Employment Protection and Firm-Provided Training in Dual Labour Markets,” *Labour Economics*, 69, 101972.
- BURDETT, K. AND D. MORTENSEN (1998): “Wage Differentials, Employer Size, and Unemployment,” *International Economic Review*, 39, 257–73.
- BUSCH, C., I. GALVEZ, E. GONZÁLEZ-AGUADO, AND L. VISSCHERS (2025): “Dual Labor Markets, Unemployment, and Career Mobility,” *Working Paper*.
- CABRALES, A., J. J. DOLADO, AND R. MORA (2017): “Dual Employment Protection and (lack of) On-the-Job Training: PIAAC Evidence for Spain and Other European Countries,” *SERIEs*, 8, 345–371.
- CAHUC, P., O. CHARLOT, AND F. MALHERBET (2016): “Explaining the Spread of Temporary Jobs and its Impact on Labor Turnover,” *International Economic Review*, 57, 533–572.
- CAHUC, P., F. POSTEL-VINAY, AND J.-M. ROBIN (2006): “Wage bargaining with on-the-job search: Theory and evidence,” *Econometrica*, 74, 323–364.
- CARD, D., A. R. CARDOSO, J. HEINING, AND P. KLINE (2018): “Firms and Labor Market Inequality: Evidence and Some Theory,” *Journal of Labor Economics*, 36, S13 – S70.
- CARD, D., J. HEINING, AND P. KLINE (2013): “Workplace Heterogeneity and the Rise of West German Wage Inequality,” *The Quarterly Journal of Economics*, 128, 967–1015.
- CARD, D., J. ROTHSTEIN, AND M. YI (2025): “Location, location, location,” *American Economic Journal: Applied Economics*, 17, 297–336.
- CATTANEO, M. D., R. K. CRUMP, M. H. FARRELL, AND Y. FENG (2024): “On binscatter,” *American Economic Review*, 114, 1488–1514.
- CIANI, E. AND G. DE BLASIO (2015): “Getting stable: an evaluation of the incentives for permanent contracts in Italy,” *IZA Journal of European Labor Studies*, 4, 6.
- CONDE-RUIZ, J. I., M. GARCÍA, L. A. PUCH, AND J. RUIZ (2023): “Reforming Dual Labor Markets: “Empirical” or “Contractual” Temporary Rates?” Tech. rep., FEDEA.

- DOLADO, J. J. (2017): *European Union Dual Labour Markets: Consequences and Potential Reforms*, Cambridge University Press, 73–112.
- DOLADO, J. J., S. ORTIGUEIRA, AND R. STUCCHI (2016): “Does Dual Employment Protection affect TFP? Evidence from Spanish Manufacturing Firms,” *SERIEs*, 7, 421–459.
- ELIAS, F. AND H. REDONDO (2025): “Escaping Temporary Traps: Evidence for Young and Adult Workers,” .
- FACCINI, R. (2014): “Reassessing labour market reforms: Temporary contracts as a screening device,” *The Economic Journal*, 124, 167–200.
- GAGLIARDUCCI, S. (2005): “The Dynamics of Repeated Temporary Jobs,” *Labour Economics*, 12, 429–448.
- GALACHO, E. R. (2006): “Las reformas laborales en España (1977-2002),” *Filosofía, política y economía en el Laberinto*, 7–22.
- GARCIA-LOUZAO, J., L. HOSPIDO, AND A. RUGGIERI (2023): “Dual Returns to Experience,” *Labour Economics*, 80, 102290.
- GARCÍA-PÉREZ, J. I., I. MARINESCU, AND J. VALL CASTELLO (2019): “Can fixed-term contracts put low skilled youth on a better career path? Evidence from Spain,” *The Economic Journal*, 129, 1693–1730.
- GARCÍA-PÉREZ, J. I. AND F. MUÑOZ-BULLÓN (2011): “Transitions into Permanent Employment in Spain: An Empirical Analysis for Young Workers,” *British Journal of Industrial Relations*, 49, 103–143.
- GATHMANN, C. AND U. SCHÖNBERG (2010): “How General is Human Capital? A Task-Based Approach,” *Journal of Labor Economics*, 28, 1–49.
- GÜELL, M. AND B. PETRONGOLO (2007): “How Binding are Legal Limits? Transitions from Temporary to Permanent Work in Spain,” *Labour Economics*, 14, 153–183.
- GUNER, N., E. KAYA, AND V. SÁNCHEZ-MARCOS (2024): “Labor market institutions and fertility,” *International Economic Review*, 65, 1551–1587.
- HECKMAN, J. J., S. URZUA, AND E. VYTLACIL (2006): “Understanding instrumental variables in models with essential heterogeneity,” *The review of economics and statistics*, 88, 389–432.
- HERSHBEIN, B. J. (2012): “Graduating High School in a Recession: Work, Education, and Home Production,” *The BE Journal of Economic Analysis & Policy*, 12.

- KAHN, L. B. (2010): “The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy,” *Labour Economics*, 17, 303–316.
- KAMBOUROV, G. AND I. MANOVSKII (2009): “Occupational Mobility and Wage Inequality,” *The Review of Economic Studies*, 76, 731–759.
- LLULL, J. AND R. A. MILLER (2018): “Internal Migration and Work Experience in Dual Labor Markets,” *Unpublished Manuscript*.
- LOPES, M. (2020): “Job Security and Fertility Decisions,” *Available at SSRN 3543204*.
- MANNING, A. (2021): “Monopsony in Labor Markets: A Review,” *ILR Review*, 74, 3–26.
- MANNING, A. AND B. PETRONGOLO (2017): “How local are labor markets? Evidence from a spatial job search model,” *American Economic Review*, 107, 2877–2907.
- MARBACH, M. AND D. HANGARTNER (2020): “Profiling compliers and noncompliers for instrumental-variable analysis,” *Political Analysis*, 28, 435–444.
- MARINESCU, I. AND R. RATHELOT (2018): “Mismatch unemployment and the geography of job search,” *American Economic Journal: Macroeconomics*, 10, 42–70.
- NIETO, A. (2022): “Can Subsidies to Permanent Employment Change Fertility Decisions?” *Labour Economics*, 78, 102219.
- OREOPOULOS, P., T. VON WACHTER, AND A. HEISZ (2012): “The Short and Long-Term Career Effects of Graduating in a recession,” *American Economic Journal: Applied Economics*, 4, 1–29.
- PESOLA, H. (2011): “Labour Mobility and Returns to Experience in Foreign Firms,” *The Scandinavian Journal of Economics*, 113, 637–664.
- POSTEL-VINAY, F. AND J.-M. ROBIN (2002): “Equilibrium wage dispersion with worker and employer heterogeneity,” *Econometrica*, 70, 2295–2350.
- RICA, S. D. L. AND A. IZA (2005): “Career planning in Spain: Do fixed-term contracts delay marriage and parenthood?” *Review of Economics of the Household*, 3, 49–73.
- ROCA, J. D. L. AND D. PUGA (2017): “Learning by Working in Big Cities,” *The Review of Economic Studies*, 84, 106–142.
- SCHWANDT, H. AND T. VON WACHTER (2019): “Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Data Sets,” *Journal of Labor Economics*, 37, S161–S198.

WACHTER, T. V. AND S. BENDER (2006): “In the Right Place at the Wrong Time: The Role of Firms and Luck in Young Workers’ Careers,” *American Economic Review*, 96, 1679–1705.

Appendix

A Supplementary Sections

A.1 Mincer Regression Results

For comparability with previous studies on returns to experience (Roca and Puga, 2017; Garcia-Louzao et al., 2023; Arellano-Bover and Saltiel, 2021), we estimate the contribution of contract-specific experience to earnings growth using a Mincer-type regression. We account for differential returns to experience by explicitly modeling combinations of experience accumulated in fixed-term and open-ended contracts. We estimate the following equation by OLS:

$$\ln w_{irt} = \beta_1 exp_{it}^{FT} + \beta_2 (exp_{it}^{FT} \times exp_{it}) + \beta_3 exp_{it}^{OE} + \beta_4 (exp_{it}^{OE} \times exp_{it}) + X'_{it} \boldsymbol{\Omega} + \sigma_r + \psi_t + \varepsilon_{irt}, \quad (6)$$

where exp_{it}^{FT} and exp_{it}^{OE} denote the worker's experience accumulated until period t in fixed-term and in open-ended contracts, respectively. The variable exp_{it} is the total experience of individual i up to period t . X_{it} is a vector of time-varying individual and job characteristics, including gender and occupation-skill group interacted with educational attainment, sector fixed-effects, age, age squared, and an interaction of tenure with a fixed-term contract indicator, σ_r is a province fixed effect, ψ_t is a year-month fixed-effect, and ε_{ict} is the error term.

Instead of the typical quadratic form of homogeneous returns to experience, equation (6) considers the product between overall experience and contract-specific experience. This interaction captures that the moment at which workers accumulate experience in each type of contract matters. In other words, the returns to an extra year of lower-quality experience at the beginning of the career may differ from the returns at mid-career.

The estimates are reported in Appendix Table A.1.1. Ignoring the distinction between fixed-term and open-ended contracts, Column (1) shows that an additional year of experience is associated with a 2.5% increase in earnings for a worker with ten years of experience. Column (2) decomposes experience by the type of contract under which it was accumulated. While the coefficients on linear experience are similar across contract types, differences in earnings trajectories arise from the interaction terms. In the early years, returns to experience are comparable across contracts, but the growth rate associated with fixed-term experience is lower thereafter. For a worker with ten years of experience, an additional year in a fixed-term contract is associated with a 3.0% increase in earnings, compared to 4.5% for experience accumulated in an open-ended contract.

Although this specification acknowledges that the value of accumulated experience in each type of contract might differ, it ignores the potential sorting of workers into each

type of contract. For instance, if high-ability workers are over-represented in open-ended positions, the coefficients of Column (2) might reflect that more able workers tend to enjoy higher earnings irrespective of contract type. Previous work has addressed this concern by including worker-fixed effects, as in Column (3). The worker-fixed effect slightly attenuates the gap between fixed-term and open-ended contract returns, but the overall pattern remains unchanged. For a worker with ten years of experience, an additional year in a fixed-term position is associated with a wage growth of 4.6% as compared to 5.6% if this experience was accumulated in a permanent contract. These findings are consistent with (Garcia-Louzao et al., 2023), who document lower returns to experience accumulated in fixed-term contracts relative to permanent contracts and show that this gap is unlikely to be driven by firm heterogeneity or match quality. However, the fixed-effects approach employed both in their paper and in Column (3) remains limited in its ability to fully address selection concerns.³²

As we show in Section 4, our descriptive estimates here have, however, no causal interpretation. Instead, they reflect that more able workers are (i) more likely to enter an open-ended contract and (ii) enjoy faster earnings growth irrespective of contract type, a form of selection that is not captured by the fixed-effects approach.

³²One alternative that Garcia-Louzao et al. (2023) implement later on is to instrument experience and tenure using their deviations relative to the average computed within the contract and match the history of the worker. Additionally, they exploit supplementary instruments based on regional variations in the availability of subsidies for hiring workers under open-ended contracts (OEs). In this paper, we leverage another form of variation using precise high-frequency data available in Spanish administrative records.

Table A.1.1: Wage growth in fixed-term and open-ended contracts (Mincer regression)

	ln earnings		
	(1)	(2)	(3)
exp	0.051*** (0.001)		
$exp^2/1000$	-1.314*** (0.032)		
exp_{FT}		0.064*** (0.001)	0.0794*** (0.001)
exp_{OEC}		0.056*** (0.001)	0.0706*** (0.001)
$exp \times exp_{FTC}/1000$		-3.373*** (0.063)	-3.312*** (0.055)
$exp \times exp_{OEC}/1000$		-1.049*** (0.039)	-1.446*** (0.031)
Obs.	16,266,496	16,266,496	16,255,262
R^2	0.475	0.478	0.754
Controls	Yes	Yes	Yes
Individual FEs	No	No	Yes

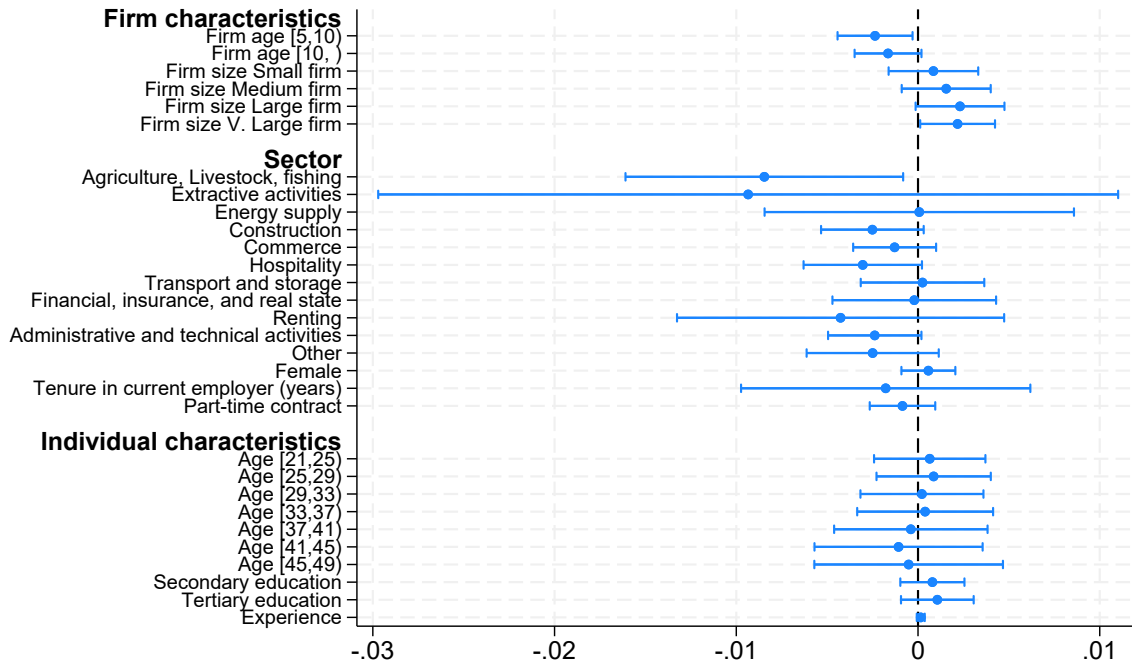
Notes: exp , exp_{FT} , and exp_{OE} denote total experience, experience accumulated in fixed-term contracts, and experience accumulated in open-ended contracts, respectively. Controls include gender, occupation-skill group, interactions with educational attainment, age and age squared, as well as interactions between tenure and an indicator for current fixed-term contract status. All specifications additionally include sector, province, and time fixed effects. Standard errors are clustered at the individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A.2 Independence test

We argue that the number of new permanent positions available when a worker's contract is about to end is effectively random from the worker's perspective and thus exogenous. While we cannot test it directly, we assess the validity of the independence assumption by examining whether the instrument correlates with observable individual or firm characteristics.

In Figure A.2.1, we present the coefficients from a regression of the standardized version (mean zero and variance one) of the instrument $\log OE_{-i,r,t+1}$ on our usual set of control variables, including year, month, province, and sector fixed effects, individual-level characteristics, and leads and lags of the number of new permanent and new temporary contracts – mirroring the specification of our first stage. As an exception, to ease the interpretation of the coefficients on education, we do not interact education with age here. The different panels of the figure report the coefficient estimates for different groups of observable characteristics.

Figure A.2.1: Effect of individual, firm characteristics and sector on $\log OE_{t+1}$



Notes: All regressions include leads and lags of the $\log$ of open-ended contracts, year, month, and province fixed effects. For this exercise we standardize the instrument (mean zero and standard deviation one). Confidence intervals are constructed using standard errors clustered at the province $\times$ year level.

Panel (a) shows no significant relationship between the instrument and worker characteristics, such as age, experience, or education. Panel (b) explores the correlation between the instrument and the worker's sector at baseline. Although its correlation with the indicator for extractive activities is statistically significant, the effect is mini-

mal: the number of new contracts is 3% of a standard deviation below the mean. Finally, Panel (c) examines the correlation with firm characteristics, firm age and size. Again, the correlation between these dimensions and the number of new open-ended contracts is negligible. The results suggest that, conditional on our standard set of fixed effects, the degree of selection is quite limited.

An additional concern is that our instrument may be correlated with the quality of OEs. Certain firms might offer many OEs in a particular month, potentially differing in quality from those in other months. To address this, we provide additional evidence showing that the instrument is not correlated with key job characteristics, including the average duration of new OEs, the average required skill level, and average initial wages. We calculate the average initial wages, average occupation skill level, and average tenure of new OEs at the province, year, and month level. We estimate these outcomes using the following equation:

$$y_{rt} = \beta \log OE_{-i,r,t+1} + \sum_{k \neq 1} \alpha_k \log OE_{-i,r,t+k} + \mu_r + \psi_t + \varepsilon_{rt}, \quad (7)$$

where y_{rt} represents the outcome variable measured at the province, year, and month level. The term $\log OEC_{-i,r,t+1}$ denotes the first lead of the log number of new open-ended contracts, while we additionally control for up to 24 leads and lags of this variable. Similarly, $\log FTC_{-i,r,t+k}$ represents the log number of new fixed-term contracts. Finally, X_{it} includes the same individual control variables as in our main specification.

Table A.2.1: Effect of new open-ended contracts on the composition of new contracts

	(1)	(2)	(3)
	Earnings	Occupation	Tenure
$\log OEC_{-i,r,t+1}$	-20.45 (19.68)	0.046 (0.037)	-0.045 (0.075)
Obs.	8,706	8,706	8,706
R2	0.165	0.267	0.431
Time FE	Yes	Yes	Yes
Region FE	Yes	Yes	Yes

Notes: The table reports the coefficients of the average starting earnings, average occupation skill level, and average tenure of new OECs at the province, year, and month level on the log number of new open-ended contracts in $t + 1$. The additional controls are the same as in our baseline specification. Robust standard errors clustered at the province $\times$ year level are reported in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A.2.1 presents the results on the composition of OECs using our instrumental variable. The findings indicate no significant relationship across any examined outcomes, suggesting that our instrument captures only variations in the quantity of OECs, not their composition.

A.3 Robustness: Between vs. Within earnings change

Panel (a) of Figure 4 shows that workers exposed to an increase in the number of new open-ended contracts at the time of contract expiration experience a large initial rise in earnings. This gain attenuates sharply over the following years, falling from 0.043 at $h = 1$ to 0.012 at $h = 12$ and 0.013 at $h = 60$, though it remains statistically significant throughout. To identify the source of the initial gain and its subsequent decay, we decompose the earnings response into between-firm growth, within-firm growth, and the employment margin.

Earnings growth between period t , the month of contract expiration, and period $t+h$ is measured as $(e_{t+h} - e_t)/e_t$. Workers are employed at t by construction, so the denominator is strictly positive; the measure equals -1 when the worker has zero earnings at $t+h$. Representing the total earnings change as a telescoping sum of monthly changes scaled by baseline earnings, and defining $\Delta_s \equiv (e_{t+s} - e_{t+s-1})/e_t$,

$$\frac{e_{t+h} - e_t}{e_t} = \sum_{s=1}^h \Delta_s = \underbrace{\sum_{s \in \mathcal{W}} \Delta_s}_{\text{Within}} + \underbrace{\sum_{s \in \mathcal{B}} \Delta_s}_{\text{Between}} + \underbrace{\sum_{s \in \mathcal{E}} \Delta_s}_{\text{Employment}},$$

where $\mathcal{W}$, $\mathcal{B}$, and $\mathcal{E}$ partition the months $s = 1, \dots, h$. A month belongs to $\mathcal{W}$ if the worker is employed at the same firm as in $s-1$, to $\mathcal{B}$ if employed at a different firm, and to $\mathcal{E}$ if either month records zero earnings. The within component aggregates wage growth realized inside employment spells, including spells at firms the worker joins after t ; the between component collects the earnings change realized across job transitions; and the employment component collects the losses from separations as capturing non-employment.

We estimate the reduced-form specification of $\log OECC_{-i,r,t+1}$ on each component separately. Because the three components sum to total earnings growth by construction, the corresponding coefficients sum to the coefficient on the total, and presented in Tables A.3.1 and 2. Figure A.3.1 plots the reduced-form estimates over the full set of horizons.

Table 2 presents the 2SLS results, the employment margin dominates the total at every horizon, accounting for between 57 and 88 percent of the estimated effect. It is 0.608 of the 0.725 total at $h = 1$, 0.176 of the 0.201 total at $h = 12$, and 0.170 of the 0.224 total at $h = 60$, and it is the only component statistically distinguishable from zero at horizons beyond six months.

The between-firm component is concentrated entirely on impact: it is 0.089 at $h = 1$, declines thereafter, and is indistinguishable from zero by $h = 48$, turning marginally negative at $h = 60$. This is inconsistent with a gradual job-ladder process in which

converted workers climb toward better-paying firms over time.

The within-firm component is significant only at the two shortest horizons, at 0.028 at $h = 1$ and 0.044 at $h = 6$. Beyond that it is not statistically distinguishable from zero, though the point estimates drift upward and reach 0.053 at $h = 24$ and 0.058 at $h = 60$. These magnitudes are not trivial relative to the total, roughly a quarter to a third of it, but the standard errors at long horizons, 0.038 and 0.076 respectively, are too large to distinguish substantial within-firm growth from none.

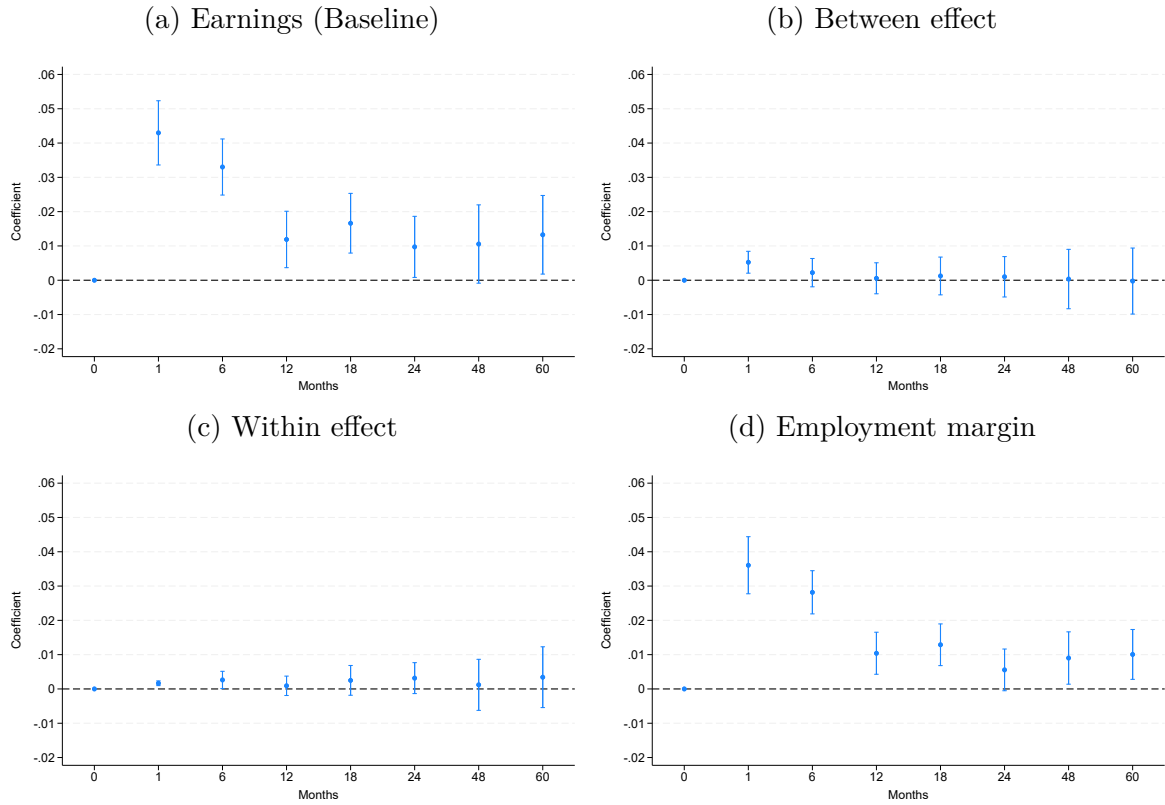
The decomposition nonetheless locates the shortfall. Essentially all of the estimated earnings gain operates through the employment margin, meaning that treated workers earn more because they are more likely to be working, not because the work they do pays better. Neither mobility to higher-paying firms nor wage growth within the conversion firm contributes detectably beyond the first six months, so the experience advantage that conversion confers does not translate into measurably higher pay conditional on employment.

Table A.3.1: Earnings and Employment Effects by Horizon: Between-Firm and Within-Firm Components (Reduced Form)

	Horizon h (months)						
	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$	$h = 48$	$h = 60$
Total earnings	0.043*** (0.005)	0.033*** (0.004)	0.012*** (0.004)	0.017*** (0.004)	0.010** (0.005)	0.011* (0.006)	0.013** (0.006)
Between-firm component	0.005*** (0.002)	0.002 (0.002)	0.001 (0.002)	0.001 (0.003)	0.001 (0.003)	0.000 (0.004)	−0.000 (0.005)
Within-firm component	0.002*** (0.000)	0.003** (0.001)	0.001 (0.001)	0.002 (0.002)	0.003 (0.002)	0.001 (0.004)	0.003 (0.005)
Employment margin	0.036*** (0.004)	0.028*** (0.003)	0.010*** (0.003)	0.013*** (0.003)	0.006* (0.003)	0.009** (0.004)	0.010*** (0.004)

Notes: Reduced-form estimates from equation (4). The dependent variable in each row is a component of earnings growth since period 0: within-firm growth, between-firm growth, and the employment margin. Components sum to total earnings growth by construction. Sample restrictions and controls are as in Figure 4. Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.3.1: Earnings Decomposition, Reduced Form

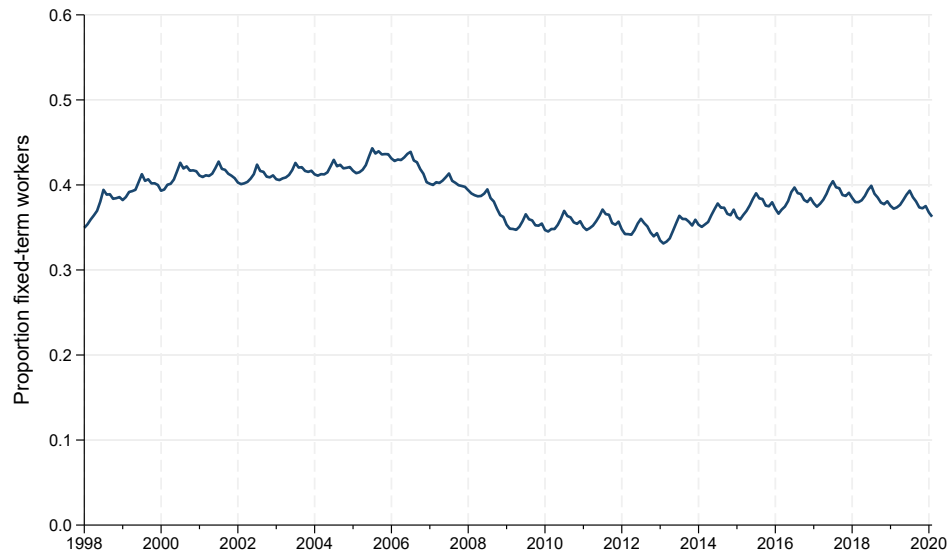


Notes: The sample consists of workers in the last month of a fixed-term position in event period $h = 0$, with at least 0.3 but less than 2 years of tenure. Period 2000–2017. The coefficients correspond to the effect of the first lead of the log number of new permanent contracts ($\log OECC$) on each outcome. All regressions control for the leads and lags of $\log OECC$ and the log number of new fixed-term contracts. Additional controls: year and month FE, province FE, sector FE, gender, interactions of age FE and educational attainment, and experience and experience squared at baseline. Standard errors are clustered at the province $\times$ year level.

B Supplementary Figures

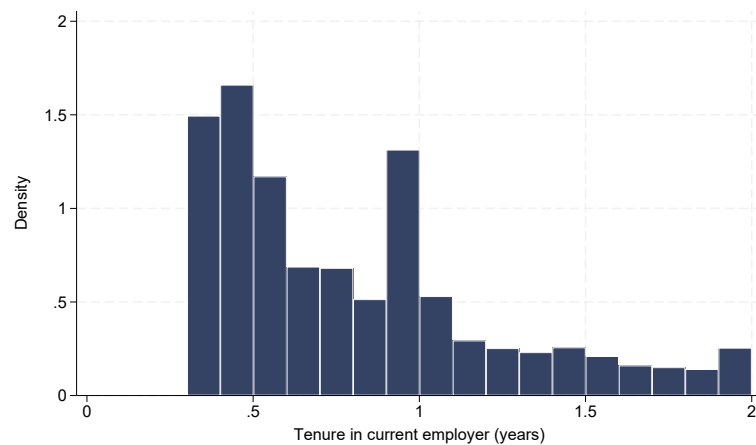
B.1 Fixed-term contracts in Spain

Figure B.1.1: Proportion of workers in fixed-term contracts, by year



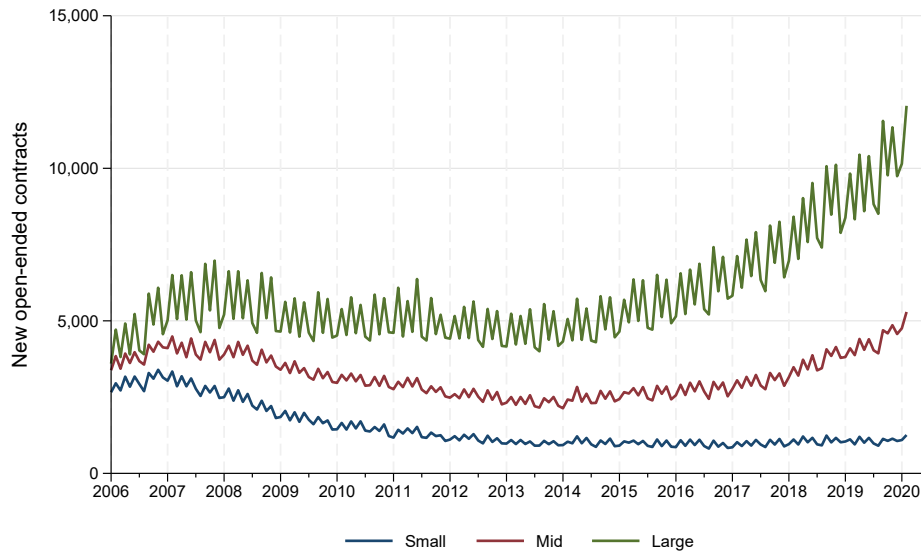
Notes: Proportion of workers under a fixed-term from 2000 to 2021.

Figure B.1.2: Maximum tenure at the expiration of fixed-term contracts



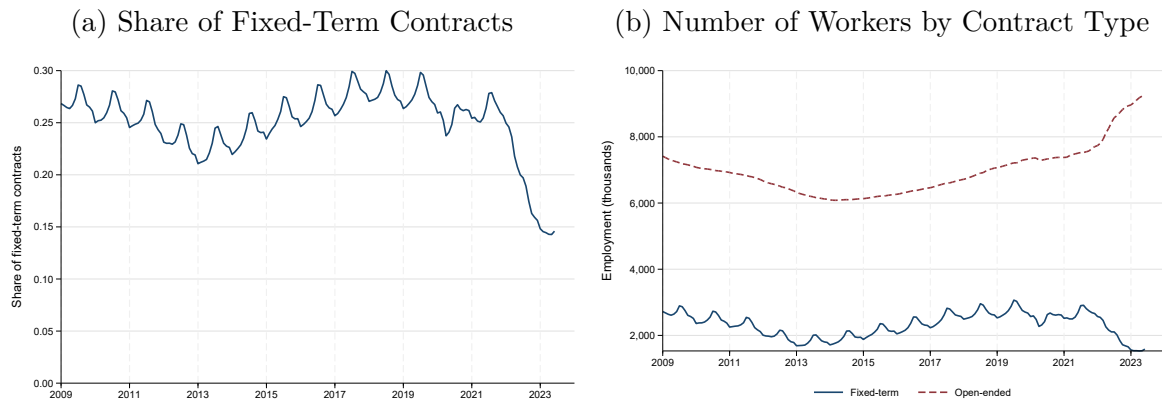
Notes: Distribution of maximum tenure in fixed-term contracts 2000-2021.

Figure B.1.4: New open-ended contracts by firm size



Notes: Count of new open-ended contracts by firm size, with firms categorized into terciles based on the annual firm size distribution.

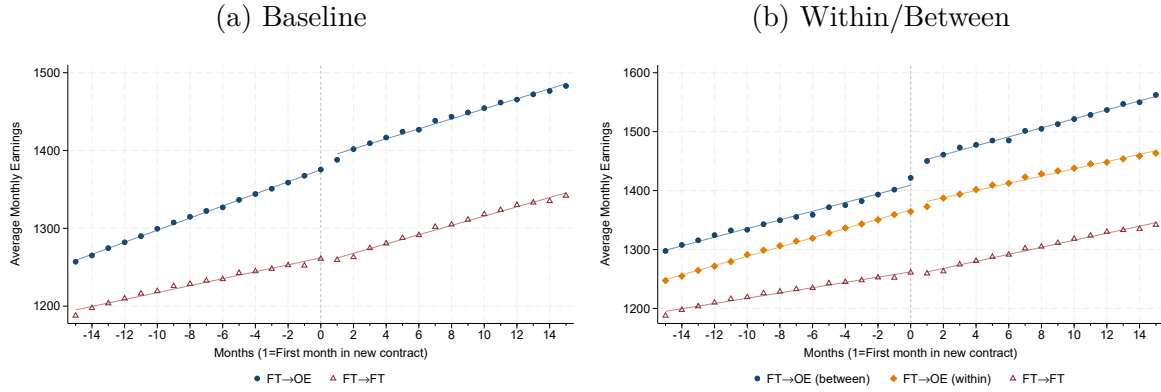
Figure B.1.3: Descriptive statistics on fixed-term contracts from Social Security records



Notes: Panel (a) reports the share of fixed-term contract workers, defined as the number of workers in fixed-term contracts divided by the total number of workers in fixed-term and open-ended contracts, restricting to full-time employment. Panel (b) shows the number of workers in fixed-term and open-ended contracts from January 2012 to May 2024, also restricted to full-time employment. The data are drawn from Social Security affiliation records, which are available from 2012 onward. Source: BBDD Estadísticas TGSS.

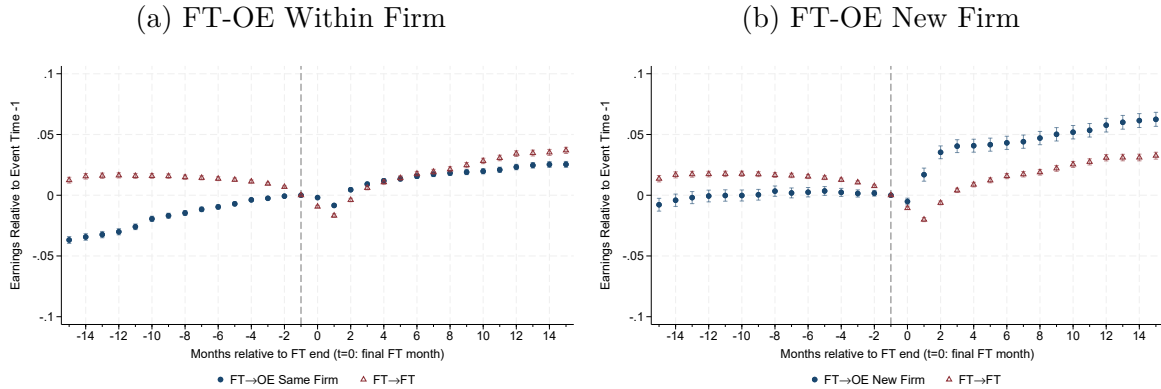
B.2 Selection: Additional Results

Figure B.2.1: Average earnings before and after a contract change, by destination contract



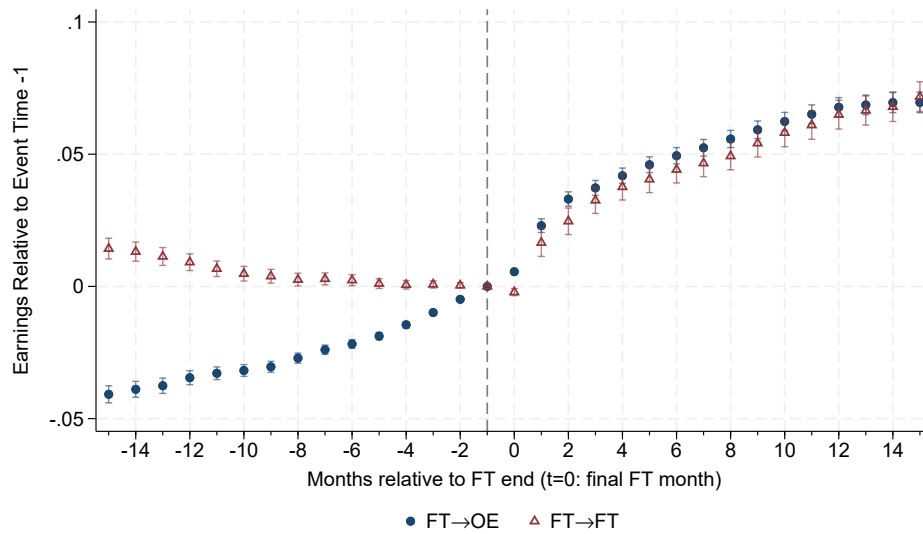
Notes: The figure shows Spanish workers' average earnings from 2000 to 2020 in the final month of a fixed-term contract and 15 periods before and after that transition. Workers are categorized into two groups based on the subsequent contract: FT → OEC (transitioning to an open-ended contract) or FT → FT (transitioning to another fixed-term contract). Panel (a) presents the average earnings of workers transitioning to an open-ended or fixed-term contract in event time 1. Panels (b) distinguish between transitions to an open-ended contract in a different firm (FT → OEC-Same Firm) and transitions to an open-ended contract within the same firm (FT → OEC-New Firm).

Figure B.2.2: Evolution of earnings before and after switching within or to a new firm (Within vs Between transitions)



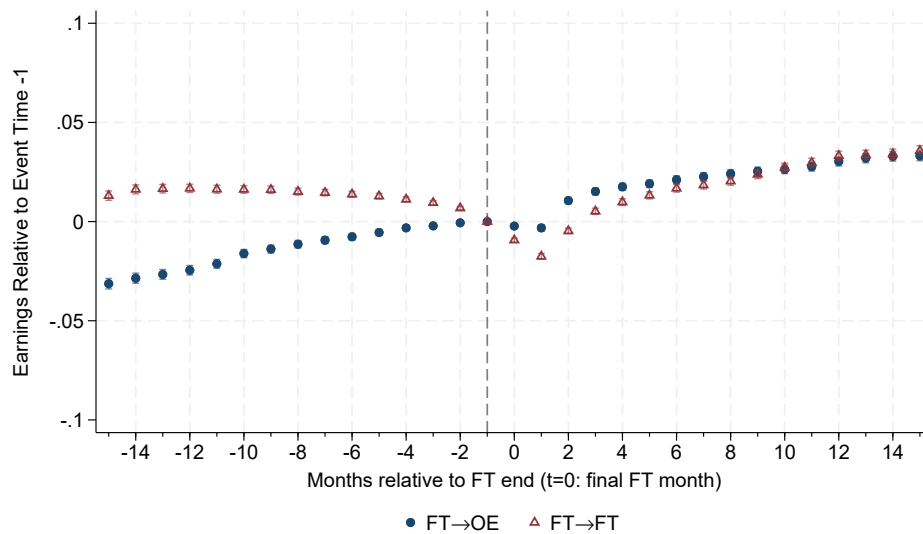
Notes: The figure shows event time coefficients estimated from equation 2 for workers transitioning to open-ended contracts or a new fixed-term contract. The exercise is described in the notes of Figure 2. Panel (a) presents fixed-term to fixed-term transitions (FT-FT) along with fixed-term to open-ended transitions (FT-OE) that occur *within* the same firm. Panel (b) presents FT-FT transitions and FT-OE transitions to a *new* firm. All specifications include interactions between event time and educational attainment and sector.

Figure B.2.3: Evolution of earnings before and after switching within or to a new firm



Notes: The figure shows event time coefficients estimated from equation 2 for workers transitioning to open-ended contracts or a new fixed-term contract. The exercise is described in the notes of Figure 2. The sample is restricted to the first contract transaction for a worker.

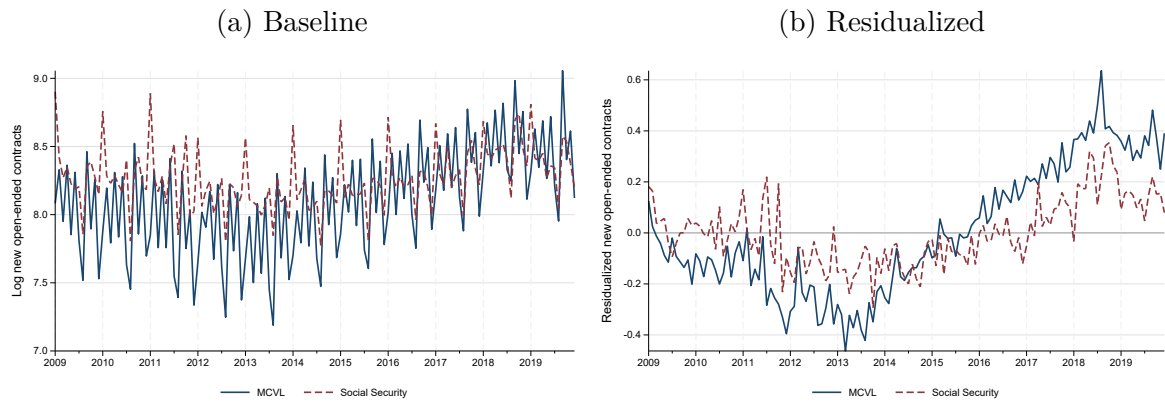
Figure B.2.4: Event study of earnings around a worker's first contract transition



Notes: The figure shows event time coefficients estimated from equation 2 for workers transitioning to open-ended contracts or a new fixed-term contract. The exercise is described in the notes of Figure 2. The sample is restricted to the first contract transition for a worker.

B.3 Social Security records

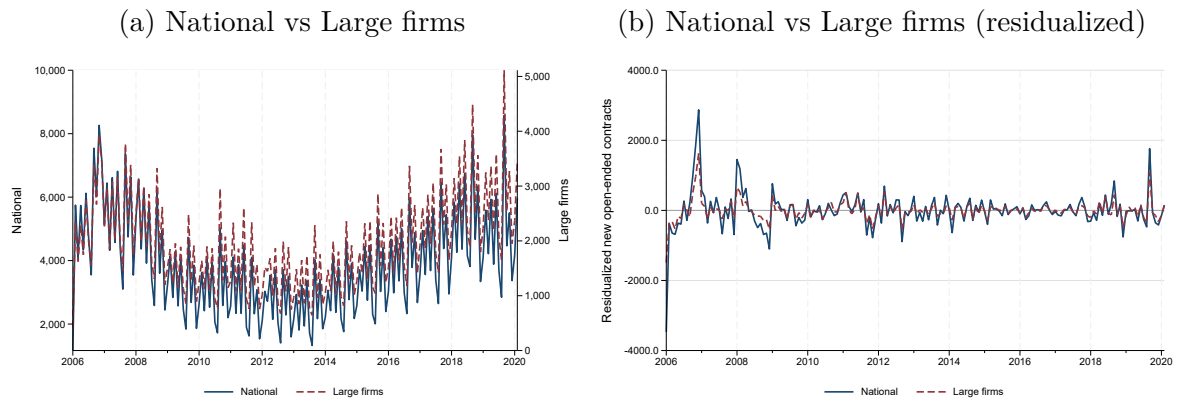
Figure B.3.1: New OEC and Total OEC from MCVL and SS records



Notes: Sum of NewOEC obtained from Social Security records (OEC_{Total}) and from the MCVL. Panel (a) displays the monthly sum of New open-ended contracts from both data sources. Panel (b) residualizes the sum of new Open-Ended contracts (OEC) by subtracting the variation explained by month-fixed effects.

B.4 OEC variation

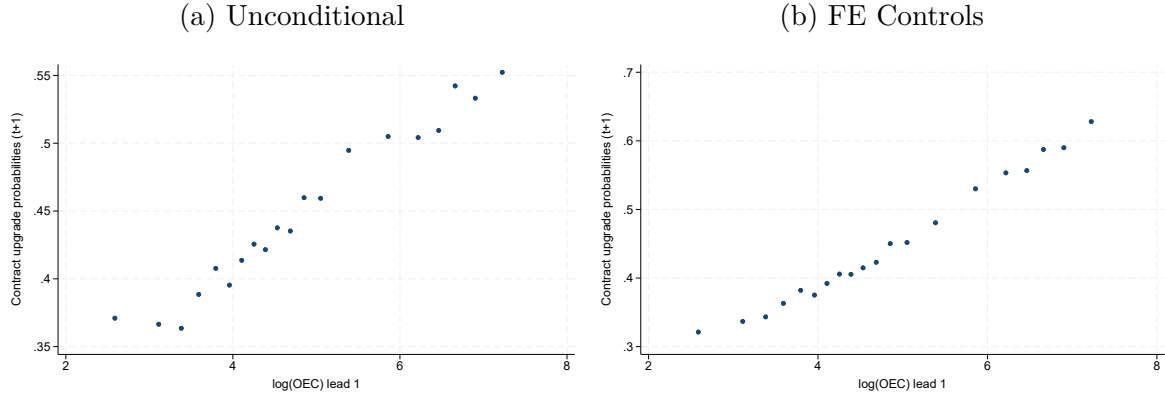
Figure B.4.1: New OECs by firm/province size



Notes: A firm posting a new open-ended contract is classified as large if it falls in the top decile of the firm-size distribution (i.e., above the 90th percentile). Panel (a) reports the national count of newly created open-ended contracts, restricting the sample to large firms. Panel (b) presents residualized series obtained by netting out year and month fixed effects.

B.5 Binscatter

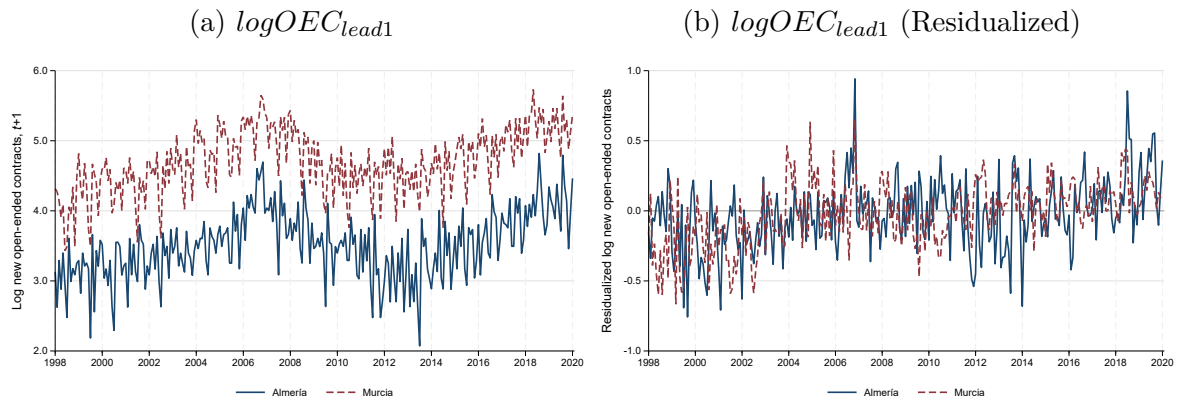
Figure B.5.1: Binned scatter plots of the $OEC_{-i,r,t+1}$ on the probability of contract upgrade in $t + 1$



Notes: This figure shows a binscatter plot with the dependent variable indicating whether a worker on an ending fixed-term contract transitions to an open-ended contract in the period $t + 1$. The independent variable is the first lead of the logarithm of the number of new open-ended contracts in the same month-year and province as the worker's ending contract. The plot is a covariate-adjusted binscatter proposed by Cattaneo et al. (2024). Panel (a) presents the unconditional relationship, while Panel (b) controls for the sector, year, month, and province fixed effects. The number of bins is set to 20.

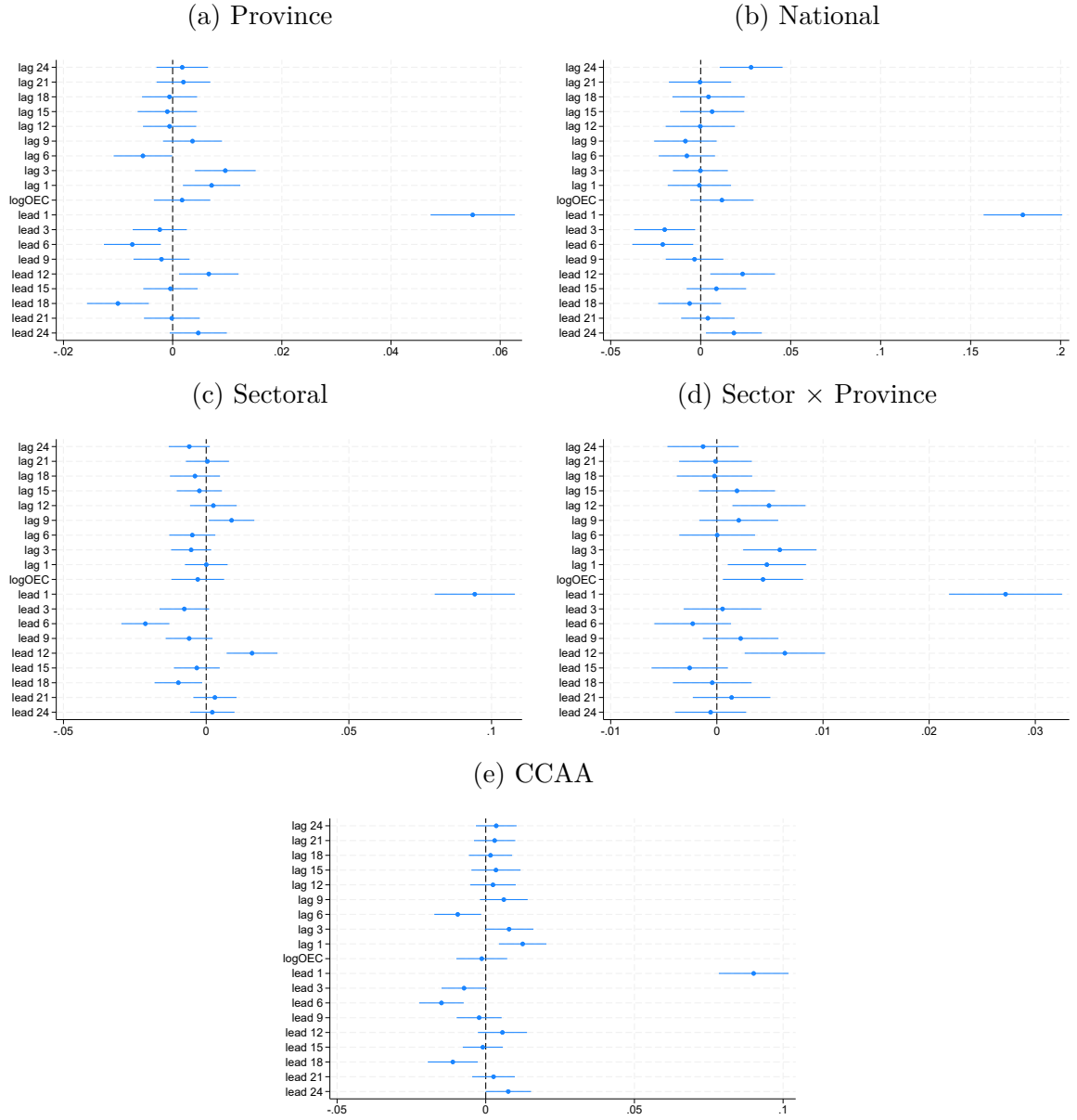
B.6 First Stage

Figure B.6.1: LogOEC lead1: Province Instrument



Notes: Panel (a) plots the log of new OECs by province and year-month from 2000 to 2020 for Almería and Murcia. Panel (b) presents the residualized log of new OECs after removing province, year, and month fixed effects.

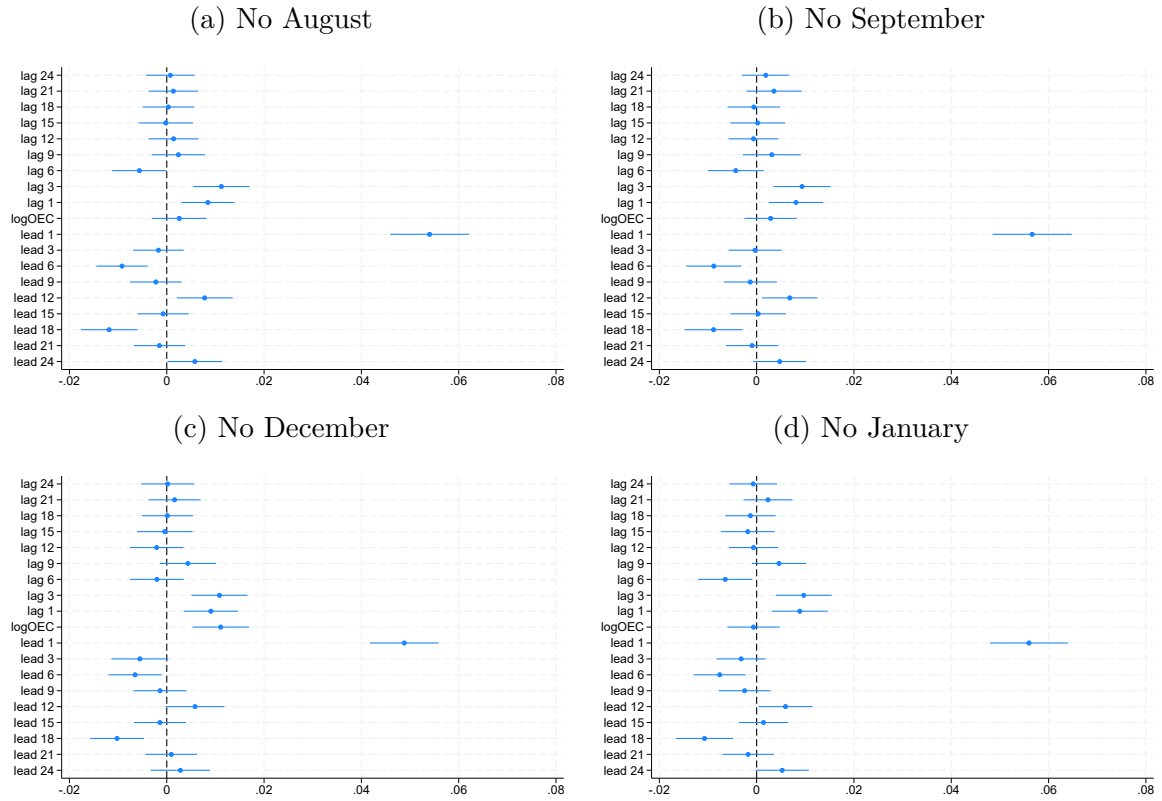
Figure B.6.2: First stage: National, Province, Sectoral, and Sector $\times$ Province Instrument



Notes: The sample consists of workers in the last month of a fixed-term contract at event period $h = 0$ with tenure between 0.3 and 2 years. Coefficients measure the effect of leads and lags of log new open-ended contracts on the probability of converting to an open-ended contract at $t+1$. Each panel varies the level of aggregation at which the instrument is constructed: (a) province $\times$ month, baseline; (b) national $\times$ month; (c) sector $\times$ month; (d) province $\times$ sector $\times$ month; (e) autonomous community $\times$ month. All specifications include year and month fixed effects, province fixed effects, sector fixed effects, gender, interactions of age and educational attainment fixed effects, experience and experience squared, contract part-time indicator and leads and lags of log new fixed-term contracts. Standard errors clustered at the province $\times$ year level.

B.7 Robustness: Job Seasonality

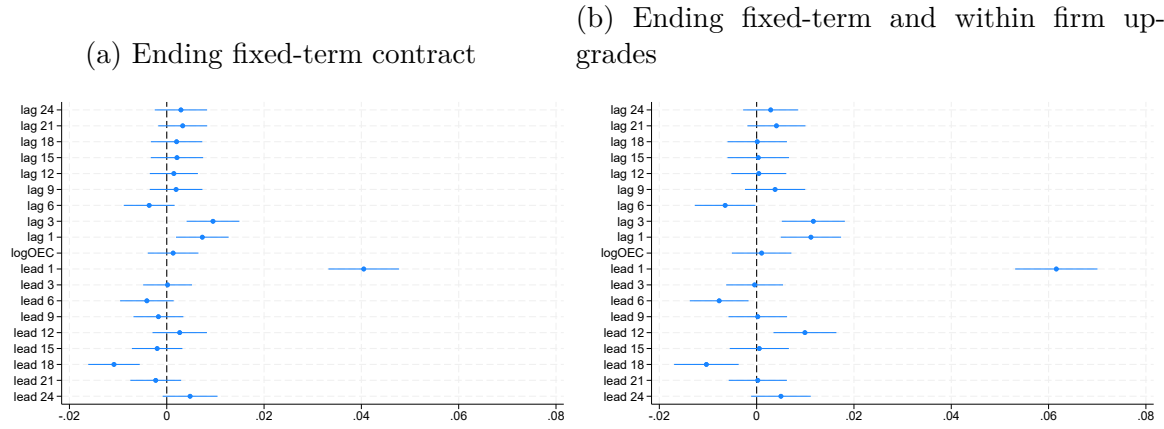
Figure B.7.1: Regional Instrument:
Removing specific months



Notes: Sample and specification as in Figure 3. Each panel replicates the baseline excluding observations in one particular month: (a) August; (b) September; (c) December; (d) January. Standard errors clustered at the province $\times$ year level.

B.8 Robustness: Reason of dismissal

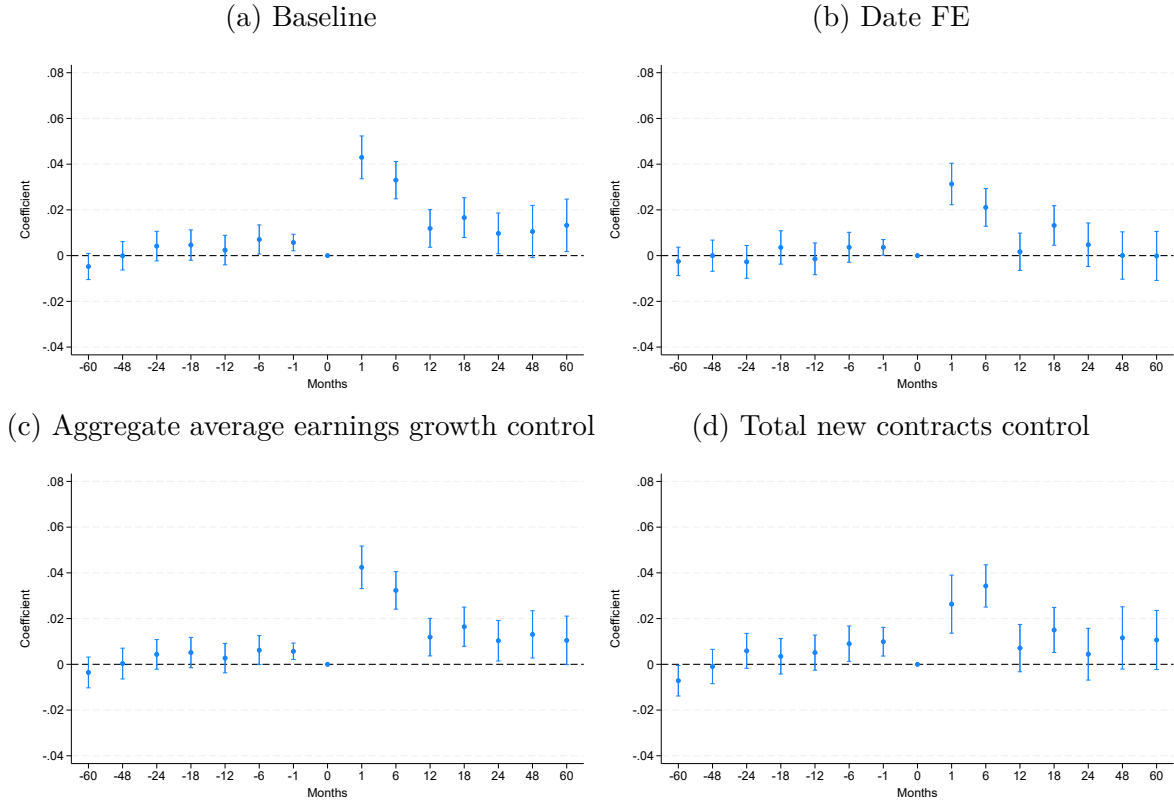
Figure B.8.1: First Stage: Provincial Instrument for Exogenous Termination of Fixed-Term Contracts



Notes: Sample and specification as in Figure 3. Panel (a) restricts to workers separated due to fixed-term contract expiration. Panel (b) additionally includes workers converted to open-ended contracts within the same firm. Standard errors clustered at the province $\times$ year level.

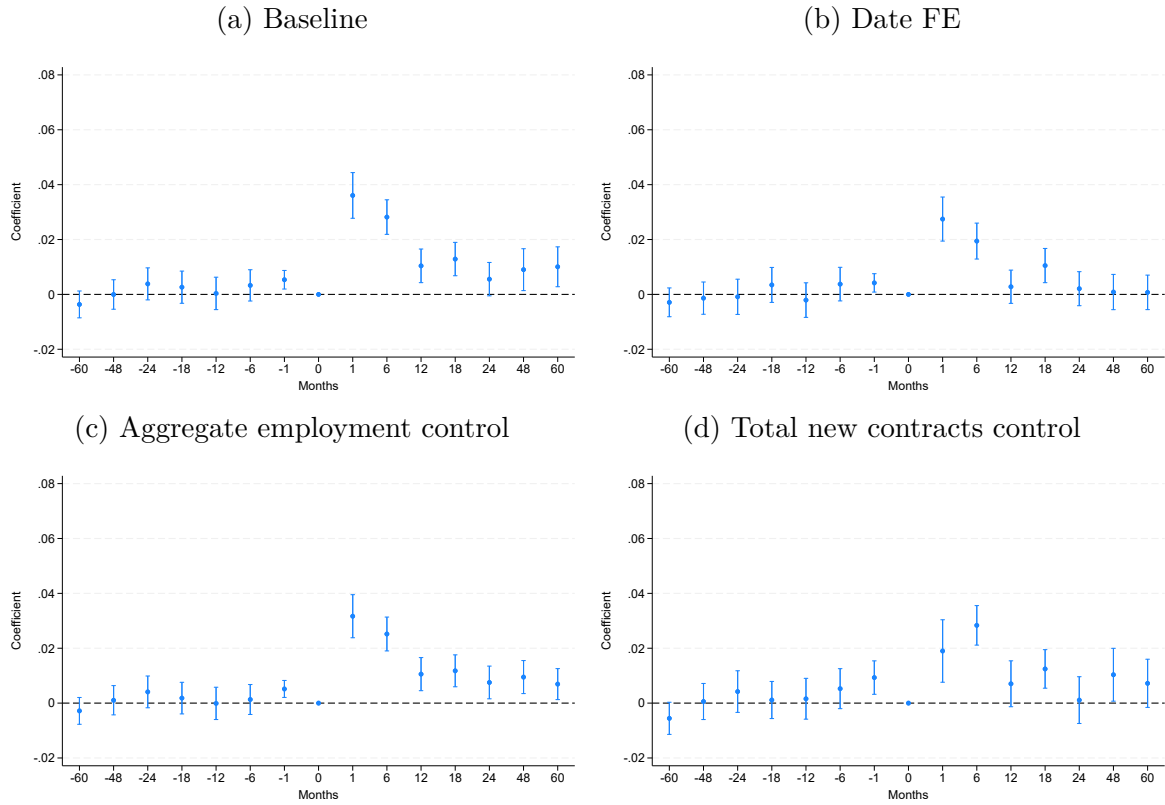
B.9 Robustness: Alternative Reduced-Form Specifications

Figure B.9.1: Effect of OEC regional shock on earnings growth:
Alternative Specifications



Notes: Impact of the first lead of the *log* of new open-ended in the province on individual earnings growth. Baseline sample restrictions and empirical specification are described in Figure 4 notes. Panel (a) presents the baseline specification from Figure 4 based on equation 4. Panel (b) adds month x year fixed effects. Panel (c) includes the aggregate average outcome as control, following the specification of equation 4. Panel (d) controls for the *log* of total new contracts (sum of fixed-term and open-ended). Standard errors are clustered at the province $\times$ year level.

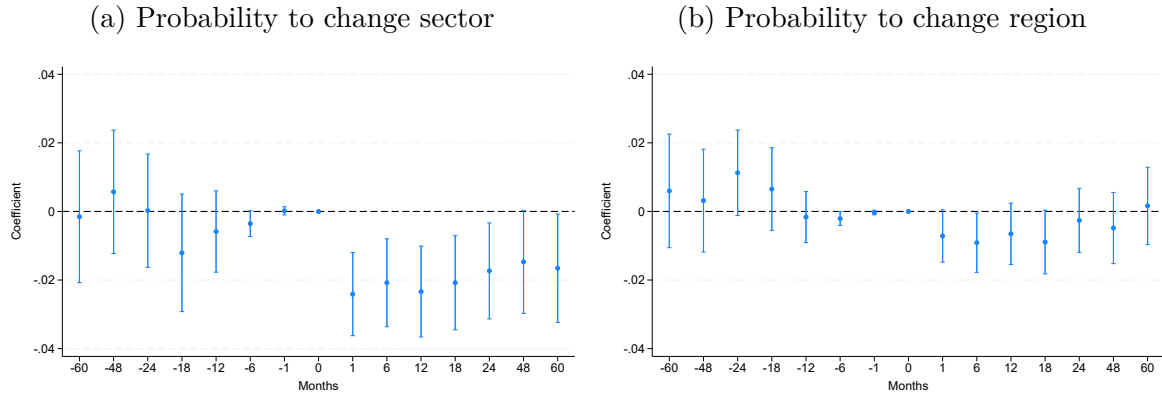
Figure B.9.2: Effect of OEC regional shock on employment:
Alternative Specifications



Notes: Impact of the first lead of the *log* of new open-ended in the province on individual employment probability. Baseline sample restrictions and empirical specification are described in Figure 4 notes. Panel (a) presents the baseline specification from Figure 4 based on equation 4. Panel (b) adds month x year fixed effects. Panel (c) includes the aggregate average outcome as control, following the specification of equation 4. Panel (d) controls for the *log* of total new contracts (sum of fixed-term and open-ended). Standard errors are clustered at the province $\times$ year level.

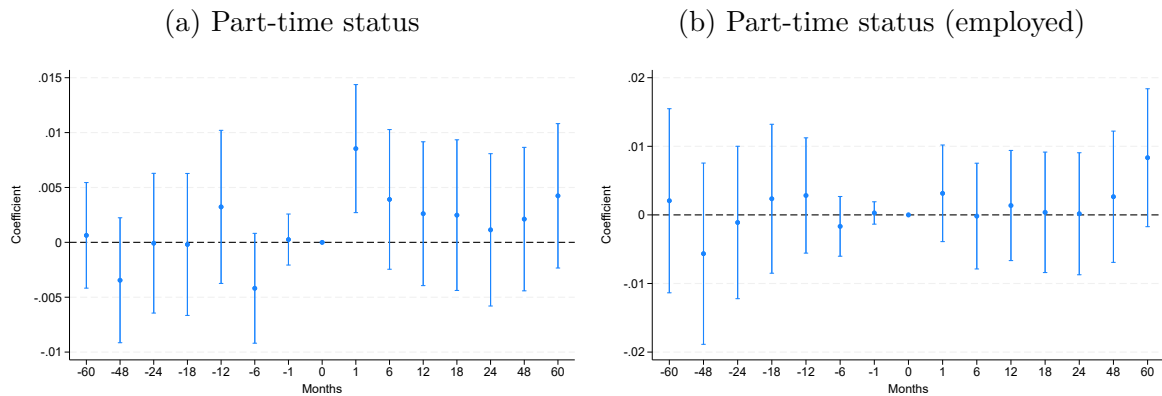
B.10 Robustness: Additional outcomes

Figure B.10.1: Effect of new open-ended contracts in the region on workers' mobility (Employed sample)



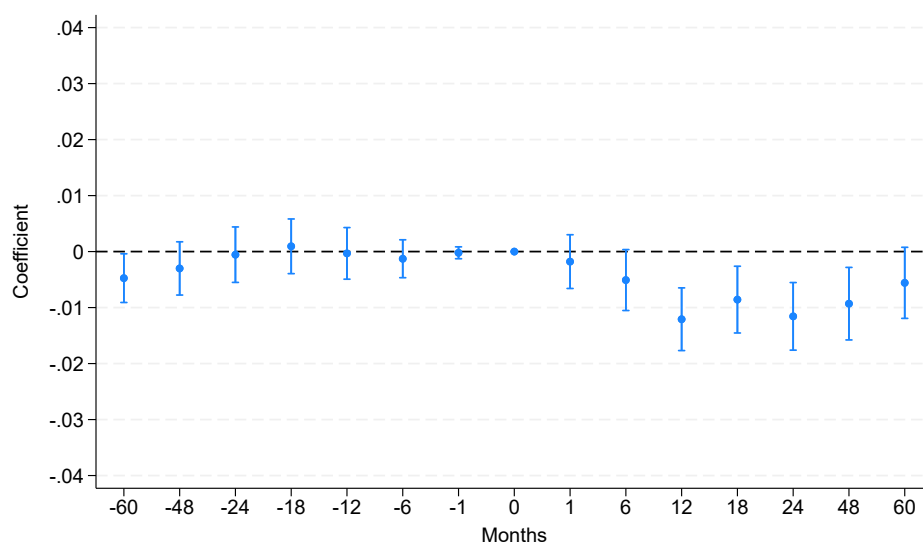
Notes: Baseline sample restrictions and empirical specification are described in Figure 4 notes. For each period, we restrict the estimation sample to workers who are employed in that period (unbalanced sample). Standard errors are clustered at the province $\times$ year level.

Figure B.10.2: Effect of new open-ended contracts in the region on workers' part-time work



Notes: Baseline sample restrictions and empirical specification are described in Figure 4 notes. Panel (a) shows the effect of our instrument on part-time status on each period. In Panel (b), we restrict the estimation sample to workers employed for each period. Standard errors clustered at the province $\times$ year level.

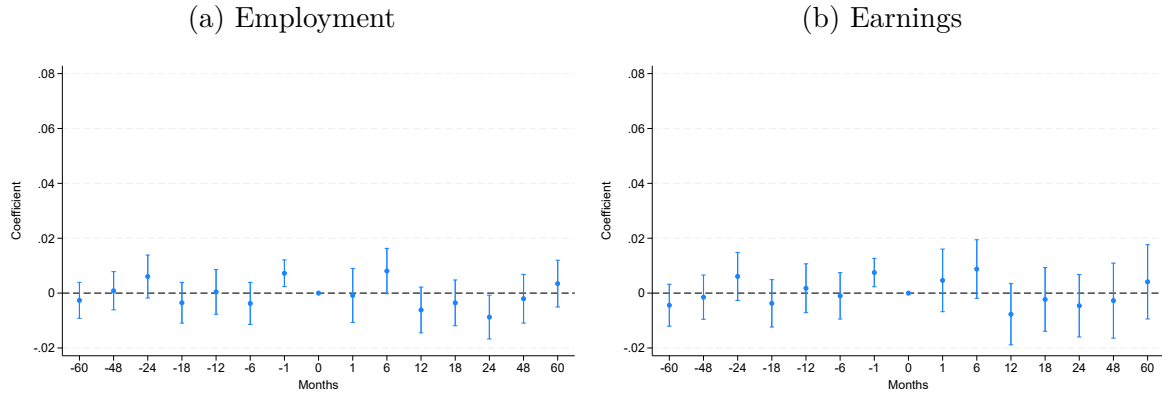
Figure B.10.3: Effect of new open-ended contracts in the region on workers' occupational mobility



Notes: Baseline sample restrictions and the empirical specification are described in the notes to Figure 4. The outcome measures the effect of new open-ended contracts on an indicator for whether the worker is employed in an occupation different from that in the baseline period. Standard errors clustered at the province $\times$ year level.

B.11 Robustness: Exclusion Restriction

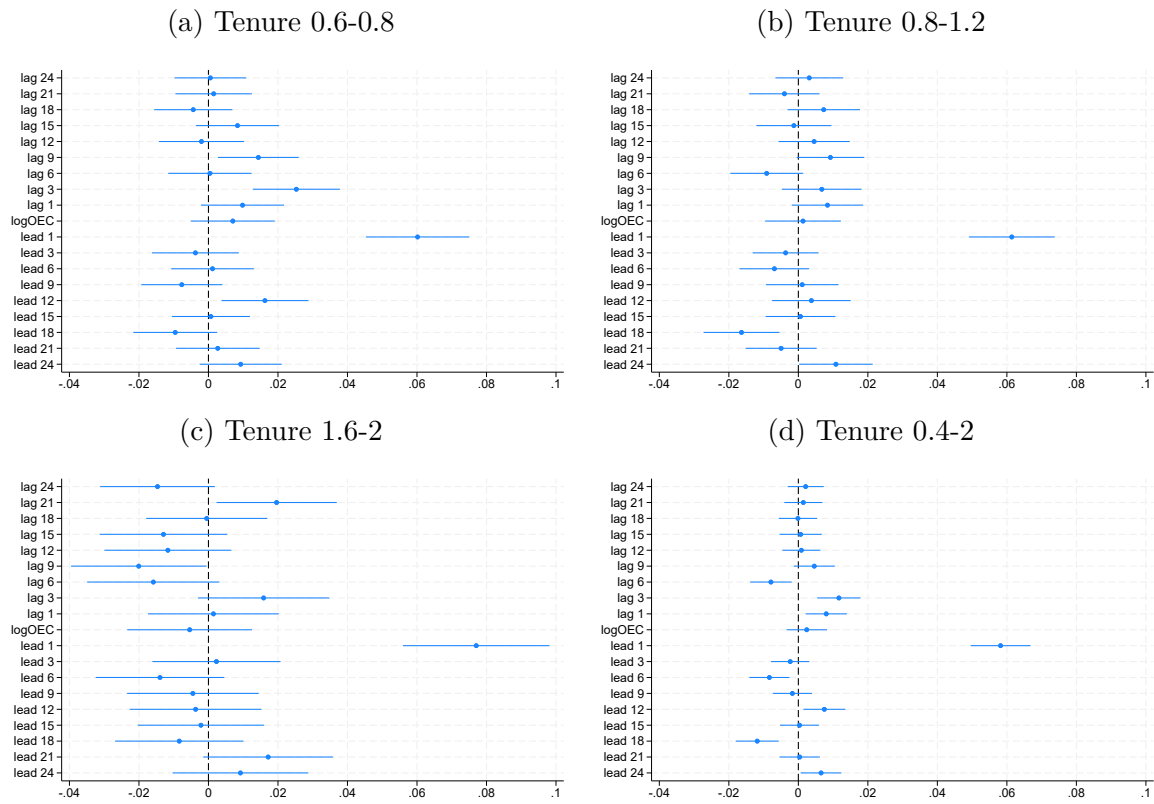
Figure B.11.1: Effect of OEC regional shock on employment and earnings: restriction to fixed-term employment



Notes: The sample consists of workers who were in the last month of a fixed-term contract in event period $h = 0$, with at least 0.3 years of tenure. We restrict the sample to those workers who *do not* start an open-ended position in period $t + h$. The coefficients correspond to the effect of the first lead of the *log* of new open-ended contracts in the province on the probability of employment (Panel a) and on earnings (Panel b). Additional controls: year and month FE, province FE, sector FE, gender, interactions of age FE and educational attainment, experience and experience squared at baseline as well as leads and lags of the *log* of new fixed-term contracts. Standard errors clustered at the province $\times$ year level.

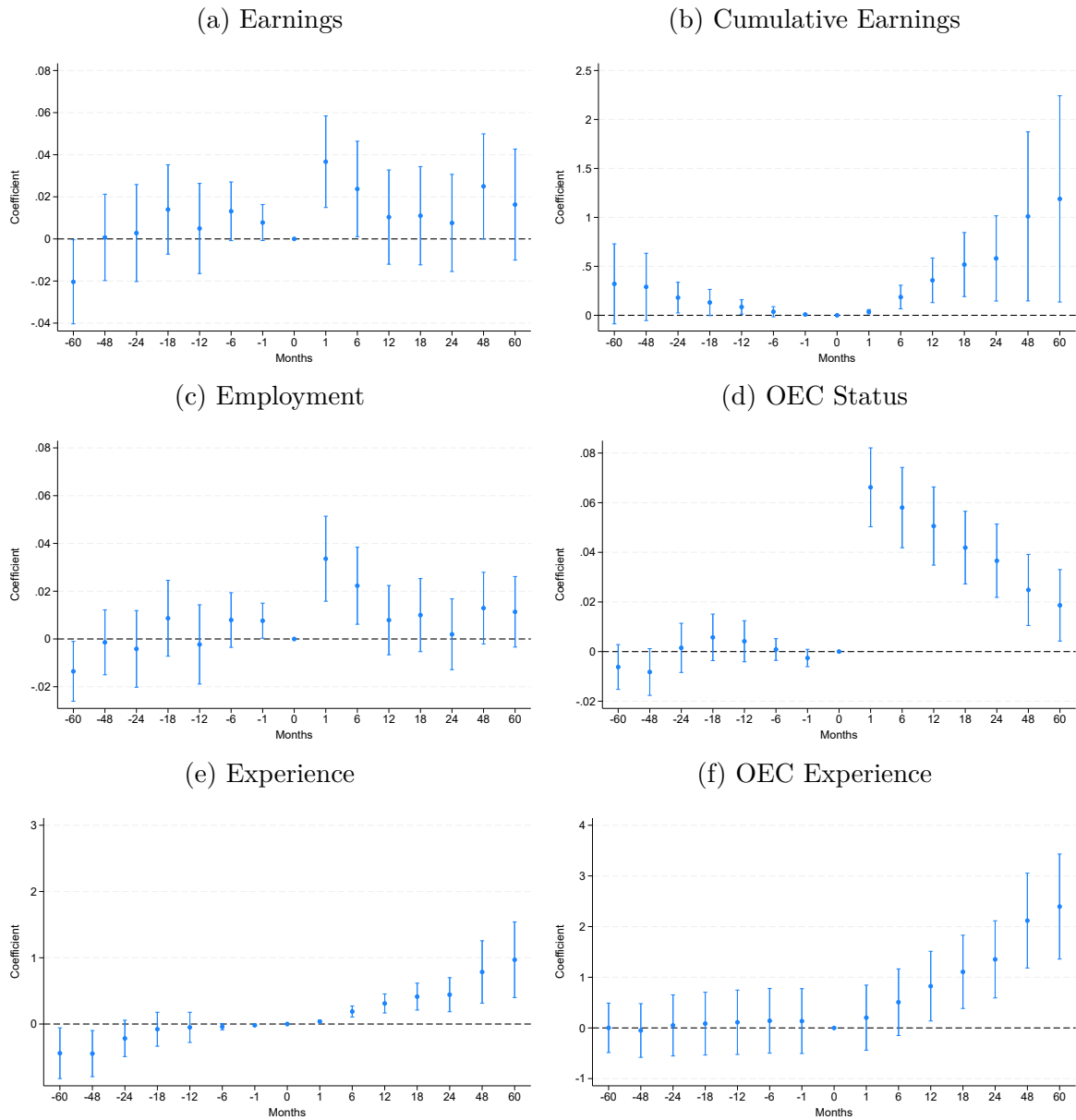
B.12 Alternative Tenure Restrictions

Figure B.12.1: First stage: Provincial Instrument. Alternative tenure restrictions



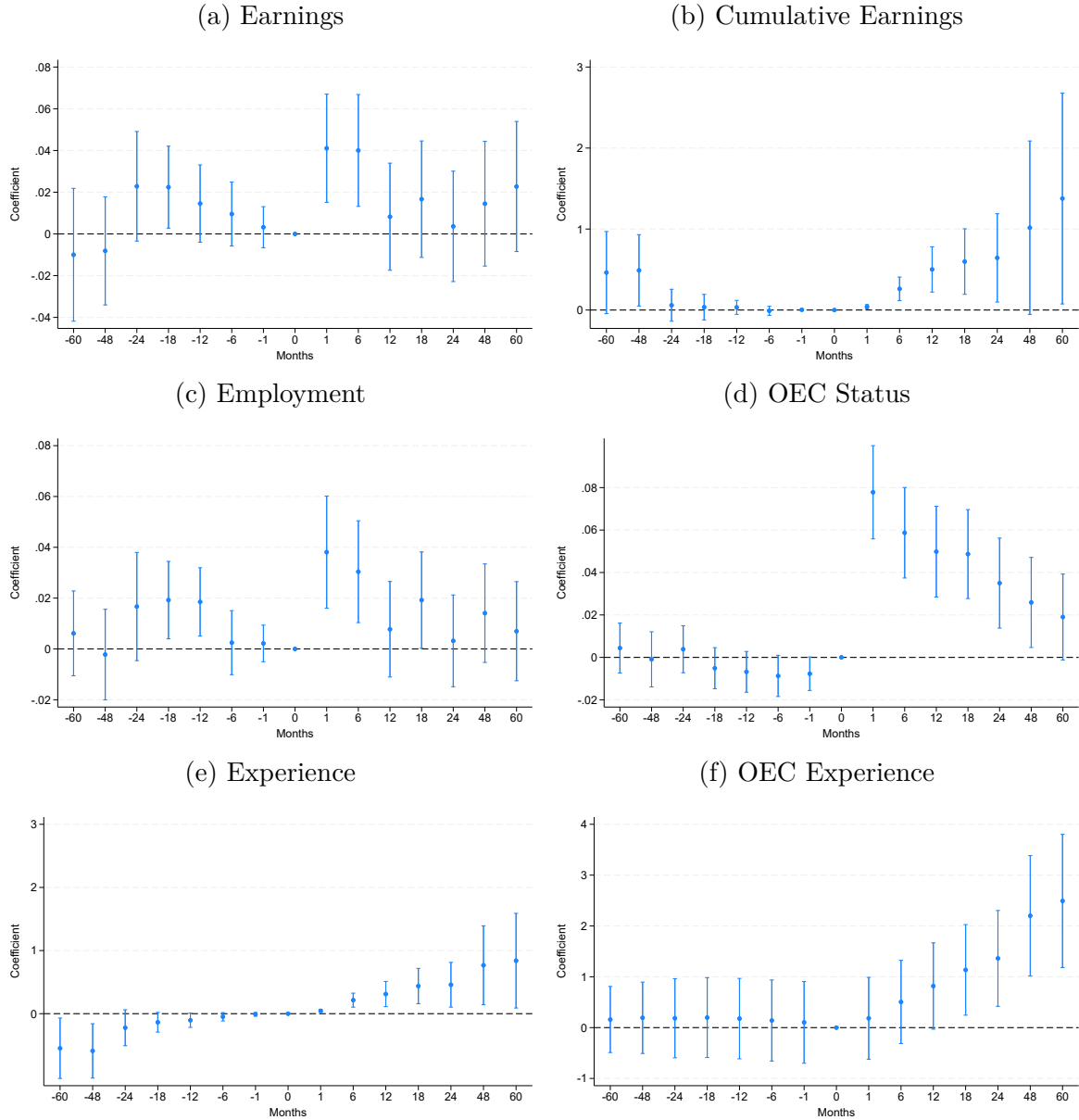
Notes: The baseline sample restrictions and empirical specification are detailed in the notes for Figure 3. Panel (a) restricts the sample to workers with tenure ranging from 0.6 to 0.8 years in the expiring fixed-term contract. Panels (b)-(d) restrict the sample to workers with tenure from 0.8-1.2, 1.6-2, and 0.4-2 years, respectively. Standard errors are clustered at the province $\times$ year level. Standard errors clustered at the province $\times$ year level.

Figure B.12.2: Effect of OEC regional shock on earnings. Tenure 0.6-0.8 years



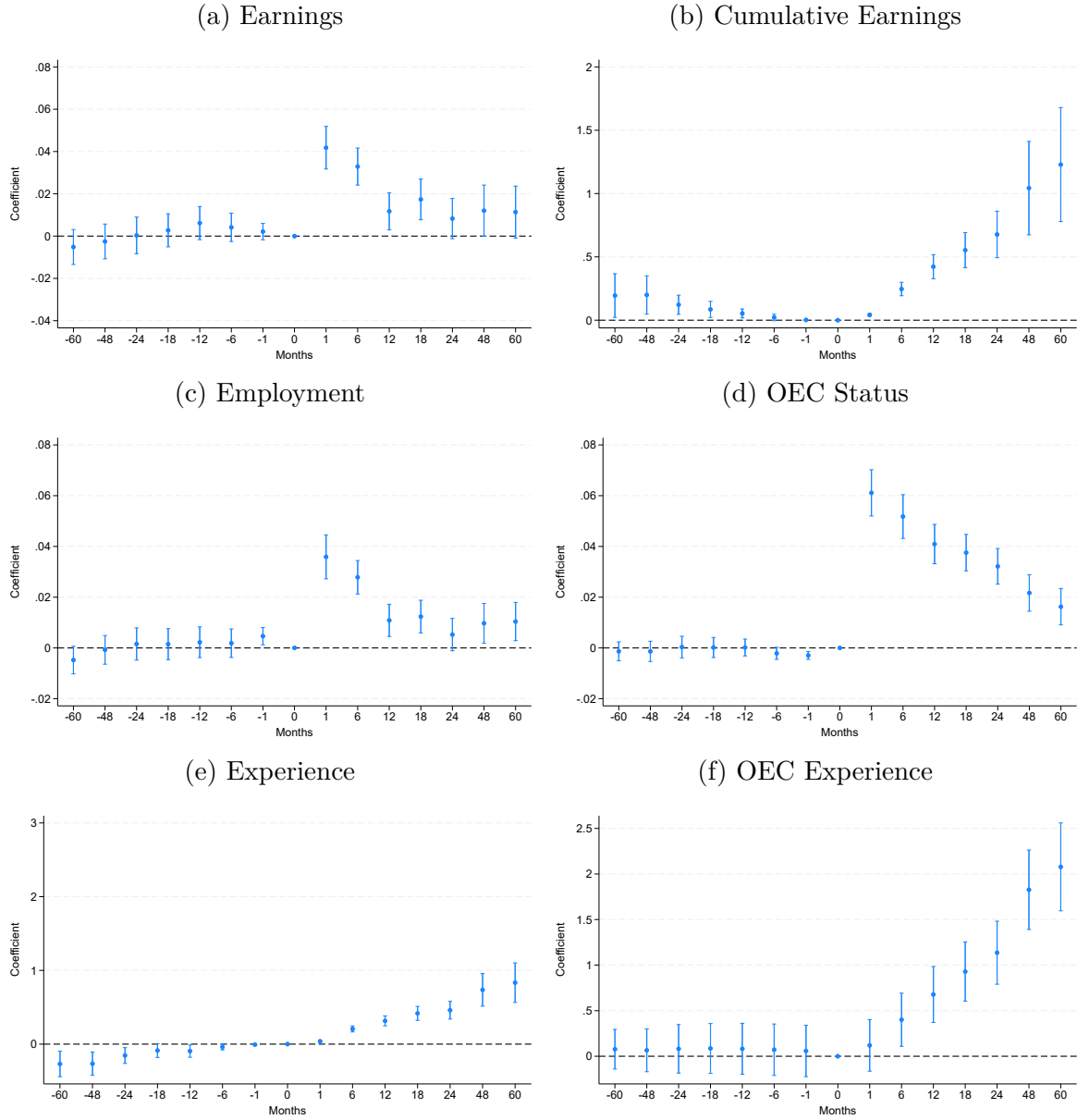
Notes: The sample consists of workers in the last month of a fixed-term position in event period $h = 0$, with at least 0.6 but less than 0.8 years of tenure. Period 2000-2017. The coefficients correspond to the effect of the first lead of the $\log$ number of new permanent contracts ($\log OEC$) on each outcome. All regressions control for the leads and lags of $\log OEC$ and the $\log$ of the number of new fixed-term contracts. Additional controls: year and month FE, province FE, sector FE, gender, interactions of age FE and educational attainment, experience, and experience squared at baseline. Standard errors clustered at the province $\times$ year level.

Figure B.12.3: Effect of OEC regional shock on earnings. Tenure 1.6-2 years



Notes: The sample consists of workers in the last month of a fixed-term position in event period $h = 0$, with at least 1.6 but less than 2 years of tenure. Period 2000-2017. The coefficients correspond to the effect of the first lead of the $\log$ number of new permanent contracts ($\log OEC$) on each outcome. All regressions control for the leads and lags of $\log OEC$ and the $\log$ of the number of new fixed-term contracts. Additional controls: year and month FE, province FE, sector FE, gender, interactions of age FE and educational attainment, experience, and experience squared at baseline. Standard errors clustered at the province $\times$ year level.

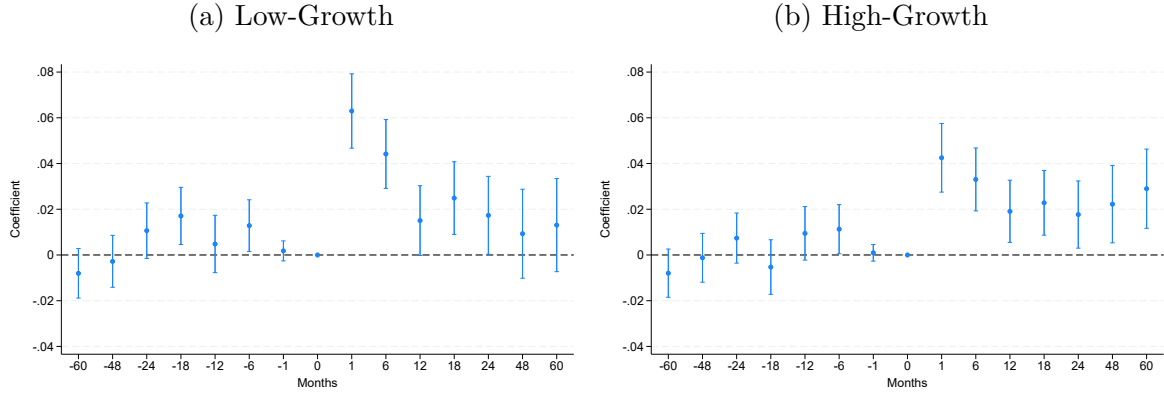
Figure B.12.4: Effect of OEC regional shock on earnings. Tenure 0.4-2 years



Notes: The sample consists of workers in the last month of a fixed-term position in event period $h = 0$, with at least 0.4 but less than 2 years of tenure. Period 2000-2017. The coefficients correspond to the effect of the first lead of the $\log$ number of new permanent contracts ($\log OEC$) on each outcome. All regressions control for the leads and lags of $\log OEC$ and the $\log$ of the number of new fixed-term contracts. Additional controls: year and month FE, province FE, sector FE, gender, interactions of age FE and educational attainment, experience, and experience squared at baseline. Standard errors clustered at the province $\times$ year level.

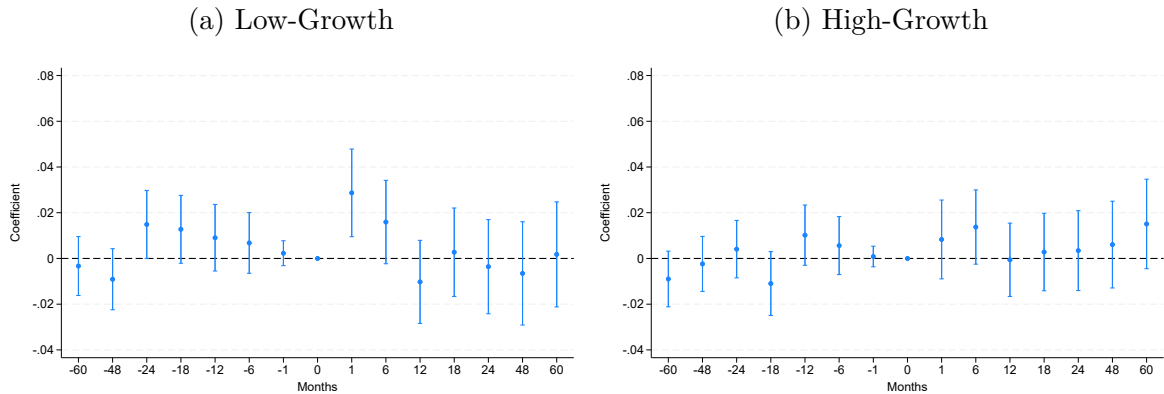
B.13 Mechanism

Figure B.13.1: Reduced form estimation by firm's growth category



Notes: Reduced-form estimates based on equation (4). Each panel plots the reduced-form coefficient on the instrument at each month relative to fixed-term contract expiration, separately for workers whose initial firm is below (Low-Growth) or above (High-Growth) the median of the firm-growth distribution. Firms are classified according to their baseline leave-one-out average monthly earnings growth, computed excluding each worker's own contribution to the firm-level average. Confidence intervals are constructed using standard errors clustered at the province $\times$ year level.

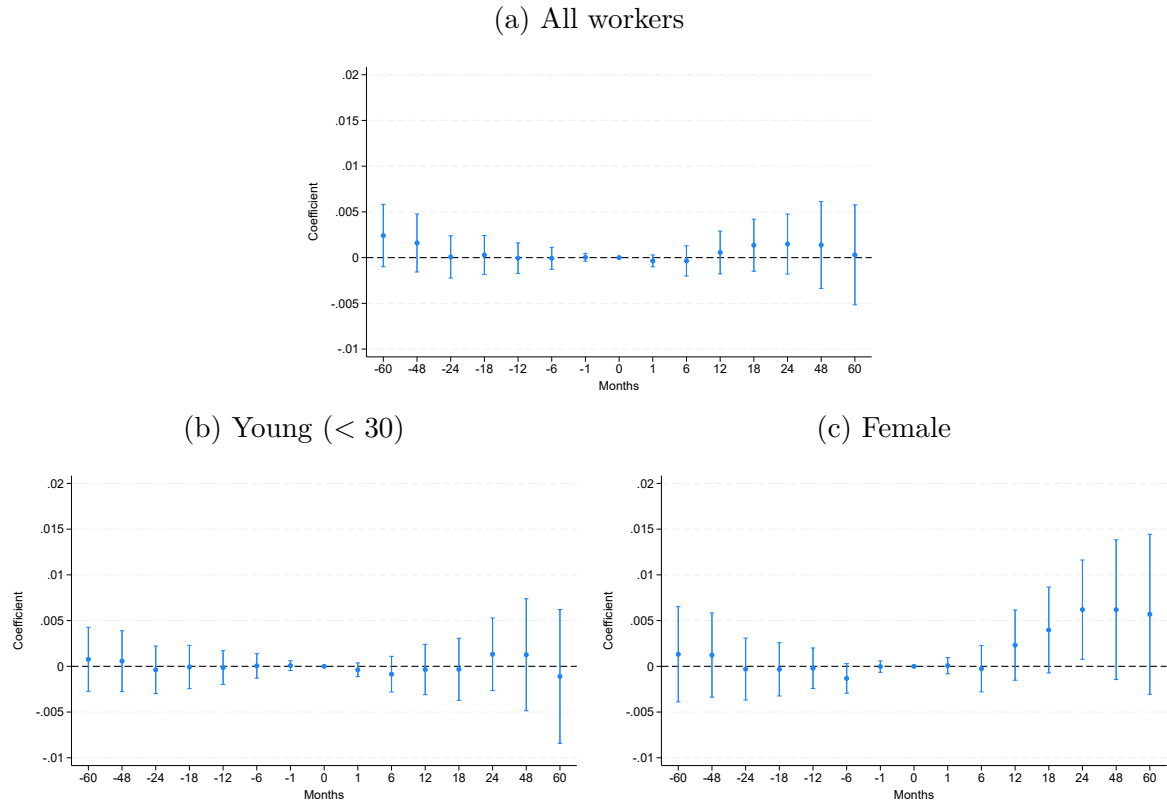
Figure B.13.2: Exclusion restriction by firm's growth category



Notes: Reduced-form estimates based on equation (4). The sample is restricted to workers who are not upgraded to a permanent contract in the month following the expiration of their fixed-term contract ($t + 1$). Firms are classified according to their baseline leave-one-out average monthly earnings growth, computed excluding each worker's own contribution to the firm-level average. Confidence intervals are constructed using standard errors clustered at the province $\times$ year level. Because the instrument is intended to shift subsequent earnings only through conversion, the absence of a systematic reduced-form response among non-upgraded workers in either firm-growth category is consistent with the exclusion restriction. This is suggestive that the dynamics in Figure B.13.1 operate through conversion rather than through instrument-correlated shocks common to high-growth firms; we treat it as supporting evidence for the identifying assumption rather than as a definitive test.

B.14 Fertility

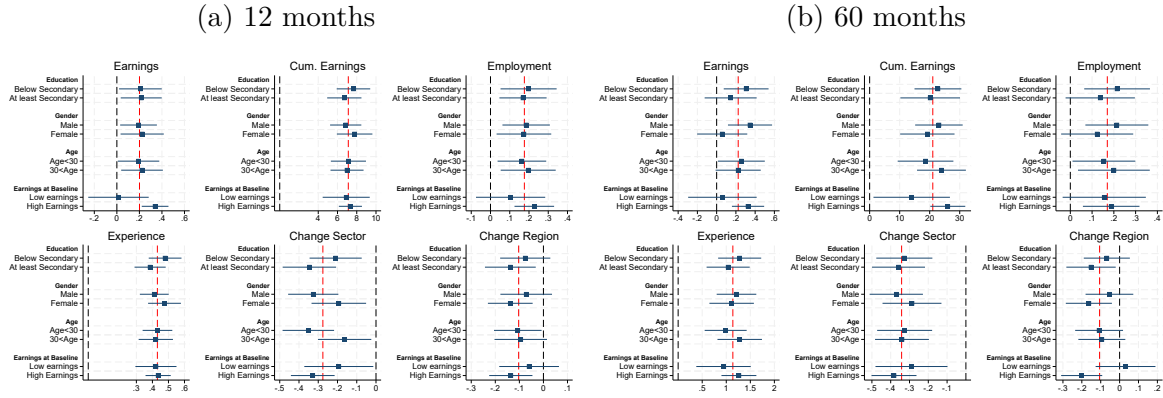
Figure B.14.1: Effect of open-ended contract status on the probability of an additional birth



Notes: Reduced-form estimates based on equation (4). Sample restrictions and controls are as in Figure B.13.1. The outcome is an indicator for having had a child, measured relative to baseline. Panel (a) reports estimates for the full sample, Panel (b) restricts to workers aged 30 or younger at baseline, and Panel (c) restricts to female workers. Confidence intervals are constructed using standard errors clustered at the province $\times$ year level.

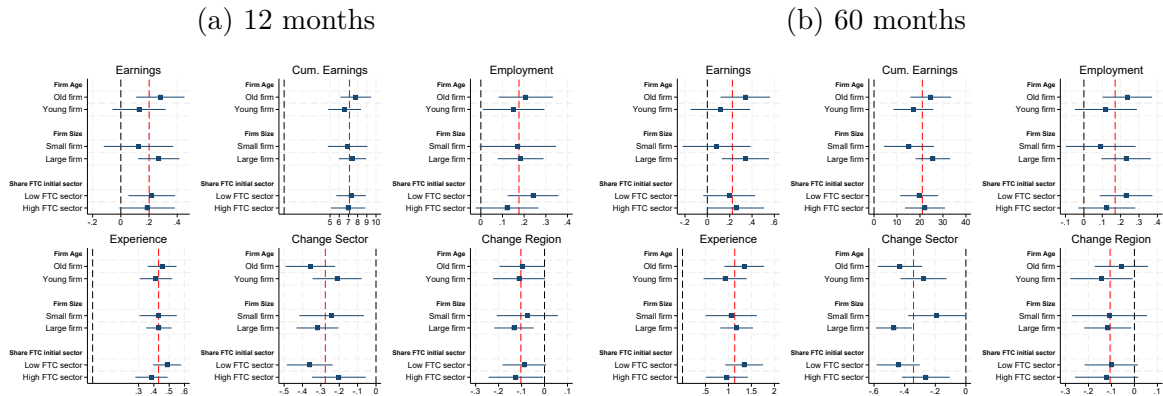
B.15 Heterogeneity

Figure B.15.1: Heterogeneity of the Effect of Permanent Contracts on Worker Careers (Worker Characteristics)



Notes: IV estimated coefficients based on equation (4). Sample restrictions and controls are identical to those in Figure 4. The dashed line represents the coefficients in Table 1. Confidence intervals use standard errors clustered at the province $\times$ year level.

Figure B.15.2: Heterogeneity of the Effect of Permanent Contracts on Worker Careers (Firm Characteristics)



Notes: IV estimated coefficients based on equation (4). Sample restrictions and controls are identical to those in Figure 4. The dashed line represents the coefficients in Table 1. Confidence intervals use standard errors clustered at the province $\times$ year level.

C Supplementary Tables

C.1 Descriptive Statistics

Table C.1.1: Characteristics of workers in fixed-term contracts

	2004	2008	2012	2016
Age group				
<24	0.207	0.174	0.116	0.112
24-35	0.487	0.458	0.433	0.388
36-50	0.262	0.316	0.373	0.400
>50	0.044	0.052	0.079	0.099
Foreign	0.137	0.234	0.205	0.176
Female	0.429	0.457	0.500	0.489
Part-time	0.192	0.198	0.308	0.317
Occupations				
Very high skilled occupations	0.050	0.059	0.083	0.080
High-skilled occupations	0.070	0.081	0.100	0.095
Medium-high skilled occupations	0.117	0.126	0.142	0.134
Medium low skilled occupations	0.475	0.479	0.431	0.419
Low-skilled occupations	0.288	0.255	0.244	0.272

Notes: Characteristics of workers employed under fixed-term contracts.

Table C.1.2: Descriptive statistics of the estimation sample

	Mean	Standard Deviation
Age	30.12	7.46
Female	0.42	0.49
Education		
<i>Below Secondary</i>	0.57	0.49
<i>Secondary</i>	0.23	0.42
<i>Tertiary</i>	0.20	0.40
Tenure	0.81	0.42
Experience	5.94	5.16
Daily Earnings (EUR 2009)	39.32	16.34
Instrument ($\log OEC_{-i,r,t+1}$)	4.77	1.35
Net Instrument ($\log OEC_{-i,r,t+1}$)	0.00	0.19
Obs.	701,761	

Notes: Descriptive statistics for the estimation sample, which consists of native workers aged 18-49 years who were in the last month of a fixed-term contract between 2000 and 2017. The net instrument is the mean and standard deviation of the first lead of log new open-ended contracts net of all controls included in the first stage.

Table C.1.3: Descriptive statistics of the complete sample

	Mean	Standard Deviation
Age	34.51	7.74
Female	0.46	0.49
Education		
<i>Below Secondary</i>	0.45	0.49
<i>Secondary</i>	0.25	0.44
<i>Tertiary</i>	0.29	0.46
Tenure	4.17	4.79
Experience	9.61	6.87
Daily Earnings (EUR 2009)	54.44	33.14
Monthly number of workers	311,150	41,517
Obs.	80,972,294	

Notes: Descriptive statistics for the complete sample, which consists of workers aged 18-49 years between 1998 and 2017.

C.2 IV Estimates: Additional Results

Table C.2.1: Effect of Permanent Contracts on Worker Careers (OLS)

	(1)	(2)	(3)	(4)	(5)	(6)
	Earnings	Cum. Earnings	Employment	Experience	Change Region	Change Sector
<i>Panel A: Short-term effects (12 months)</i>						
$p_{i,t+1}$	0.276*** (0.005)	5.388*** (0.065)	0.254*** (0.004)	0.388*** (0.004)	-0.095*** (0.002)	-0.319*** (0.003)
Obs.	701,761	701,761	701,761	701,761	701,761	701,761
Mean Dep.	-0.223	7.739	0.696	6.574	0.118	0.354
<i>Panel B: Long-term effects (60 months)</i>						
$p_{i,t+1}$	0.123*** (0.004)	17.327*** (0.265)	0.092*** (0.003)	0.991*** (0.016)	-0.111*** (0.002)	-0.312*** (0.002)
Obs.	701,761	701,761	701,761	701,761	701,761	701,761
Mean Dep.	-0.264	39.204	0.563	9.110	0.201	0.553

Notes: OLS estimates of Equation (4). Sample and controls as in Figure 4. Standard errors clustered at the province $\times$ year level in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2.2: Effect of Permanent Contracts on Earnings (Robustness)

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: 12-month horizon</i>					
$p_{i,t+1}$	0.201*** (0.067)	0.196*** (0.068)	0.202*** (0.067)	0.200*** (0.068)	0.221*** (0.065)
Obs.	701,761	701,761	701,761	701,761	701,761
R^2	0.099	0.099	0.100	0.100	0.103
<i>Panel B: 60-month horizon</i>					
$p_{i,t+1}$	0.224** (0.097)	0.222** (0.098)	0.224** (0.098)	0.220** (0.099)	0.235** (0.096)
Obs.	701,761	701,761	701,761	701,761	701,761
R^2	0.230	0.231	0.232	0.231	0.233
Occupation $\times$ Month FE	No	Yes	No	No	No
Occupation $\times$ Year FE	No	No	Yes	No	No
Sector $\times$ Month FE	No	No	No	Yes	No
Sector $\times$ Year FE	No	No	No	No	Yes

Notes: The table reports IV estimates based on equation 4 for earnings measured 12 and 60 months after the expiration of the fixed-term contract. The sample restrictions and set of controls are identical to those used in the reduced-form specification described in the notes to Figure 4. Column (1) presents the baseline specification. Columns (2) and (3) include occupation $\times$ month and occupation $\times$ year fixed effects, respectively, while Columns (4) and (5) incorporate sector $\times$ month and sector $\times$ year fixed effects. Robust standard errors, clustered at the province $\times$ year level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2.3: Effect of switching to an OEC on OEC Experience

	(1)	(2)	(3)
	OEC Status	Experience	OEC Experience
<i>Panel A: Short-term effects (12 months)</i>			
$p_{i,t+1}$	0.680*** (0.035)	0.431*** (0.036)	0.873*** (0.191)
Obs.	701,761	701,761	701,761
Mean Dep.	0.315	6.574	1.407
<i>Panel B: Long-term effects (60 months)</i>			
$p_{i,t+1}$	0.257*** (0.056)	1.137*** (0.169)	2.757*** (0.251)
Obs.	701,761	701,761	701,761
Mean Dep.	0.332	9.110	2.780

Notes: IV estimates of Equation (4). Sample and controls as in Figure 4. Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2.4: Effect of switching to an OEC on firm mobility

	(1)	(2)	(3)	(4)
	12 months	18 months	24 months	48 months
<i>Outcome: Firm mobility (counts)</i>				
$p_{i,t+1}$	-0.823*** (0.107)	-1.117*** (0.138)	-1.320*** (0.165)	-1.660*** (0.267)
Obs.	701,761	701,761	701,761	701,761

Notes: IV estimates of Equation (4). The outcome is the cumulative number of firm changes between baseline and h months after fixed-term contract expiration, for $h \in \{12, 18, 24, 48\}$. Sample and controls as in Figure 4. Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Heterogeneity IV

Table C.3.1: Heterogeneity of the Treatment Effects: Firm characteristics (12 months)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	First Stage	Earnings	Cum. Earnings	Employment	Experience	Change Sector	Change Region
Firm's age							
Old firm	0.063***	0.280***	7.800***	0.207***	0.454***	-0.357***	-0.097*
Obs. 332,337	(0.006)	(0.088)	(0.853)	(0.064)	(0.048)	(0.069)	(0.051)
Young firm	0.055***	0.129	6.580***	0.151**	0.413***	-0.212***	-0.111*
Obs. 369,424	(0.005)	(0.096)	(0.914)	(0.073)	(0.054)	(0.068)	(0.058)
Firm's size							
Small firm	0.045***	0.125	6.916***	0.171*	0.428***	-0.241***	-0.075
Obs. 336,447	(0.004)	(0.126)	(1.102)	(0.090)	(0.062)	(0.090)	(0.068)
Large firm	0.074***	0.269***	7.451***	0.183***	0.433***	-0.319***	-0.133***
Obs. 365,314	(0.006)	(0.075)	(0.754)	(0.054)	(0.042)	(0.058)	(0.044)
Share of FTC at initial sector							
Low FTC	0.065***	0.219***	7.296***	0.241***	0.487***	-0.360***	-0.088*
Obs. 312,936	(0.006)	(0.084)	(0.825)	(0.060)	(0.047)	(0.064)	(0.049)
High FTC	0.055***	0.186*	6.965***	0.121	0.384***	-0.202***	-0.125**
Obs. 388,825	(0.004)	(0.100)	(0.950)	(0.074)	(0.054)	(0.075)	(0.061)

Notes: This table reports first-stage and IV estimates of the effect of upgrading to an open-ended contract on earnings and employment twelve months after the expiration of a fixed-term contract, separately by firm characteristics. Column (1) reports the first-stage coefficient on the leave-out instrument, $\log OEC_{-i,r,t+1}$. Columns (2)–(7) report the corresponding second-stage estimates, where the endogenous variable is an indicator for holding an open-ended contract in $t + 1$, instrumented as described in Equation 4. All specifications include the full set of leads and lags of the instrument and the baseline controls X_{it} . Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3.2: Heterogeneity of the Treatment Effects: Firm characteristics (60 months)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	First Stage	Earnings	Cum. Earnings	Employment	Experience	Change Sector	Change Region
Firm's age							
Old firm	0.063***	0.340***	24.712***	0.236***	1.345***	-0.431***	-0.057
Obs. 332,337	(0.006)	(0.113)	(4.543)	(0.069)	(0.221)	(0.074)	(0.059)
Young firm	0.055***	0.116	17.116***	0.120	0.927***	-0.277***	-0.144**
Obs. 369,424	(0.005)	(0.136)	(4.425)	(0.086)	(0.239)	(0.077)	(0.069)
Firm's size							
Small firm	0.045***	0.083	15.211***	0.092	1.060***	-0.190**	-0.109
Obs. 336,447	(0.004)	(0.156)	(5.563)	(0.097)	(0.285)	(0.095)	(0.083)
Large firm	0.074***	0.341***	25.644***	0.230***	1.177***	-0.470***	-0.116**
Obs. 365,314	(0.006)	(0.108)	(3.838)	(0.069)	(0.181)	(0.060)	(0.052)
Share of FTC at initial sector							
Low FTC	0.065***	0.196*	19.614***	0.230***	1.342***	-0.442***	-0.101*
Obs. 312,936	(0.006)	(0.118)	(4.204)	(0.073)	(0.210)	(0.072)	(0.059)
High FTC	0.055***	0.257**	22.174***	0.125	0.968***	-0.261***	-0.121*
Obs. 388,825	(0.004)	(0.129)	(4.446)	(0.079)	(0.232)	(0.080)	(0.070)

Notes: This table reports first-stage and IV estimates of the effect of upgrading to an open-ended contract on earnings and employment sixty months after the expiration of a fixed-term contract, separately by firm characteristics. Column (1) reports the first-stage coefficient on the leave-out instrument, $\log OEC_{-i,r,t+1}$. Columns (2)–(7) report the corresponding second-stage estimates, where the endogenous variable is an indicator for holding an open-ended contract in $t + 1$, instrumented as described in Equation 4. All specifications include the full set of leads and lags of the instrument and the baseline controls X_{it} . Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3.3: Heterogeneity of the Treatment Effects: Worker characteristics (12 months)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	First Stage	Earnings	Cum. Earnings	Employment	Experience	Change Sector	Change Region
Education							
Below Secondary	0.054***	0.209**	7.659***	0.198***	0.479***	-0.210***	-0.076
Obs. 403,345	(0.004)	(0.097)	(0.889)	(0.074)	(0.052)	(0.069)	(0.053)
At least Secondary	0.067***	0.216**	6.709***	0.169***	0.386***	-0.346***	-0.137**
Obs. 298,416	(0.007)	(0.093)	(0.908)	(0.063)	(0.050)	(0.071)	(0.054)
Gender							
Male	0.058***	0.192**	6.853***	0.185***	0.412***	-0.326***	-0.072
Obs. 406,060	(0.005)	(0.083)	(0.829)	(0.063)	(0.046)	(0.067)	(0.055)
Female	0.061***	0.225**	7.781***	0.174**	0.475***	-0.193***	-0.138***
Obs. 295,701	(0.005)	(0.098)	(0.941)	(0.072)	(0.052)	(0.072)	(0.047)
Age							
Age < 30	0.058***	0.192**	7.135***	0.163**	0.431***	-0.352***	-0.107**
Obs. 397,968	(0.005)	(0.094)	(0.931)	(0.064)	(0.047)	(0.068)	(0.050)
Age > 30	0.060***	0.223**	6.992***	0.196***	0.421***	-0.163**	-0.094*
Obs. 303,793	(0.005)	(0.094)	(0.876)	(0.073)	(0.055)	(0.071)	(0.056)
Earnings at baseline							
Low Earnings	0.047***	0.014	6.905***	0.105	0.422***	-0.193**	-0.060
Obs. 323,757	(0.004)	(0.137)	(1.248)	(0.092)	(0.065)	(0.091)	(0.063)
High Earnings	0.071***	0.339***	7.341***	0.228***	0.436***	-0.328***	-0.136***
Obs. 378,004	(0.006)	(0.060)	(0.612)	(0.052)	(0.040)	(0.058)	(0.046)

Notes: This table reports first-stage and IV estimates of the effect of upgrading to an open-ended contract on earnings and employment twelve months after the expiration of a fixed-term contract, separately by baseline worker characteristics. Column (1) reports the first-stage coefficient on the leave-out instrument, $\log \text{OEC}_{-i,r,t+1}$. Columns (2)–(7) report the corresponding second-stage estimates, where the endogenous variable is an indicator for holding an open-ended contract in $t + 1$, instrumented as described in Equation 4. All specifications include the full set of leads and lags of the instrument and the baseline controls X_{it} . Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3.4: Heterogeneity of the Treatment Effects: Worker characteristics (60 months)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	First Stage	Earnings	Cum. Earnings	Employment	Experience	Change Sector	Change Region
Education							
Below Secondary	0.054***	0.307***	22.746***	0.215***	1.279***	-0.327***	-0.068
Obs. 403,345	(0.004)	(0.119)	(3.984)	(0.077)	(0.228)	(0.076)	(0.062)
At least Secondary	0.067***	0.146	20.165***	0.137*	1.039***	-0.358***	-0.151**
Obs. 298,416	(0.007)	(0.138)	(5.067)	(0.081)	(0.229)	(0.072)	(0.067)
Gender							
Male	0.058***	0.347***	23.073***	0.214***	1.213***	-0.370***	-0.054
Obs. 406,060	(0.005)	(0.117)	(4.038)	(0.074)	(0.210)	(0.072)	(0.064)
Female	0.061***	0.058	19.222***	0.123	1.112***	-0.287***	-0.163***
Obs. 295,701	(0.005)	(0.133)	(4.618)	(0.084)	(0.236)	(0.080)	(0.062)
Age							
Age < 30	0.058***	0.257**	18.617***	0.154**	0.988***	-0.326***	-0.109*
Obs. 397,968	(0.005)	(0.124)	(4.733)	(0.074)	(0.223)	(0.075)	(0.065)
Age > 30	0.060***	0.224*	23.917***	0.200**	1.280***	-0.340***	-0.094
Obs. 303,793	(0.005)	(0.120)	(4.151)	(0.084)	(0.236)	(0.073)	(0.063)
Earnings at baseline							
Low Earnings	0.047***	0.060	14.007**	0.156	0.943***	-0.289***	0.031
Obs. 323,757	(0.004)	(0.182)	(6.496)	(0.098)	(0.289)	(0.098)	(0.081)
High Earnings	0.071***	0.326***	26.027***	0.188***	1.263***	-0.382***	-0.201***
Obs. 378,004	(0.006)	(0.085)	(2.973)	(0.066)	(0.185)	(0.061)	(0.055)

Notes: This table reports first-stage and IV estimates of the effect of upgrading to an open-ended contract on earnings and employment sixty months after the expiration of a fixed-term contract, separately by baseline worker characteristics. Column (1) reports the first-stage coefficient on the leave-out instrument, $\log OEC_{-i,r,t+1}$. Columns (2)–(7) report the corresponding second-stage estimates, where the endogenous variable is an indicator for holding an open-ended contract in $t + 1$, instrumented as described in Equation 4. All specifications include the full set of leads and lags of the instrument and the baseline controls X_{it} . Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Social Security records

Table C.4.1: Relationship between New Open-Ended Contracts and Total OEC from Social Security records

	(1)	(2)	(3)
	$\log NewOEC_{prov}$		
$\log OEC_{total}$	1.084*** (0.006)	1.083*** (0.005)	1.309*** (0.138)
Obs.	6,697	6,697	6,697
R^2	0.810	0.907	0.941
Time FE	No	Yes	Yes
Region FE	No	No	Yes

Notes: The table presents the regression coefficients of the logarithm of new open-ended Contracts by province from the MCVL on the logarithm of total OEC registered in the population records of the Social Security between January 2009 and March 2020. Column (1) presents the baseline relationship between these variables. Columns (2) and (3) additionally control for year-month and province-fixed effects, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Compliers

Table C.5.1: Average Characteristics of Always-Takers, Compliers, and Never-Takers

Panel A: Worker characteristics				
	At least Secondary Education	Age<30	Above median Initial Earnings	Female
Whole sample	0.423 (0.001)	0.578 (0.001)	0.500 (0.001)	0.417 (0.001)
Complier	0.448 (0.030)	0.603 (0.031)	0.576 (0.031)	0.410 (0.031)
Always-taker	0.509 (0.001)	0.568 (0.001)	0.545 (0.001)	0.460 (0.001)
Never-taker	0.383 (0.001)	0.581 (0.001)	0.477 (0.001)	0.397 (0.001)

Panel B: Firm characteristics			
	Large firm	Young firm	High sectoral share FTC
Whole sample	0.477 (0.001)	0.425 (0.001)	0.552 (0.001)
Complier	0.515 (0.030)	0.273 (0.031)	0.500 (0.029)
Always-taker	0.521 (0.001)	0.413 (0.001)	0.435 (0.001)
Never-taker	0.455 (0.001)	0.435 (0.001)	0.606 (0.001)

Population shares: complier 0.019 (0.001), always-taker 0.306 (0.001), never-taker 0.675 (0.001)

Notes: Descriptive statistics for complier and non-complier subpopulations based on individuals whose treatment status (the probability of being upgraded to an open-ended contract) is affected by the creation of new open-ended contracts. The instrument is constructed as the residualized count of new open-ended contracts, with individuals classified as one when this residualized count is positive. Standard errors are obtained by bootstrap.

C.6 Mechanism: lock-in

Table C.6.1: Employment Lock-In and Earnings Divergence: Heterogeneity by Firm Size

	Panel A: $h = 12$ months				Panel B: $h = 60$ months			
	Median split		Top-quartile split		Median split		Top-quartile split	
	Small	Large	Below p75	Top quartile	Small	Large	Below p75	Top quartile
Earnings	0.125 (0.126)	0.269*** (0.075)	0.125 (0.082)	0.468*** (0.114)	0.083 (0.156)	0.341*** (0.108)	0.148 (0.104)	0.519*** (0.159)
Cumulative earnings	6.916*** (1.102)	7.451*** (0.754)	7.039*** (0.759)	7.726*** (1.084)	15.211*** (5.563)	25.644*** (3.838)	17.496*** (3.708)	33.890*** (5.801)
Employment	0.171* (0.090)	0.183*** (0.054)	0.134** (0.060)	0.319*** (0.082)	0.092 (0.097)	0.230*** (0.069)	0.138** (0.067)	0.293*** (0.098)
Experience (years)	0.428*** (0.062)	0.433*** (0.042)	0.427*** (0.043)	0.452*** (0.064)	1.060*** (0.285)	1.177*** (0.181)	1.049*** (0.199)	1.458*** (0.273)
Change region	-0.075 (0.068)	-0.133*** (0.044)	-0.075 (0.046)	-0.198*** (0.065)	-0.109 (0.083)	-0.116** (0.052)	-0.089 (0.056)	-0.179** (0.078)
Change sector	-0.241*** (0.090)	-0.319*** (0.058)	-0.221*** (0.060)	-0.464*** (0.089)	-0.190** (0.095)	-0.470*** (0.060)	-0.274*** (0.064)	-0.578*** (0.089)
Observations	336,447	365,314	518,688	183,073	336,447	365,314	518,688	183,073

Notes: IV estimates of equation (4). Firms are classified by baseline employment. Small (Large) firms are those below (above) the sample median. Very large firms are those in the top quartile of the firm-size distribution; Below p75 includes all remaining firms. Sample restrictions and controls are as in Figure 4. Robust standard errors clustered at the province $\times$ year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6.2: Employment Lock-In and Earnings Divergence: Heterogeneity by Firm Growth (year FE)

	Panel A: $h = 12$ months		Panel B: $h = 60$ months	
	Below median	Above median	Below median	Above median
<i>First stage</i>				
$\log OEC_{t+1}$	0.069*** (0.006)	0.066*** (0.007)	0.069*** (0.006)	0.066*** (0.007)
<i>Earnings and employment</i>				
Monthly earnings	0.300*** (0.112)	0.216** (0.103)	0.218 (0.151)	0.409*** (0.136)
Cumulative earnings	8.636*** (1.207)	5.980*** (0.958)	26.555*** (5.741)	24.084*** (4.887)
Employment	0.215*** (0.077)	0.180** (0.082)	0.148* (0.087)	0.278*** (0.090)
Experience (years)	0.505*** (0.065)	0.358*** (0.057)	1.327*** (0.269)	1.064*** (0.248)
<i>Mobility</i>				
Change sector	-0.357*** (0.085)	-0.281*** (0.081)	-0.500*** (0.090)	-0.327*** (0.084)
Change region	-0.165** (0.066)	-0.083 (0.065)	-0.234*** (0.079)	-0.033 (0.078)
Mean dep. (earnings)	-0.189	-0.240	-0.265	-0.240
Observations	207,880	205,290	207,880	205,290

Notes: IV estimates of equation 4. Firms are classified by their leave-one-out average monthly earnings growth at baseline, computed excluding each worker's own contribution to the firm-level average. Firm earnings growth is measured as the average growth in monthly earnings after residualizing interactions of age, gender, and education and year fixed-effects. Low-growth (high-growth) firms are those below (above) the sample median. Sample restrictions and control variables are identical to those in Figure 4. Robust standard errors clustered at the province $\times$ year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6.3: Employment Lock-In and Earnings Divergence: Heterogeneity by Firm Growth (Large firms)

	Panel A: $h = 12$ months		Panel B: $h = 60$ months	
	Below median	Above median	Below median	Above median
<i>First stage</i>				
$\log OEC_{t+1}$	0.072*** (0.008)	0.059*** (0.007)	0.072*** (0.008)	0.059*** (0.007)
<i>Earnings and employment</i>				
Monthly earnings	0.182 (0.131)	0.382*** (0.135)	0.158 (0.175)	0.521*** (0.180)
Cumulative earnings	8.113*** (1.313)	7.571*** (1.350)	26.983*** (6.560)	30.603*** (6.635)
Employment	0.133 (0.095)	0.255** (0.106)	0.076 (0.103)	0.289** (0.123)
Experience (years)	0.466*** (0.073)	0.427*** (0.077)	1.127*** (0.307)	1.251*** (0.332)
<i>Mobility</i>				
Change sector	-0.334*** (0.101)	-0.141 (0.117)	-0.465*** (0.109)	-0.302*** (0.113)
Change region	-0.060 (0.074)	-0.165* (0.091)	-0.110 (0.090)	-0.159 (0.107)
Mean dep. (earnings)	-0.178	-0.215	-0.209	-0.308
Observations	129,643	130,996	129,643	130,996

Notes: IV estimates of equation 4. Firms are classified by their baseline leave-one-out average monthly earnings growth, computed excluding each worker's own contribution to the firm-level average. The sample is restricted to firms with at least five workers per year. Firm earnings growth is measured as the average growth in monthly earnings after residualizing interactions of age, gender, and education. Low-growth (high-growth) firms are those below (above) the sample median. Sample restrictions and controls are identical to those in Figure 4. Robust standard errors clustered at the province $\times$ year level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6.4: Effect of Permanent Contracts on Worker Careers (Leave-one-out sample)

	(1)	(2)	(3)	(4)	(5)	(6)
	Earnings	Cum. Earnings	Employment	Experience	Change Region	Change Sector
<i>Panel A: Short-term effects (12 months)</i>						
$p_{i,t+1}$	0.255*** (0.076)	7.169*** (0.735)	0.193*** (0.057)	0.423*** (0.042)	-0.114** (0.046)	-0.319*** (0.060)
Obs.	413,170	413,170	413,170	413,170	413,170	413,170
Mean Dep.	-0.214	7.842	0.713	6.450	0.125	0.378
<i>Panel B: Long-term effects (60 months)</i>						
$p_{i,t+1}$	0.318*** (0.111)	25.326*** (3.750)	0.215*** (0.070)	1.199*** (0.183)	-0.121** (0.055)	-0.410*** (0.064)
Obs.	413,170	413,170	413,170	413,170	413,170	413,170
Mean Dep.	-0.253	40.086	0.579	9.049	0.209	0.574

Notes: IV estimates of Equation (4). $p_{i,t+1}$ indicates whether worker i holds an open-ended contract in $t + 1$, the month immediately following fixed-term contract expiration. Sample and controls as in Figure 4. Standard errors clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.7 Two instruments

Table C.7.1: Effect of Permanent Contracts on Worker Careers (Endogenous Employment)

	First Stage		Second Stage			
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Contract</i>	<i>OEC</i> _{<i>t</i>+1}	<i>Employed</i> _{<i>t</i>+1}	Earnings +12	Earnings +60	
$p_{i,t+1}$			0.133** (0.067)	0.092 (0.076)	0.110 (0.081)	0.109 (0.095)
$e_{i,t+1}$				0.222** (0.089)		0.004 (0.110)
$\log OEC_{-i,r,t+1}$	0.090*** (0.004)	0.016*** (0.005)				
$\log TNC_{-i,r,t+1}$	0.017*** (0.006)	0.120*** (0.008)	0.026** (0.011)		0.001 (0.013)	
Obs	725,992	725,992	725,992	725,992	725,992	725,992

Notes: The table reports the first-stage and IV estimates from Equation 4. Sample restrictions and controls are identical to those in the reduced-form specification described in the notes to Figure 4, except that the specification here additionally includes leads and lags of the leave-out measures for new OECs and total contracts. New OECs are measured at the month-autonomous-community level. Columns (1) and (2) present the first-stage results for OEC status and employment in period $t + 1$, respectively. Columns (3) and (4) report the second-stage estimates for changes in earnings 12 periods later, where the endogenous variables—OEC status and employment in $t + 1$ —are instrumented using the leave-out measures of new OECs and total contracts. Columns (5) and (6) repeat the analysis using earnings 60 months later as the dependent variable. Columns (3) and (5) include only one endogenous variable (OEC status), whereas Columns (4) and (6) include both endogenous variables, OEC status and employed in $t + 1$. The observations are 725,992 slightly larger than baseline. Robust standard errors, clustered at the province $\times$ year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.